



Marvell® QLogic® Control Suite CLI

3400/8400/4500/41000 Series Adapters and Adapters Based on 57xxx Controllers

User's Guide

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Preface

This guide describes how to install, start, and use QLogic® Control Suite™ CLI, a primary CLI for configuring Marvell FastLinQ®41000/45000 Series Adapters and adapters based on 57xxx controllers. QLogic Control Suite CLI can also be used to list and view information about the legacy network adapters (that is, adapters not based on the 57xxx/41000/45000 Series Controllers).

NOTE

Throughout this guide, the QLogic Control Suite CLI management tool is also referred to as *QCS CLI*.

NOTE

Marvell supports QConvergeConsole® GUI as the *only* GUI management tool across all adapters. QLogic Control Suite GUI is no longer supported for the 3400/8400 Series Adapters and adapters based on 57xxx controllers, and has been replaced by the QConvergeConsole GUI management tool. QConvergeConsole GUI provides single-pane-of-glass GUI management for all Marvell QLogic adapters.

In Windows® environments, when you run the QCS CLI and Management Agents Installer, it will uninstall QLogic Control Suite (QCS) GUI (if installed on the system) and any related components from your system. To obtain the new GUI, download QLogic Control Suite CLI for your adapter from the Marvell Web site, www.marvell.com.

Intended Audience

This guide is designed for users who are managing Marvell 3400/8400/41000/45000 Series Adapters and adapters based on 57xxx controllers using QLogic Control Suite CLI.

What Is in This Guide

This preface specifies the intended audience, lists related documents, explains typographic conventions used in this guide, and provides technical support information.

The remainder of the guide is organized into the following chapters and appendices:

- [Chapter 1 Introduction](#) describes the components, architecture, and structure of QLogic Control Suite CLI.
- [Chapter 2 Installing and Launching](#) lists the QLogic Control Suite CLI requirements, package contents, and supported OS and adapters. It provides QCS CLI and management agent installation instructions, as well as how to launch the CLI and add a host.
- [Chapter 3 QCS CLI Commands](#) provides details about commands in QLogic Control Suite CLI, including a description, syntax, keywords, and examples for each command.
- [Chapter 4 QCS CLI Usage](#) covers using interactive mode and provides task-oriented information using QLogic Control Suite CLI.
- [Appendix A Security Considerations](#) provides security information for QLogic Control Suite CLI.
- [Appendix B Configuration File Examples](#) provides samples of the team, multi-boot agent, iSCSI boot, FCoE boot, NIC partition, and SR-IOV configuration files used with QLogic Control Suite CLI.
- [Appendix C Third-party Licenses](#) details license information for third-party software used in QLogic Control Suite CLI.
- [Appendix D Exit Codes](#) lists the exit codes that indicate the level of success for each command.
- [Appendix E Revision History](#) contains a list of changes made to this guide since the last revision.

Included also are a glossary of terms used and an index to help you quickly find the information you need.

Related Materials

For additional information, refer to following document that is available from the Marvell Web site, www.marvell.com:

- QLogic Control Suite CLI *Read Me* and *Release Notes* documents contain information about installation, OS support, and known issues.

For information about downloading documentation from the Marvell Web site, see “[Downloading Updates and Documentation](#)” on page xiv.

Documentation Conventions

This guide uses the following documentation conventions:

- **NOTE** provides additional information.
- Text in [blue](#) font indicates a hyperlink (jump) to a figure, table, or section in this guide, and links to Web sites are shown in [underlined blue](#). For example:
 - ❑ [Table 9-2](#) lists problems related to the user interface and remote agent.
 - ❑ See “[Installation Checklist](#)” on page 6.
 - ❑ For more information, visit www.marvell.com.
- Text in **bold** font indicates user interface elements such as a menu items, buttons, check boxes, or column headings. For example:
 - ❑ Click the **Start** button, point to **Programs**, point to **Accessories**, and then click **Command Prompt**.
 - ❑ Under **Notification Options**, select the **Warning Alarms** check box.
- Text in `Courier` font indicates a file name, directory path, or command line text. For example:
 - ❑ To return to the root directory from anywhere in the file structure:
Type `cd /root` and press ENTER.
 - ❑ Issue the following command: `sh ./install.bin`
- Key names and key strokes are indicated with UPPERCASE:
 - ❑ Press CTRL+P.
 - ❑ Press the UP ARROW key.
- Text in *italics* indicates terms, emphasis, variables, or document titles. For example:
 - ❑ For a complete listing of license agreements, refer to the applicable *Software End User License Agreement*.

- ❑ What are *shortcut keys*?
- ❑ To enter the date, type *mm/dd/yyyy* (where *mm* is the month, *dd* is the day, and *yyyy* is the year).
- Topic titles between quotation marks identify related topics within this guide.
- CLI command syntax conventions include the following:
 - ❑ Plain text indicates items that you must type as shown. For example:
 - **cfg iSCSIBoot CDUMP=Enable**
 - ❑ < > (angle brackets) indicate a variable whose value you must specify. For example:
 - <serial_number>

NOTE

For CLI commands only, variable names are always indicated using angle brackets instead of *italics*.

- ❑ [] (square brackets) indicate an optional parameter. For example:
 - [<file_name>] means specify a file name, or omit it to select the default file name.
- ❑ | (vertical bar) indicates mutually exclusive options; select one option only. For example:
 - on|off
 - 1|2|3|4
- ❑ ... (ellipsis) indicates that the preceding item may be repeated. For example:
 - x... means *one* or more instances of x.
 - [x...] means *zero* or more instances of x.
- ❑ ⋮ (vertical ellipsis) within command example output indicate where portions of repetitious output data have been intentionally omitted.

- ❑ () (parentheses) and { } (braces) are used to avoid logical ambiguity. For example:
 - a|b c is ambiguous
 - { (a|b) c } means a or b, followed by c
 - { a| (b c) } means either a, or b c

Technical Support

Customers should contact their authorized maintenance provider for technical support of their Marvell QLogic products. Technical support for QLogic-direct products under warranty is available with a Marvell support account.

To set up a support account (if needed) and submit a case:

1. Go to www.marvell.com.
2. Click **Support**.
3. Scroll down if needed, and then click **Ethernet Adapters**.
4. Under Related Links, click **Customer Support**.
5. Do one of the following:
 - ❑ If you do not have an account, click **Sign Up**.
 - ❑ If you have an account, enter your **User Name** and **Password**, and then click **Log In**.

Downloading Updates and Documentation

The Marvell Web site provides periodic updates to product firmware, software, and documentation.

To download QLogic firmware, software, and documentation:

1. Go to www.marvell.com.
2. Point to **Support**, and then under **Support by Product**, click **Ethernet Adapters**.
3. On the Driver Downloads page, click **Marvell Drivers**.
4. Under **Marvell Drivers**, click **NICs and HBA drivers**.
5. Click the corresponding button to search **by Model** or **by Operating System**.
6. To define a search, click an item in each selection column, and then click **Go**.
7. Locate the firmware, software, or document you need, and then click the item's name or icon to download or open the item.

1 Introduction

The QLogic Control Suite (QCS CLI) utility is a console application that you can run from a Windows command prompt or Linux terminal console. Use QLogic Control Suite CLI to manage Marvell 3400/8400/41000/45000 Series Adapters and adapters based on 57xxx controllers on both local and remote computer systems.

This chapter provides the following information to introduce QLogic Control Suite CLI:

- [QCS CLI Components](#)
- [“QCS CLI Architecture” on page 2](#)
- [“QCS CLI Structure” on page 5](#)
- [“Linux Feature Limitations” on page 8](#)

QCS CLI Components

The Marvell 3400/8400/41000/45000 Series Adapters and adapters based on 57xxx controllers comprise distinct components that are manageable elements. The manageable elements are organized in an object-oriented hierarchical manner in QLogic Control Suite CLI. Each object has an associated description for the following:

- Properties or data that the object exposes
- Whether or not the properties or data are configurable
- Whether or not actions can be performed on the object
- What relationships exist between objects

QCS CLI Architecture

Illustrations in this section show the architecture of QLogic Control Suite CLI for Windows and Linux.

Windows System Architecture

[Figure 1-1 on page 3](#) shows the relationship between the Ethernet controller and driver. A host may have multiple chips, each chip may have multiple physical ports, and each physical port may have multiple functions. Each function may contain Network Driver Interface Specification (NDIS) driver only, virtual bus driver (VBD) and NDIS, iSCSI, or Fibre Channel over Ethernet (FCoE).

Linux Systems Architecture

[Figure 1-2 on page 4](#) shows the relationship between the Ethernet controller and driver. A host may have multiple chips, each chip may have multiple physical ports, and each physical port may have multiple functions. Each function may contain a Layer 2 device only, Layer 2 device, iSCSI, or FCoE.

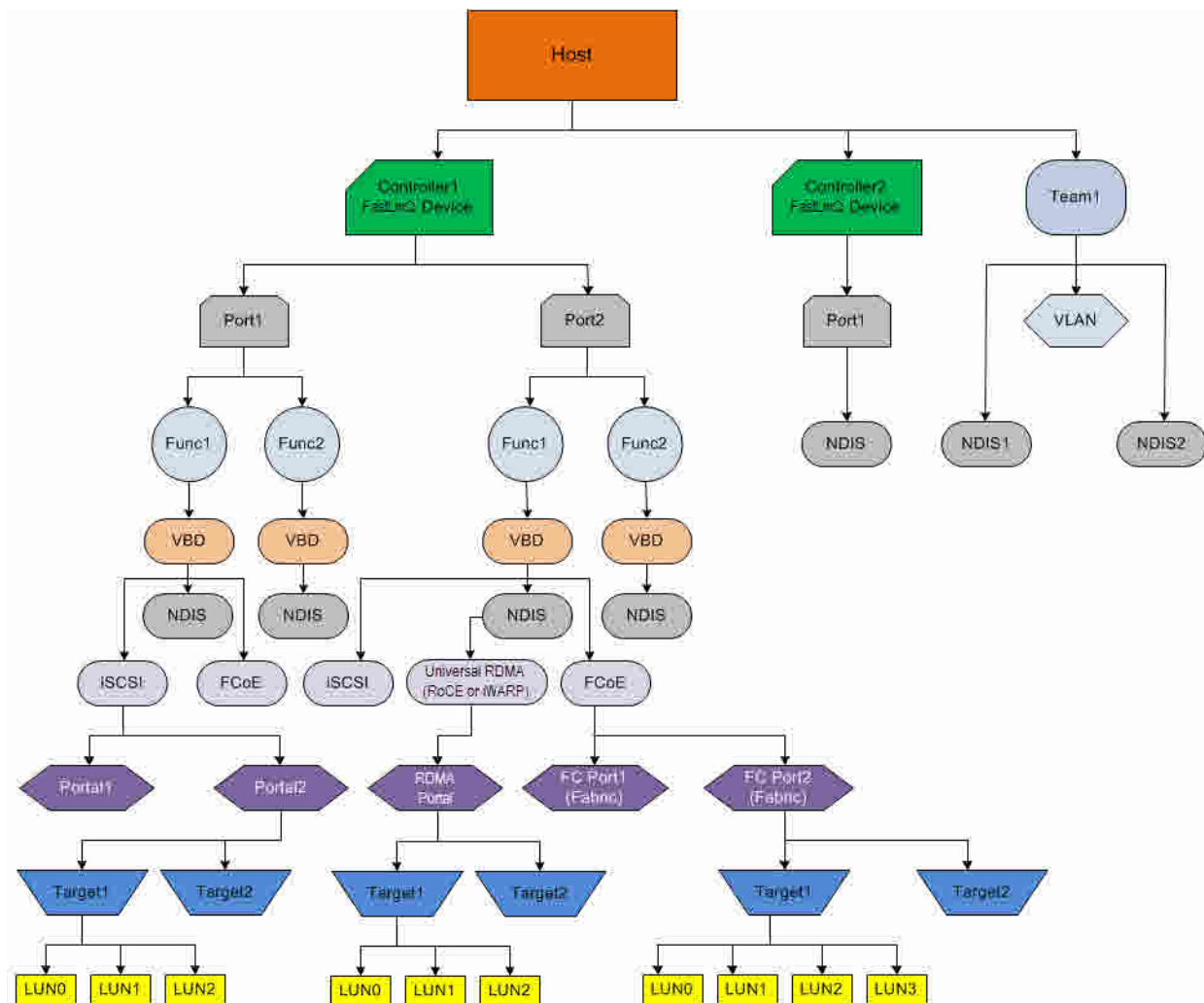


Figure 1-1. QLogic Control Suite CLI Windows Architecture

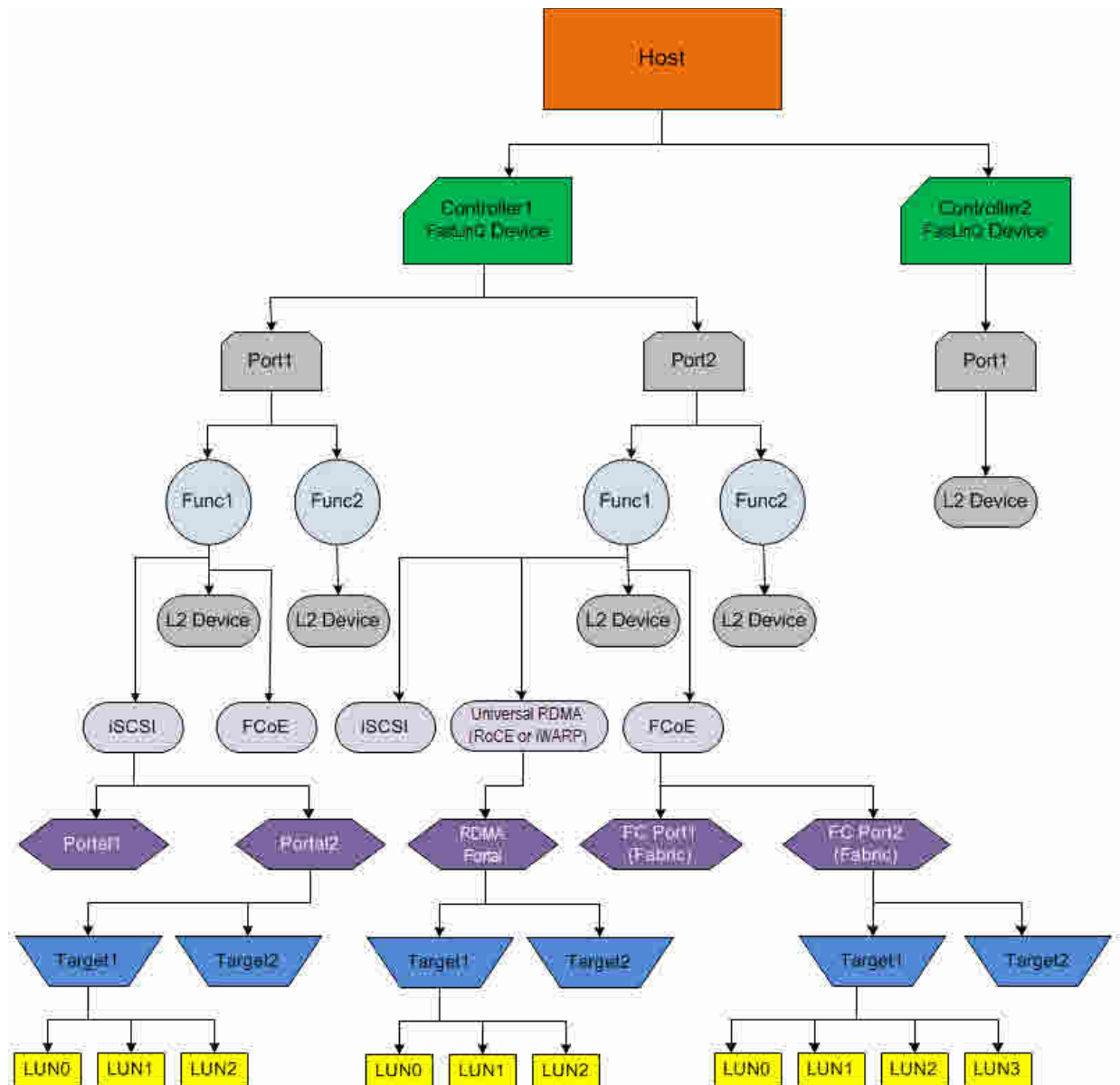


Figure 1-2. QLogic Control Suite CLI Linux Architecture

QCS CLI Structure

Information about the structure of QLogic Control Suite CLI includes the following:

- [QCS CLI Modes](#)
- [QCS CLI Elements \(Targets\)](#)
- [QCS CLI Hierarchy](#)
- [Command Categories](#)
- [Client Layout Design](#)

QCS CLI Modes

The QLogic Control Suite CLI application uses the console and can be run in either of two modes: interactive or noninteractive.

- In the **interactive** mode, users make connections to local and remote hosts, and manage them by entering commands in the QCS CLI window. The output is displayed as either information or an error message. The interactive mode is not menu based. You must enter a correct command to list specific types of targets before you can manage those targets.
- In the **noninteractive mode (also called CLI mode)**, QCS CLI can take a command as an input argument and run the command to produce the appropriate effect and output, and then return a meaningful exit code. This mode is ideal for scripting CLI commands.

QCS CLI Elements (Targets)

Each manageable element is known as a *target*. (Target, in this context, does not refer to the FCoE or iSCSI target, which is usually referred to as a storage system or LUN.) When you issue a command to list a specific target or manageable element, all the targets matching the criteria defined in the query are listed and the first element is selected by default. When you issue a command to list a specific type of target, the focus shifts from the target that you selected before the query was run to the first target in the current output.

Example 1:

Suppose the currently selected target is a host. If you issue the command to list FCoE devices, all matching FCoE devices are listed and the first FCoE device in the list is automatically selected for management, as shown in [Figure 1-3](#). If you instead want to manage the NDIS device that was originally selected, you must first list all of the NDIS devices, and then select the appropriate instance to manage it, as shown in [Figure 1-4](#).

NOTE

This example does not apply if you issue a command to list the information regarding the currently selected target.

```

Administrator: Command Prompt - qcscli
m630-lt\host\0>
m630-lt\host\0>list hosts
C      Status      Name      OS      Platform
0      connected   m630-lt   Windows Server 2012 R2   64 bit <x64>
1      connected   WIN-IUDGFC3I4II   Windows Server 2012 R2   64 bit <x64>
m630-lt\host\0>list -f -bdf fcoe
C      MAC      DevType Name
0      5CF9DD9E4ECC FCoE      Host      m630-lt
1      5CF9DD9E48B3 FCoE      -[0002] QLogic BCM57810 10 Gigabit Ethernet FCoE Adapter #38
- [0003] QLogic BCM57810 10 Gigabit Ethernet FCoE Adapter #35
2      D4AE52772D35 FCoE      Host      WIN-IUDGFC3I4II
3      D4AE52772D37 FCoE      -[0003] QLogic BCM57800 10 Gigabit Ethernet FCoE Adapter #38
- [0002] QLogic BCM57800 10 Gigabit Ethernet FCoE Adapter #39
m630-lt\FCoE\0>_
  
```

Figure 1-3. FCoE Device Selected

```

Administrator: Command Prompt - qcscli
0      5CF9DD9E4ECC FCoE      Host      m630-lt
1      5CF9DD9E48B3 FCoE      -[0002] QLogic BCM57810 10 Gigabit Ethernet FCoE Adapter #38
- [0003] QLogic BCM57810 10 Gigabit Ethernet FCoE Adapter #35
2      D4AE52772D35 FCoE      Host      WIN-IUDGFC3I4II
3      D4AE52772D37 FCoE      -[0003] QLogic BCM57800 10 Gigabit Ethernet FCoE Adapter #38
- [0002] QLogic BCM57800 10 Gigabit Ethernet FCoE Adapter #39
m630-lt\FCoE\0>list -f bdf ndis
C      B:D.F      DevType Name
0      01:00.00      Host      m630-lt
1      01:00.01      NDIS      -[0019] QLogic BCM57810 10 Gigabit Ethernet <NDIS UBD Client> #37
2      03:00.00      NDIS      -[0020] QLogic BCM57810 10 Gigabit Ethernet <NDIS UBD Client> #38
3      03:00.01      NDIS      -[0013] Broadcom NetXtreme Gigabit Ethernet #3
4      03:00.02      NDIS      -[0010] Broadcom NetXtreme Gigabit Ethernet #2
5      03:00.03      NDIS      -[0012] Broadcom NetXtreme Gigabit Ethernet #4
6      04:00.00      NDIS      -[0011] QLogic BCM57810 10 Gigabit Ethernet <NDIS UBD Client> #35
7      04:00.01      NDIS      -[0017] QLogic BCM57810 10 Gigabit Ethernet <NDIS UBD Client> #36
8      01:00.00      Host      WIN-IUDGFC3I4II
9      01:00.01      NDIS      -[0016] QLogic BCM57800 10 Gigabit Ethernet <NDIS UBD Client> #38
10     01:00.02      NDIS      -[0014] QLogic BCM57800 10 Gigabit Ethernet <NDIS UBD Client> #39
11     01:00.03      NDIS      -[0013] QLogic BCM57800 Gigabit Ethernet <NDIS UBD Client> #40
12     01:00.00      NDIS      -[0015] QLogic BCM57800 Gigabit Ethernet <NDIS UBD Client> #41
- [0010] Intel(R) PRO/1000 PI Server Adapter
m630-lt\ndis\0>_
  
```

Figure 1-4. Focus Changed to NDIS Device

Example 2:

Suppose QCS CLI has connections to two hosts, where each host has two adapters. When you issue the command to list the adapters, all adapters are listed, grouped by host. The output shows four entries: instance 0 through 3. To select one of the adapters, issue the `select <instance_number>` command.

QCS CLI Hierarchy

QCS CLI always displays the prompt with the current focus in a hierarchical manner starting with the host. When you enter a command, output is shown, the prompt changes to the first target, starting with the host and ending with current target following a hierarchical structure. To visualize the hierarchical structure of the Converged Network Adapter, see [“Windows System Architecture” on page 2](#).

The QCS CLI interactive mode has no backward or forward levels. You must explicitly define the target that you want to manage. If the new target is different than the currently selected target, you must first list the specific target type, and then select the appropriate target.

Command Categories

All QCS CLI commands can be broadly divided in four categories:

- Commands to display information of the target
- Commands to modify the configuration of the target
- Commands to display the statistical counters for the target
- Commands to perform a diagnostic test on the target

After you make configuration changes using QCS CLI, you must refresh the host to display the most up-to-date information. For a host refresh, issue a command to list the hosts, select the appropriate host, and then issue the `refresh` command. You can then change the focus back to the target that you were managing.

Client Layout Design

Start the QCS CLI client Windows executable (`QCSCLI.exe`) in the command window and add a host. [Figure 1-5](#) shows an example.

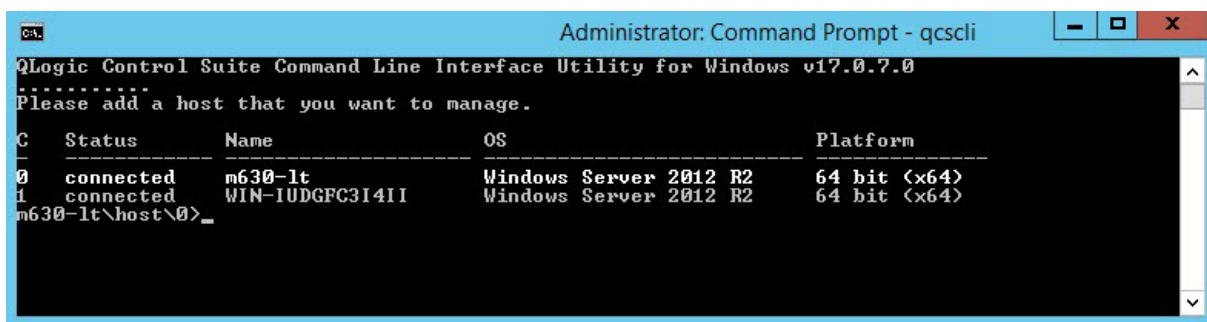


Figure 1-5. QLogic Control Suite CLI Window

The CLI application starts in a command line window. The QCS CLI window is similar to and is managed in the same way you manipulate any Windows command line window. You can resize, minimize, and close the window using the mouse. When you close the QCS CLI window, the connection to the managed host is lost.

Linux Feature Limitations

Using QLogic Control Suite CLI on a Linux system has the following limitations on the available features:

- Data center bridging netlink interface (DCBNL) is not supported for inbox drivers for Red Hat® Enterprise Linux® (RHEL®) 6.2 and SUSE® Linux Enterprise Server (SLES®) 11 SP2 and later.
- RHEL 5 (Linux kernel version 2.6.18-398 and earlier) and SLES 10 (Linux kernel version 2.6.16.60 and earlier) do not support data center bridging exchange (DCBX) due to lack of kernel support.
- Red Hat does not support the iSCSI DataDigest login option.
- The iSCSI multipath input/output (MPIO) login option is not available because MPIO for Linux iSCSI operates differently than Windows. Linux requires that two iSCSI sessions be established first to the same target LUN, and then the multipaths must be started so the SCSI layer treats the device mapper disk as one disk. Because these requirements have nothing to do with the actual iSCSI login procedure, MPIO is not available during login.
- SLES 10 (Linux kernel version 2.6.16.60 and earlier) does not support the iSCSI offload features due to lack of kernel support.
- Configuration options are not available if the device interface is down.
- The network test, on-chip CPU test, and cable analysis are not currently supported.
- The multifunction mode does not currently support the loopback PHY or loopback MAC test in 3400/8400 Series Adapters.
- Bandwidth: (QL41000 Series adapters (25G/40G/100G)): for *best practice*, set the bandwidth to 0 on storage functions.
- The iSCSI ping test may pass or fail based on the real time network traffic and bandwidth. If few pings are successful and few fail, this limitation could be expected. If the iSCSI ping test fails repeatedly, this limitation can be an issue.

- For adapters based on 57xxx controllers, some of the most recent kernels (for example, in SLES 15) have a security feature to prevent user space access to physical memory above 1MB (IIRC). Consequently, the firmware upgrade tool may not function properly.

To resolve this issue, do one of the following:

- ❑ Add `iomem=relaxed` in the kernel command line during boot time.
- ❑ Update `grub.cfg` file:
 1. Change the `GRUB_CMDLINE_LINUX_DEFAULT` parameter to `iomem=relaxed` in the `/etc/default/grub.cfg` file.
The location of this file varies depending on the Linux distribution.
 2. Issue the following command:

```
run cmd grub2-mkconfig -o /boot/grub2/grub.cfg
```
 3. Reboot the server for the changes to take effect.

2 Installing and Launching

Install the QLogic Control Suite CLI management application using the QCS CLI installer package. The installer package for Windows OS is based on the Microsoft® MSI installation technology. The managed host can be either Windows® or Linux®.

NOTE

Marvell supports QConvergeConsole GUI as the *only* GUI management tool across all adapters. QLogic Control Suite GUI is no longer supported for the 3400/8400 Series Adapters and adapters based on 57xxx controllers, and has been replaced by the QConvergeConsole GUI management tool. QConvergeConsole GUI provides single-pane-of-glass GUI management for all Marvell adapters.

In Windows® environments, when you run the QCS CLI and Management Agents Installer, it will uninstall QLogic Control Suite (QCS) GUI (if installed on the system) and any related components from your system. To obtain the new GUI, download QLogic Control Suite CLI for your adapter from the Marvell Web site, www.marvell.com.

This chapter provides the following major sections with information about QCS CLI installation and configuration:

- [“Before You Begin” on page 11](#)
- [“Downloading QLogic Control Suite CLI” on page 12](#)
- [“Downloading and Installing QCS CLI and Management Agents” on page 12](#)
- [“Launching QLogic Control Suite” on page 15](#)
- [“Launching QLogic Control Suite” on page 15](#)
- [“Adding a Host” on page 16](#)

Before You Begin

Before you begin installing QCS CLI and related applications, understand the requirements, package contents, and supported operating systems and adapters as described in this section.

- [QCS CLI Requirements](#)
- [Supported OS](#)
- [Supported Adapters](#)

QCS CLI Requirements

The section lists the hardware, software, and server required for installing and running QLogic Control Suite CLI. QCS CLI requires the following:

- Appropriate device driver for the NIC installed on the system that is to be managed by QCS CLI.
- For the FCoE and iSCSI interface, the appropriate FCoE and iSCSI device driver to correctly discover the protocol.
- For managing iSCSI on Linux hosts, Open-iSCSI and SCSI generic (sg) utilities installed on the Linux host.
- For managing FCoE on Linux hosts, Open-FCoE installed on the Linux host.
- To discover FCoE LUNs on Linux hosts, sg3_utils RPM installed on Linux host.
- The appropriate QLMAPI module.

Hardware Requirements

The minimum hardware requirement for hosts include:

- One or more Marvell 3400/8400/41000/45000 Series Adapters and adapters based on 57xxx controllers.
- Single or multiprocessor system: Pentium III with 450MHz or greater to install the QCS CLI client software component.
For a list of supported operating systems, see the QCS CLI *Read Me* file.
- Video card capable of handling 256K colors and screen resolution of 800×600 pixel required; 16K colors and 1024×768 pixels is recommended.
- About 200MB of disk space

Software Requirements

QLogic Control Suite CLI requires no specific software components. All required software components are included in the QCS CLI installer package and are installed as part of the installation.

Server Agent Requirements

Multiple operating system support allows control of heterogeneous environments. The QConvergeConsole GUI server agent works with the following OSs:

- CentOS®
- Citrix XenServer®
- Microsoft Windows Server
- Novell® SLES
- Oracle® Linux—Unbreakable Enterprise Kernel (UEK)
- Red Hat Linux Advanced Server and Enterprise Server
- Ubuntu® Server

NOTE

For details on OS versions, see the QConvergeConsole *Read Me*, which you can download from the Marvell Web site (see [“Downloading Updates and Documentation” on page xiv](#)).

Supported OS

For a list of supported OS versions for QLogic Control Suite CLI, see the *Read Me* file.

Supported Adapters

QLogic Control Suite CLI supports Marvell 3400/8400/41000/45000 Series Adapters and adapters based on 57xxx controllers.

Downloading QLogic Control Suite CLI

To download QLogic Control Suite, follow the procedure in [“Downloading Updates and Documentation” on page xiv](#).

Downloading and Installing QCS CLI and Management Agents

To manage the Marvell 3400/8400/41000/45000 Series Adapters and adapters based on 57xxx controllers on a local or remote host, qlnxremote (the management agent used by the host adapters) must already be installed on the host.

To connect to a remote host, the management agent used by any of the host adapters must be installed and running on the host. The QLNXRemote management agent is not automatically installed by the QCS CLI and Agents installer and must be downloaded and installed separately.

This section provides procedures for either downloading and installing QCS CLI and the management agents together, or just the agents alone:

- [QCS CLI and Agent Installation on Windows and Linux](#)
- [Agent Installation on Windows and Linux](#)
- [Agent Installation on Ubuntu](#)

QCS CLI and Agent Installation on Windows and Linux

To download and install the management agents and installer on Windows and Linux (all supported versions):

1. Go to www.marvell.com.
2. Point to **Support**, and then under **Support by Product**, click **Ethernet Controllers**.
The Ethernet Adapters and Controllers Support page appears.
3. Under Related Links, click **QLogic Driver Download**.
4. Click the corresponding button to search **by Model** or **by Operating System**.
5. To define a search, click an item in each selection column, and then click **Go**.
6. Under **Management Tools**, locate the Windows or Linux QCS CLI, and then click the item's name or icon to download the item.

The following table lists the files contained in each installer for QLogic Control Suite CLI.

Windows QCS CLI Installer Files (in <code>Windows_QCS_xx.x.xx.zip</code>)	Linux QCS CLI Installer Files (in <code>Linux_QCS_x.x.xx.zip</code>)
<code>Release.txt</code>	<code>QCSCLI_ReadmeLnx.txt</code>
<code>Version.txt</code>	<code>QCSCLI_ReleaseLnx.txt</code>
<code>SilentMgmtApps.txt</code>	<code>QCS-CLI-xx.x.xx-0.i386.rpm</code>
<code>setup.exe</code>	<code>QCS-CLI-xx.x.xx-0.x86_64.rpm</code>

7. Download the appropriate agent for the host OS that the adapter is installed on, either the **Windows QCS CLI and Agents Installer** or the **Linux RPM Package**. The Linux Red Hat Package Manager (RPM) package contains QLogic Control Suite CLI and QLogic QConvergeConsole remote procedure call (RPC) agents that are common to Marvell 3400/8400/41000/45000 Series Adapters and adapters based on 57xxx controllers.

8. Install the QCS CLI and agent by running the appropriate Windows installer or Linux package. For more detailed information, refer to the *Read Me* and *Release Notes*.

Agent Installation on Windows and Linux

You can also install just the QLNXRemote agents from the Marvell Web site. For `qlremote_fca`, see “[Launching QLogic Control Suite](#)” on page 15 for download and installation instructions.

To install agents on Windows and Linux (all supported versions):

1. Go to www.marvell.com.
2. Point to **Support**, and then under **Driver Downloads**, click **NICs and HBA Drivers**.
3. On the Downloads and Documentation page, click **Adapters**.
4. For each adapter on the host server, download the following:
 - ☐ RPM package (Linux only)
 - ☐ *Read Me*
 - ☐ *Release Notes*
5. Install the agent by running the Linux RPM Package. Refer to the *Read Me* and *Release Notes* for detailed information about running the SuperInstaller.

NOTE

Although QLogic Control Suite CLI does not run on a Windows Nano Server shell, you can install the QLNXRemote agent on a Nano Server system and manage it remotely through QCS CLI using the `addhost` ([Add Host](#)) and `removehost` ([Remove Host](#)) commands. For QCS CLI to connect to the remote QLNXRemote Nano agent, you must disable the firewall on the Nano system. For instructions on installing the QLNXRemote agent on a Nano Server, refer to the *QConvergeConsole Windows Agent Installers Readme*. (This Readme is located inside the QCS CLI for Windows download available on www.marvell.com).

Agent Installation on Ubuntu

To install QLNXRemote on Ubuntu:

1. To install `alien`, issue the following command:

```
apt-get install alien
```

2. To install rpcbind, issue the following command:

```
apt-get install rpcbind
```
3. Issue the `alien` command as follows:

```
alien -i -scripts -k ./<installer name.rpm>
```
4. To install libtc18.4 (for qlnxremote agents only), issue the following command:

```
apt-get install libtc18.4
```

If the `apt-get` command is not available or installed, install the RPM by issuing the following command:

```
rpm --ignorearch -ivh <package_name>.rpm
```
5. To disable the firewall, issue the following command:

```
ufw disable
```

NOTE

For more detailed Firewall configuration, go to:
<https://help.ubuntu.com/community/UFW>

Launching QLogic Control Suite

To launch QLogic Control Suite CLI:

- On a Windows client, launch QCS CLI by double-clicking the `QCSCLI.exe` file installed in the `C:\Program Files\QLogic Corporation\QCS` folder.
- On a Linux client, launch QCS CLI by executing the `QCSCLI` file installed in the `/opt/QLogic_Corporation/QCS` folder.

To close QLogic Control Suite CLI:

- In the QCS CLI command window, type `q` at the prompt.
- In the QCS CLI command window, click the  button in the upper-right corner.

Adding a Host

QLogic Control Suite CLI allows you to add one or more local or remote hosts for the management.

Adding a Local Host Using QCS CLI

When you open QCS CLI, it automatically connects to a local host. The local host is displayed at the top of the QCS CLI window, and the local host is automatically selected as the current object.

Adding a Remote Host Using QCS CLI

To add a remote host:

1. Open the QLogic Control Suite client.
The connection to the local host is automatically made and selected as the current object.
2. Issue the following command:

```
addhost <localhost> | [ <remote host name | remote host IP>  
-p <password> ] [-protocol <rpc | local>] [-persist]
```

Where the `-protocol` keyword indicates one of the following options:

- ☐ `rpc`: Uses the remote procedure call protocol to connect to QLogic Control Suite.
 - ☐ `local`: Uses the local protocol to connect to QLogic Control Suite on the local system. The `local` option does not require an agent.
3. On the remote host, enter the agent password with administrative privilege. You can also use a user account with domain administrator privilege. A password is required to execute write operations such as configuration changes.
 4. (Optional) Include the `-persist` keyword to have QCS CLI remember the entry for this host. If you select this option, you are not required to add the IP address or host name of the remote host.

3 QCS CLI Commands

This chapter provides the details about the QLogic Control Suite CLI commands, including:

- [General Syntax and Keywords](#)
- [“Command List” on page 19](#)
- [“Commands by Item View” on page 23](#)
- [“Command Details” on page 26](#)
- [“Legacy Commands” on page 144](#)

General Syntax and Keywords

QLogic Control Suite CLI portable operating system interface (POSIX)-compliant command syntax and keywords are as follows:

Syntax **QCSCLI [-t <target type>]**
 [-f <target ID format>]
 [-i <target ID>]
 [-r <IP address>]
 [-p <password>]
 [-protocol <rpc | local>]
 [-persist] <command string>

NOTE

For an explanation of how brackets, bars, parentheses, and braces are used in QCS CLI command syntax, see the *CLI command syntax conventions* bullet under [Documentation Conventions](#).

Keywords **-t**
Specifies the target type. This keyword must be followed by the <target type>, which is either the VBD (Windows only), NDIS (Linux only), iSCSI, iSCSI portal, iSCSI target, team, VNIC, FCoE, FCoE target, physical adapter, physical port, host, system, or Layer 2 NIC (displays as a part of an interactive command for a Linux host) target type.

-f

Specifies the format of the `<target ID>` used in the `-i` option. The `<target ID format>` can be MAC, BDF, or NAME. Use MAC or BDF to select a device of a `<target type>`. Use NAME to select either a team or a virtual adapter.

-i

Specifies the target identified by the `<target ID>`. The `<target ID>` can be the Ethernet MAC address (using the NDIS device's MAC address for Ethernet and using the iSCSI device's MAC address for iSCSI hardware offload and using the teaming MAC address for VDB), the PCI bus-device-function (BDF) number, iSCSI target name, or the name of a team or virtual adapter or host name.

-r

Specifies an IP address of a host to be accessed. If the `-r` option is not specified, QCS CLI tries to read a persistent host file and connects to all hosts in that file. When the `-r` option is specified, only the specified host is connected and `<command string>` applies only to the specified host.

-p

Specifies the agent password; required to make any configuration changes.

-protocol

Specifies one of the following protocol types:

rpc

Uses the remote procedure call protocol to connect to QLogic Control Suite.

local

Uses the local protocol to connect to QLogic Control Suite on the local system. The `local` option does not require an agent.

-persist

Indicates that the host information is saved to the persistent hosts file when QCS CLI exits.

Notes

The `<command string>` includes the command, its options, parameters, and values for the command. You must specify the `<command string>` within double quotes, but if the `<command string>` contains only one command without any option or command argument, the use of double quotes is optional.

If any name or parameter contains special character such as `^`, `&`, and so on, you must specify the special character within double quotes; for example, `"^"`.

Command List

[Table 3-1](#) lists alphabetically the QCS CLI commands, briefly describes each, and provides a link to more details.

Table 3-1. QCS CLI Commands

Command	Description	See Section
add	Adds team configuration from a file (Windows only)	Add (Team Configuration)
adddiscoveryportal	Adds a discovery portal to the host	Add Discovery Portal
addhost	Adds a remote host for management	Add Host
addisnsserver	Adds the IP address or DNS name of an iSNS server to the list of iSNS servers	Add iSNS Server
addtarget	Manually configures a target and optionally persists that target	Add Target
bootcfg	Configures MBA, FCoE, or iSCSI boot	Boot Configuration
cablediag	Runs cable diagnostic tests on the selected physical port device	Cable Diagnostics
cfg	Configures parameters of the selected device	Configure Advanced Settings Configure Advanced FCoE Settings Configure Advanced Adapter Settings Configure Advanced Port Settings Configure iSCSI Boot Configure iSCSI Initiator Configure iSCSI Management Configure iSCSI Management (on a Linux Host) Configure iSCSI Secret

Table 3-1. QCS CLI Commands (Continued)

Command	Description	See Section
cfg (continued)	Configures parameters of the selected device	Configure Licenses Configure Multifunction Configure Resource Configure SR-IOV Configure System TOE Configure vLAN
createmultinpiport	Creates multiple NPIV ports	Create Multiple NPIV Ports
createnpiport	Creates an NPIV port	Create NPIV Port
diag	Configures and conducts a diagnostic test	Diagnostics
discoverhost	Searches for and adds remote hosts from a range of IP addresses	Discover Host
fallback	Falls back to primary adapters from standby	Fallback
help	Lists available commands	Help
info	Displays adapter information of the selected NIC	Information (Adapter) Information (Team)
list	Lists target items in different views	List (Targets)
listdiscoveryportals	Lists persisted target portals	List Discovery Portals
listisnsservers	Lists iSNS server addresses that are persisted by the iSCSI initiator service	List iSNS Servers
log	Logs all input and output into a file	Log
login	Logs in to an iSCSI target	Login
logout	Logs out of an iSCSI target	Logout
networkdiag	Runs network diagnostic test on the selected NDIS device	Network Diagnostics
pingtest	Runs the iSCSI ping test	Ping Test
q	Exits the program	Quit
refresh	Scans the selected system for hardware and configuration changes	Refresh

Table 3-1. QCS CLI Commands (Continued)

Command	Description	See Section
refreshall	Scans all systems for hardware and configuration changes	Refresh All
refreshdiscoveryportal	Performs a SendTargets operation to the target portal	Refresh Discovery Portal
refreshisnsserver	Refreshes the list of targets discovered from the specified iSNS server	Refresh iSNS Server
remove	Removes a team	Remove (Team)
removeallhosts	Removes all hosts from the host management list	Remove All Hosts
removediscoveryportal	Removes a discovery portal from the host	Remove Discovery Portal
removehost	Removes the specified host from the host management list	Remove Host
removeisnsserver	Removes the IP address or DNS name of the iSNS server from the persisted list of iSNS servers	Remove iSNS Server
removemultinpivport	Removes multiple NPIV ports	Remove Multiple NPIV Ports
removenpivport	Removes the specified NPIV port	Remove NPIV Port
removepersistenttarget	Removes a target from the list of persisted targets	Remove Persistent Target
removetarget	Removes a target from the list of persisted targets	Remove Target
resetsessionstats	Displays session statistics for all or the selected session	Reset Session Statistics
resetstats	Resets the statistics	Reset Statistics
restore	Restores a team configuration from a file	Restore (Team)
save	Saves a team configuration to a file	Save (Team)
select	Selects an adapter or list of available adapters	Select (Target)
sessions	Lists iSCSI sessions on the selected iSCSI adapter.	Sessions

Table 3-1. QCS CLI Commands (Continued)

Command	Description	See Section
sessionstats	Displays session statistics for all or the selected session	Session Statistics
showsel	Shows the selected target item	Show Selected (Target)
stats	Displays statistic information for the selected NIC	Statistics
unassigned	Displays adapters that are not yet part of a team	Unassigned (Adapters)
upgrade	Upgrades the firmware with the MBI file at the adapter level.	Upgrade (Firmware)
version	Displays the version of this program	Version

Commands by Item View

[Table 3-2](#) lists the applicable commands for each QCS CLI item view, and provides a link to more details.

Table 3-2. Applicable QCS CLI Commands by Item View

Item View	Applicable Commands	See Section
All	<code>addhost</code> <code>discoverhost</code> <code>help</code> <code>list</code> <code>log</code> <code>q</code> <code>removeallhosts</code> <code>removehost</code> <code>select</code> <code>showsel</code> <code>version</code>	Add Host Discover Host Help List (Targets) Log Quit Remove All Hosts Remove Host Select (Target) Show Selected (Target) Version
FCoE	<code>cfg</code> <code>createmultinpivport</code> <code>createnpivport</code> <code>info</code> <code>removemultinpivport</code> <code>removenpivport</code> <code>resetstats</code> <code>stats</code>	Configure Advanced FCoE Settings Create Multiple NPIV Ports Create NPIV Port Information (Adapter) Remove Multiple NPIV Ports Remove NPIV Port Reset Statistics Statistics
FCoE Target	<code>info</code>	Information (Adapter)
Fibre Channel Port	<code>info</code>	Information (Adapter)

Table 3-2. Applicable QCS CLI Commands by Item View (Continued)

Item View	Applicable Commands	See Section
Host	addhost adddiscoveryportal addisnsserver addtarget cfg info listdiscoveryportals listisnsservers login logout refresh refreshall refreshdiscoveryportal refreshisnsserver removediscoveryportal removeisnsserver removepersistenttarget removetarget	Add Host Add Discovery Portal Add iSNS Server Add Target Configure Advanced Settings Information (Adapter) List Discovery Portals List iSNS Servers Login Logout Refresh Refresh All Refresh Discovery Portal Refresh iSNS Server Remove Discovery Portal Remove iSNS Server Remove Persistent Target Remove Target
iSCSI	cfg info login logout pingtest resetsessionstats resetstats sessions sessionstats stats	Configure Advanced Settings Information (Adapter) Login Logout Ping Test Reset Session Statistics Reset Statistics Sessions Session Statistics Statistics
iSCSI Portal	info resetsessionstats sessionstats	Information (Adapter) Reset Session Statistics Session Statistics

Table 3-2. Applicable QCS CLI Commands by Item View (Continued)

Item View	Applicable Commands	See Section
iSCSI Target	info login logout removepersistenttarget resetsessionstats sessionstats	Information (Adapter) Login Logout Remove Persistent Target Reset Session Statistics Session Statistics
L2NIC Target (Linux only)	cfg info resetstats stats	Configure Advanced Settings Information (Adapter) Reset Statistics Statistics
LUN	info	Information (Adapter)
NDIS (Windows only)	cfg info networkdiag resetstats stats	Configure Advanced Settings Information (Adapter) Network Diagnostics Reset Statistics Statistics
Physical Adapter	cfg Advanced cfg Advanced Adapter info	Configure Advanced Settings Configure Advanced Adapter Settings Information (Adapter)
Physical Port	bootcfg cablediag cfg Advanced diag info resetstats stats	Boot Configuration Cable Diagnostics Configure Advanced Settings Diagnostics Information (Adapter) Reset Statistics Session Statistics

Table 3-2. Applicable QCS CLI Commands by Item View (Continued)

Item View	Applicable Commands	See Section
Team (Windows only)	add fallback info remove resetstats restore save stats unassigned	Add (Team Configuration) Fallback Information (Team) Remove (Team) Reset Statistics Restore (Team) Save (Team) Statistics Unassigned (Adapters)
VBD	cfg info resetstats stats	Configure Advanced Settings Information (Adapter) Reset Statistics Statistics
VNIC	info networkdiag resetstats stats	Information (Adapter) Network Diagnostics Reset Statistics Statistics

NOTE

If a command returns the `INVALID` error, it means that command is not valid for the current item view. Issue only the commands applicable to the view level that you are in, as shown in [Table 3-2](#).

Command Details

This section provides details about each QCS CLI command listed in [Table 3-1 on page 19](#) and [Table 3-2 on page 23](#), including the command description, syntax, keywords, and examples.

Add (Team Configuration)

Adds team configuration from a file. Existing team configuration is preserved and the new team or teams are added to the system from the configuration file.

This command applies only to the Windows OS and is only available if the active selection is a team or in the team view. This command is not supported on Windows Server 2012 R2.

Syntax **add**
 [-h <host>]
 <file name>

Keywords **-h**
 Specifies the host to which the teams will be added, if no team is currently selected.

Examples The following example shows the `add` command:

```
qcscli -t TEAM "add c:\team_bdf.txt"
QLogic Control Suite Command Line Interface Utility for Windows v17.0.7.0
.....
Please wait while executing Teaming operation...
Add Team succeeded.
```

Add Discovery Portal

Adds a static target portal to the list of target portals to which the iSCSI initiator service transmits SendTarget requests. The `adddiscoveryportal` command is available in the context of a host and is available only for Linux hosts.

Syntax

adddiscoveryportal

```
{-m <iSCSI HBA MAC Address>}  
{-i <TargetPortalAddress>}  
{-if <iface file name>}  
[-n <TargetPortalSocket>]  
[-mu <CHAP authentication>]  
[-u <CHAP name>]  
[-p <CHAP secret>]  
[-iu <Initiator CHAP name>]  
[-ip <Initiator CHAP secret>]  
[-ir <iSER target type>]
```

Keywords

-m

Specifies the iSCSI Host Bus Adapter MAC address.

-i

Specifies the address of the target portal.

-if

(Linux only) Specifies the name of the interface (iface) file.

-n

Specifies a port on a host to be accessed. If you do not specify a the `-n` option, the default 3260 port number is used.

-mu

Specifies that mutual CHAP authentication type is used.

-u

Specifies the CHAP name.

-p

Specifies the CHAP secret.

-iu

Specifies the initiator CHAP name.

-ip

Specifies the initiator CHAP secret.

-ir

Specifies the iSCSI Extensions for RDMA (iSER) target type.

Notes

If the CHAP name and CHAP secret are both specified, CHAP authentication is used for login.

Examples The following example shows the `adddiscoveryportal` command:

```
./QCSscli "adddiscoveryportal -m 000E1E502683 -i 193.169.101.17"
```

C	Status	DevType	Name
0	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-4734b7e04-df70000002051ffa
1	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-5844b7e04-a2b0000002852021
2	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-9db4b7e04-b3a0000005c5231b
3	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-0bc4b7e04-b020000007452692
4	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-7c94b7e04-1596c0a617d52d65
5	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-80c4b7e04-f9e6c0a618952e77
6	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-9364b7e04-5716c0a619652f4f
7	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-94d4b7e04-d706c0a619952f4f
8	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-a96898704-71aa7986a8f533a8
9	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-3a14b7e04-b5e000e7ac253579
10	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-07f4b7e04-8496c0a767e54930
11	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-8a0906-c474b7e04-229a2d3cfa354fef

Add Host

Adds a host to the managed host list by its name or IP address.

Syntax **addhost**
 < [< localhost | <local host name> | <local host IP>] | [-p <password>
 <host name | IP address>] >
 [-protocol <rpc | local>
 [-persist]

Keywords **-p**
 Specifies the agent password; required to make any configuration changes.

-protocol
 Specifies one of the following protocol types:

rpc

Uses the remote procedure call protocol to connect to QLogic Control Suite.

local

Uses the local protocol to connect to QLogic Control Suite on the local system. The `local` option does not require an agent.

-persist
 Indicates that the host information will be saved to the persistent hosts file when you close QLogic Control Suite with the `-q` command. All the hosts in the saved file will be automatically connected when QCS CLI starts. To break the operation of connecting to the persistent remote hosts, press the CTRL+BREAK keys.

Notes The variable `host name` is the name of a host to be connected. The variable `IP address` is the IP address of a host to be connected.

Examples The following example shows the `addhost` command:

```
qcscli "addhost 172.28.62.78 -p abc@USA -protocol rpc -persist"
```

QLogic Control Suite Command Line Interface Utility for Windows v17.0.7.0

...

.....

C	Status	Name	OS	Platform
1	connected	DL380G7-3809	Windows Server 2012	64 bit (x64)

Add iSNS Server

Adds the IP address or DNS name of an iSNS server to the list of iSNS servers. This command is available in the context of a host.

Syntax **addisnsserver**
 {-i <iSNS Server Address>}

Keywords **-i**
Identifies the iSNS server by its address.

Examples The following example shows the `addisnsserver` command:

```
QCScli.exe "addisnsserver -i 192.168.100.3"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v40.0.18.0
```

```
.....
```

Add Target

Manually configures a target and optionally persists that target. This command is only available in the context of a host.

Syntax **addtarget**
 {-t <TargetName>}
 {-i <TargetPortalAddress>}
 {-n <TargetPortalSocket>}
 [-f <ifaceName>]

Keywords **-t**
 Adds the specified target to the list of static targets.

-i
 Specifies the IP address of the target portal.

-n
 Specifies the socket number of the target portal.

-f
 Specifies the iSCSI interface (iface) file name to which the static target will be added. This option applies only to Linux hosts.

Examples The following example shows the `addtarget` command:

```
./QCScli "addtarget -t
iqn.2001-05.com.equallogic:0-8a0906-07f4b7e04-8496c0a767e54930 -i
192.168.100.10 -n 3260"
```

```
C    Status            DevType Name
-    -----
0    disconnected iSCSI.T
iqn.2001-05.com.equallogic:0-8a0906-07f4b7e04-8496c0a767e54930
```

Boot Configuration

Displays the current iSCSI, FCoE, or Multi-Boot Agent (MBA) boot configuration or one of its subcategory boot configurations; or saves the current iSCSI, FCoE, or MBA boot configuration to an XML file; or restores the iSCSI FCoE, or MBA boot configuration from an XML file. For a boot configuration file example, see [“iSCSI Boot Configuration XML File” on page 177](#).

The `bootcfg` command is only available in the context of a physical port.

Syntax

bootcfg

`[-t {iSCSI | fcoe | mba}]`

`[-o show {General | Initiator | Target | MPIO}]`

`[-o {save <filename.xml> | restore <filename.xml>}]`

Additional syntax for MBA boot configuration:

`QCScli.exe -t phyports -f bdf -i <BDF values> "bootcfg -t mba -o save <filename>.xml"` ' For saving mba boot configuration file

`QCScli.exe -t phyports -f bdf -i <BDF values> "bootcfg -t mba -o restore <filename>.xml"` ' For upload the updated file

Keywords

-t

Specifies the boot configuration type as either iSCSI, FCoE, or MBA.

-o show

Shows boot configuration for either General, Initiator, Target, or MPIO. (MPIO is supported only for adapters based on 57xxx controllers.).

-o save | restore

Specifies the name of the XML file to either save the boot configuration to or restore it from. On Linux, you must specify the full path to the file.

Notes

For Marvell 3400/8400/41000/45000 Series Adapters, the boot protocol is either None, `UNDI` (Universal Network Device Interface), or `iBFT` (SCSI Boot Firmware Table). Only the last saved boot protocol is shown, saved, or restored.

Examples

The following examples show the `bootcfg` command:

```
qcscli -t phyports -f BDF -i 0b:00.00 "bootcfg -t MBA -o show"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.12.0
```

```
..
```

```
MBA: General Configuration Parameters
```

```
-----
```

```
Option ROM
```

```
Enabled
```

3-QCS CLI Commands

Boot Configuration

Boot Protocol	iSCSI Boot
Boot Strap Type	Auto
Hide Setup Prompt	Disabled
Setup Key Stroke	Ctrl-S
Banner Message Timeout	5 seconds
Link Speed	10G
Pre-boot Wake on LAN	Disabled
VLAN mode	Disabled
VLAN ID(1..4094)	1
Boot Retry Count	0

```
qcscli -t phyports -f BDF -i 0b:00.00 "bootcfg -t iSCSI -o show Initiator"
```

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..

iSCSI: Initiator Configuration Parameters

IP Address	172.17.69.40
Subnet Mask	255.255.0.0
Default GateWay	172.17.69.101
Primary DNS	
Secondary DNS	
iSCSI Name	iqn.2011-12.com:qlogic.com
CHAP ID	
CHAP Secret	

```
qcscli -t phyports -f BDF -i 0b:00.00 "bootcfg -t FCoE -o show Target"
```

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..

FCoE: Target Configuration Parameters

Connect	Disabled
Port WWN	0000000000000000
Boot Lun	0
Connect	Disabled
Port WWN	0000000000000000

```

Boot Lun          0
Connect           Disabled
Port WWN          0000000000000000
Boot Lun          0
Connect           Disabled
Port WWN          0000000000000000
Boot Lun          0
Connect           Disabled
Port WWN          0000000000000000
Boot Lun          0
Connect           Disabled
Port WWN          0000000000000000
Boot Lun          0
Connect           Disabled
Port WWN          0000000000000000
Boot Lun          0
Connect           Disabled
Port WWN          0000000000000000
Boot Lun          0

```

```

C:\Automation_QCSCLI>QCScli.exe -t phyports -f bdf -i 01:00.00
"bootcfg -t mba -o save mba.xml"

```

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.....

```

C:\Automation_QCSCLI>QCScli.exe -t phyports -f bdf -i 01:00.00
"bootcfg -t mba -o restore mba.xml"

```

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.....

Linux examples:

```

# ./QCScli -t PHYPORTS -f bdf -i 0b:00.0 "bootcfg -t mba -o show
general"

```

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v30.0.10.0

semaphore_create: semaphore exists, getting semaphore!

MBA: General Configuration Parameters

Option ROM	Enabled
Boot Protocol	PXE
Boot Strap Type	Auto
Hide Setup Prompt	Disabled
Setup Key Stroke	Ctrl-S
Banner Message Timeout	5 seconds
Link Speed	AutoNeg
Pre-boot Wake on LAN	Enabled
VLAN mode	Disabled
VLAN ID(1..4094)	1
Boot Retry Count	0

```
# ./QCScli "bootcfg -t mba -o save /tmp/file.xml"
```

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The MBA boot configuration settings are saved in the specified XML file; no message is returned to indicate success or failure of the command.

```
linux-merk\devcon\1\port>bootcfg -t fcoe -o show General
```

FCoE: General Configuration Parameters

Boot Protocol	FCoE
FIP VLAN	0
Target Login Retry Count	5
Fabric Login Retry Count	5

```
linux-merk\devcon\1\port>bootcfg -t iscsi -o show General
```

iSCSI: General Configuration Parameters

Boot Protocol	None
Boot Mode	Non-Offload
TCP/IP Parameters via DHCP	Enabled
iSCSI Parameters via DHCP	Enabled
CHAP Authentication	Disabled
Mutual CHAP	Disabled
DHCP Vendor ID	QLGC ISAN
DHCP Request Timeout	60
Target Login Timeout	60
IPVersion	IPv4

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Boot Configuration

IPv4 Fallback	Disabled
Vlan	Disabled
VLANID	0
Address Redirect	Disabled

Cable Diagnostics

Runs cable diagnostic tests on the selected Marvell Network Adapter physical port device.

Syntax **cablediag**

Keywords None

Notes To stop running a test, press the CTRL+BREAK keys.

Examples The following example shows the `cablediag` command:

```
qcscli -t phyports -f BDF -i 03:00.00 "cablediag"
QLogic Control Suite Command Line Interface Utility for Windows
v17.0.12.0
..
.....
Cable Length          Status
-----
2.260000              Good
1.850000              Good
3.090000              Good
2.680000              Good

Link: 1000 Mbps Full Duplex
Cable analysis test completed successfully.
```

Configure Advanced Settings

Gets and sets advanced parameters for a device.

Syntax **cfg Advanced**

[[default] | [parameter] | [parameter=value]]

Additional syntax for VBD:

```
QCScli.exe -t VBD -f mac -i <MAC address> "cfg advanced  
\"<parameter>\"=\"<value>\""
```

Additional syntax for NDIS:

```
QCScli.exe -t NDIS -f mac -i <MAC address> "cfg advanced  
\"<parameter>\"=\"<value>\""
```

Keywords **default**

Sets all advanced parameters to their default values for an NDIS device.

parameter

Displays the current setting and all valid settings of the specified parameter.

parameter=value

Sets the specified value to the specified `parameter`, which must be one of those parameters that are displayed by the `cfg Advanced` command. The `value` must be one of the valid settings of the parameters that are displayed by the `cfg Advanced parameter` command.

Both the parameter and value are case insensitive and must be specified within the double quotes if spaces or special characters are used.

Do not use spaces around the `=` (equal sign) in `parameter=value`.

Notes Issuing the `cfg Advanced` command without options displays all advanced parameters and their current settings.

Examples The following examples show the `cfg Advanced` command:

```
qcscli -t NDIS -f MAC -i 001018AD45E0 "cfg Advanced"  
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.....  
Advanced  
-----  
Encapsulated Task Offload:           Enabled (Default)  
Flow Control:                        Rx & Tx Enabled (Default)  
Interrupt Moderation:                Enabled (Default)  
Jumbo Packet:                        1514 (Default)1
```

3-QCS CLI Commands

Configure Advanced Settings

Large Send Offload V2 (IPv4):	Enabled (Default)
Large Send Offload V2 (IPv6):	Enabled (Default)
Locally Administered Address:	Not Present (Default)
Maximum Number of RSS Queues:	4 (Default)
NUMA Node ID:	0
Priority & VLAN:	Priority & VLAN enabled (Default)
Quality of Service:	Enabled (Default)
Receive Buffers (0=Auto):	0 (Default)
Receive Side Scaling:	Enabled (Default)
Recv Segment Coalescing (IPv4):	Enabled (Default)
Recv Segment Coalescing (IPv6):	Enabled (Default)
RSS Profile:	NUMAScalingStatic (Default)
Speed & Duplex:	10 Gbps Full Duplex (Default)
SR-IOV:	Enabled
Starting RSS CPU:	0 (Default)
TCP/UDP Checksum Offload (IPv4):	Rx & Tx Enabled (Default)
TCP/UDP Checksum Offload (IPv6):	Rx & Tx Enabled (Default)
Transmit Buffers (0=Auto):	0 (Default)
Virtual Machine Queues:	Enabled (Default)
VLAN ID:	0 (Default)

```
qcscli -t FCoE -f MAC -i 000e1e5027b3 "cfg Advanced"
```

Advanced

Fabric disconnect timeout (LDTIMEOUT):	30
Target removal timeout (PDTIMEOUT):	30

```
qcscli -t FCoE -f MAC -i 000e1e5027b3 "cfg Advanced LDTIMEOUT"
```

Fabric disconnect timeout (LDTIMEOUT)

Current setting:	30
Valid setting:	5 to 300 seconds

```
qcscli -t FCoE -f MAC -i 000e1e5027b3 "cfg Advanced LDTIMEOUT=25"
```

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You must restart your computer for the new settings to take effect.

¹ For 41000, 57xxx, and 579xx Series Adapters, advance parameter settings for the Jumbo Packet specified using the agent or QCS CLI will supersede the Linux YaST settings.

VBD example:

```
C:\Automation_QCSCLI>QCScli.exe -t VBD -f mac -i 000E1ED23B11 "cfg advanced  
\"Flow Control\"=\"Tx Enabled\""
```

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.....

NDIS example:

```
C:\Automation_QCSCLI> QCScli.exe -t NDIS -f mac -i 000E1ED23B11 "cfg advanced  
\"SR-IOV\"=\"Disabled\""
```

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.....

Configure Advanced FCoE Settings

Configures FCoE settings. This command is only available if the actively selected device is an FCoE device.

Syntax **cfg Advanced**
 [[key] | [key=value]]

Additional syntax:

```
QCScli.exe -t phyports -f bdf -i < BDF values> "bootcfg -t fcoe -o save  
                 <filename>.xml"                ' For saving fcoe boot configuration file  
QCScli.exe -t phyports -f bdf -i < BDF values> "bootcfg -t fcoe -o restore  
                 <filename>.xml"                ' For upload the updated file
```

Keywords **key**
Specifies an FCoE device setting. The key is case insensitive and must be specified within double quotes if it contains any spaces or special characters.

key=value

Specifies an FCoE device key and value. The key and value are case insensitive and must be specified within double quotes if they contain any spaces or special characters. Do not include spaces around the = (equal sign).

Examples The following examples show the `cfg Advanced` command:

```
qcscli -t FCoE -f MAC -i 000e1e5027b3 "cfg Advanced"  
Advanced  
-----
```

```
Fabric disconnect timeout (LDTIMEOUT): 30  
Target removal timeout (PDTIMEOUT): 30
```

```
qcscli -t FCoE -f MAC -i 000e1e5027b3 "cfg Advanced LDTIMEOUT"  
Fabric disconnect timeout (LDTIMEOUT)  
-----
```

```
Current setting: 30  
Valid setting: 5 to 300 seconds
```

```
qcscli -t FCoE -f MAC -i 000e1e5027b3 "cfg Advanced LDTIMEOUT=25"  
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.10.0  
You must restart your computer for the new settings to take effect.
```

```
C:\Automation_QCSCLI>QCScli.exe -t phyports -f bdf -i 01:00.00  
"bootcfg -t fcoe -o save fcoe.xml"
```

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.....

C:\Automation_QCSCLI>**QCScli.exe -t phyports -f bdf -i 01:00.00**
"bootcfg -t fcoe -o restore fcoe.xml"

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.....

Configure Advanced Adapter Settings

Configures settings at the adapter level on some 41000 Series Adapters.

Syntax **cfg Advanced**
 [[key] | [key=value]]

Keywords **key**
Specifies an adapter setting. The key is case insensitive and must be specified within double quotes if it contains any spaces or special characters.
Key includes:

Port Mode

Valid values can be retrieved by issuing `cfg advanced "Port Mode"` at the adapter level.

Notes This command configures the port mode of the adapter. The valid values represent the supported port modes for the adapter in use. The adapter is configured to the specific mode after a reboot (SET operation).

<u>Value</u>	<u>Configuration</u>
1x100	Single 100G port
4x25	Four ports with 25G speed
4x10	Four ports with 10G speed

Examples The following example shows the `cfg Advanced` command for adapter settings:

```
WIN-D0ICI9TJD00\devcon\0\adapter>cfg advanced
```

```
Advanced
```

```
-----
```

```
Port Mode:                               4x25
```

```
WIN-D0ICI9TJD00\devcon\0\adapter>cfg advanced "Port Mode"
```

```
Port Mode
```

```
-----
```

```
Current setting:                          4x25
```

```
Valid settings:                           1x100 (Default)
```

```
4x10
```

```
4x25
```

```
WIN-D0ICI9TJD00\devcon\0\adapter>
```

Configure Advanced Port Settings

Configures FastLinQ Smart Auto Negotiation (SmartAN™) settings at the port level. This command is only available if the actively selected device is a physical port device. This feature is available for a selected family of adapters.

SmartAN sets the port link speed to use Smart Auto Negotiation. The SmartAN feature in the 41000 Series Adapters provides the ability to automatically set up a link between a 25G adapter and 10G switch over DAC or optics media without user intervention or manual setup.

Syntax **cfg Advanced**
 [[key] | [key=value]]

Keywords **key**

Specifies a physical port device setting. The key is case insensitive and must be specified within double quotes if it contains any spaces or special characters. Keys include:

DCBXMODE

Selects one of the following data center bridging exchange (DCBX) modes:

- ☐ IEEE
- ☐ CEE
- ☐ Dynamic

By default, this setting is disabled (no mode selected).

FEC AN Mode

Selects which forward error correction (FEC) modes are advertised as supported and requested during Auto Negotiation. This key applies only to 41000 Series Adapters and only when in Auto Negotiation mode. Options for this key include:

- ☐ None. No FEC is advertised as supported.
- ☐ 10G_Firecode: Only Fire Code FEC (FC-FEC) is advertised as supported on 10G links.
- ☐ 25G_Firecode: Only FC-FEC is advertised as supported on 25G links.
- ☐ 10G_and_25G_Firecode: Only FC-FEC is advertised as supported on 10G and 25G links.
- ☐ 25G_RS: Only Reed-Solomon FEC (RS-FEC) is advertised as supported on 25G links.

- ☐ **25G_Firecode_and_RS**: Both FC-FEC and RS-FEC are advertised as supported on 25G links (RS-FEC takes precedence over FC-FEC if both sides advertise supporting and requesting both types of FEC modes).
- ☐ **All**: Both FC-FEC and RS-FEC are advertised as supported on 10G and 25G links.

NVM Speed

Sets the default speed mode used by the port. Not all speed modes are available on each adapter.

- ☐ **AutoNeg**: Sets the speed to AutoNeg mode.
- ☐ **1G**: Sets the speed to 1GbE fixed link speed.
- ☐ **10G**: Sets the speed to 10GbE fixed link speed.
- ☐ **25G**: Sets the speed to 25GbE fixed link speed.
- ☐ **40G**: Sets the speed to 40GbE fixed link speed.
- ☐ **50G**: Selects 50GbE fixed link speed.
- ☐ **100G**: Sets the speed to 100GbE fixed link speed.

Smart AN Mode

Enables (1) or disables (0) SmartAN mode. Generally, for SmartAN mode to work, 29/31 (and 67 for the preboot link speed used for PXE/iSCSI/FCoE boot) must be set to AutoNeg mode (0). If Smart AN Mode is enabled, the NVM Speed setting is automatically set to AutoNeg.

FEC Force Mode

Sets the FEC mode forced on when a fixed speed mode is enabled. Not all FEC modes are available on each adapter.

- ☐ **None**: No FEC mode is used.
- ☐ **FC**: FC-FEC mode is used.
- ☐ **RS**: RS-FEC mode is used.
- ☐ **Auto**: The port will cycle through the applicable FEC modes for the selected fixed speed mode enabled until a link is established.

key=value

Specifies a physical port device key and value. The key and value are case insensitive and must be specified within double quotes if they contain any spaces or special characters. Do not include spaces around the = (equal sign).

Notes

Only the applicable properties are shown in the command output. For example, if SmartAN is set to 0 (disabled) and NVM Speed is set to AutoNeg, FEC AN Mode is shown but FEC Force Mode is not shown.

Examples The following example shows the `cfg Advanced` command for port settings:

```
172.28.13.128\devcon\64\port>cfg advanced
```

```
Advanced
```

```
-----
```

```
DCBXMODE:                                CEE
FEC An Mode:                             10G_Firecode
NVM Speed:                               AutoNeg (Default)
SmartAN(TM) Mode:                         0 (Default)
```

The following examples show how to view the supported range values for any property:

```
WIN-06AFM1900QA\devcon\21\port>cfg advanced
```

```
Advanced
```

```
-----
```

```
DCBXMODE:                                CEE
FEC Force Mode:                           Firecode
NVM Speed:                                25G
Smart An Mode:                            0 (Default)
```

```
localhost.localdomain\devcon\21\port>cfg advanced DCBXMode
```

```
DCBXMode
```

```
-----
```

```
Current setting:                           Dynamic

Valid settings:                            Disabled (Default)
                                              IEEE
                                              CEE
                                              Dynamic
```

```
WIN-06AFM1900QA\devcon\21\port>cfg advanced "FEC Force Mode"
```

```
FEC Force Mode
```

```
-----
```

```
Current setting:                           Firecode

Valid settings:                            None (Default)
                                              Firecode
                                              RS
                                              Auto
```

```
WIN-O6AFM1900QA\devcon\21\port>cfg advanced "NVM Speed"
```

```
NVM Speed
```

```
-----
```

```
Current setting:                25G
```

```
Valid settings:
```

```
AutoNeg (Default)
```

```
25G
```

```
WIN-O6AFM1900QA\devcon\21\port>cfg advanced "Smart An Mode"
```

```
Smart An Mode
```

```
-----
```

```
Current setting:                0
```

```
Valid settings:
```

```
value between 0 and 1
```

```
0 (Default)
```

Configure iSCSI Boot

Gets and sets iSCSI boot keys for the device. This command is only available if the actively selected device is an NDIS device and the system is on an iSCSI boot using the selected NDIS device.

Syntax **cfg iSCSIBoot**

[[key] | [key=value]]

Additional syntax:

QCScli.exe -t phyports -f bdf -i < BDF values> "bootcfg -t iscsi -o save
<filename>.xml" ' For saving iSCSI boot configuration file

QCScli.exe -t phyports -f bdf -i < BDF values> "bootcfg -t iscsi -o restore
<filename>.xml" ' For upload the updated file

Keywords **key**

Displays the current setting and all valid settings of the specified key. The **key** is case insensitive and must be specified within double quotes if it contains spaces or special characters.

key=value

Sets the specified value to the specified key. No space is allowed around the = (equals sign) in the **key=value**. Both **key** and **value** are case insensitive and must be specified within double quotes if they contain spaces or special characters. Do not use spaces around the = (equal sign) in **key=value**.

CDUMP

Sets the "iSCSI Crash Dump" to a <value> of either **Enable** or **Disable**.

Notes Issuing the **cfg iSCSIBoot** command without options displays all iSCSI boot keys and their current settings.

Examples The following example shows the **cfg iSCSIBoot** command:

```
qcscli "cfg iSCSIBoot CDUMP=Enable"
```

```
iSCSI: General Configuration Parameters
```

```
-----
```

```
Boot Protocol           None
Boot Mode               Non-Offload
TCP/IP Parameters via DHCP Enabled
iSCSI Parameters via DHCP Enabled
CHAP Authentication     Disabled
Mutual CHAP             Disabled
```

3-QCS CLI Commands

Configure iSCSI Boot

DHCP Vendor ID	QLGC ISAN
DHCP Request Timeout	60
Target Login Timeout	60
IPVersion	IPv4
IPv4 Fallback	Disabled
Vlan	Disabled
VLANID	0
Address Redirect	Disabled

iSCSI: Initiator Configuration Parameters

IP Address
Subnet Mask
Default GateWay
Primary DNS
Secondary DNS
iSCSI Name iqn.1994-02.com.qlogic.iscsi:fastlingboot
CHAP ID
CHAP Secret

iSCSI: Target Configuration Parameters

Connect	Enabled
IP Address	0.0.0.0
TCP Port	3260
Boot LUN	0
iSCSI Name	
CHAP ID	
CHAP Secret	

iSCSI: Secondary Target Configuration Parameters

Connect	Enabled
IP Address	0.0.0.0
TCP Port	3260
Boot LUN	0
iSCSI Name	
CHAP ID	
CHAP Secret	

```
C:\Automation_QCSCLI>QCScli.exe -t phyports -f bdf -i 01:00.00  
"bootcfg -t iscsi -o save iscsi.xml"
```

```
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```

```
.....
```

```
C:\Automation_QCSCLI>QCScli.exe -t phyports -f bdf -i 01:00.00  
"bootcfg -t iscsi -o restore iscsi.xml"
```

```
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```

```
.....
```

Configure iSCSI Initiator

Displays the name of the system iSCSI initiator. This command is currently valid only for Windows OS.

Syntax **cfg iSCSIInitiator**
 [value]

Keywords **value**
Sets the name of iSCSI Initiator of the system.

Examples The following example shows the `cfg iSCSIInitiator` command:

```
qcscli "cfg iSCSIInitiator"
iSCSI Initiator Name
-----
iSCSI Initiator Name      :  qlogic
```

Configure iSCSI Management

Gets and sets iSCSI management keys for the device if the actively selected device is an iSCSI device and belongs to the Marvell Network Adapter family of devices.

Syntax **cfg iSCSIMgmt**
 [[key] | [key=value]]

Keywords Both the key and value are case insensitive and must be specified within double quotes if they contain any spaces or special characters. Do not use spaces around the = (equal sign) in `key=value`.

cfg iSCSIMgmt key

Displays the current setting and all valid settings of the specified key.

cfg iSCSIMgmt key=value

Sets the specified value to the specified key.

IPV4DHCP

Enables DHCP for IPv4. The `<value>` can only be `Enable`. To disable DHCP, use `IPV4ADDR` to set a static IPv4 address; DHCP will be automatically disabled.

IPV4ADDR

Sets the static IPv4 address to the specified `<value>`. Setting the static IPv4 address automatically disables the DHCP of IPv4.

IPV4MASK

Sets the IPv4 subnet mask to the specified `<value>`.

IPV4GATEWAY

Sets the default IPv4 gateway address to the specified `<value>`.

IPV6DHCP

Enables DHCP for IPv6. The `<value>` can only be `Enable`. To disable DHCP, use `IPV6ADDR` to set the static IPv6 address; DHCP will be automatically disabled.

IPV6PRA

Sets the "Process Router Advertisement" to a `<value>` of either `Enable` or `Disable`. The value is set to `Enable` and the key is not configurable when DHCP is enabled.

IPV6ADDR

Sets the static IPv6 address to the specified `<value>`. Setting the static IPv6 address automatically disables the DHCP of IPv4.

IPV6PL

Sets the IPv6 address subnet prefix length to the specified `<value>`.

IPV6DG

Sets the IPv6 default gateway address to the specified `<value>`.

VLANID

Sets the VLAN ID for the iSCSI. The `<value>` is the range of 0–4094.

MTU

Sets the MTU for the iSCSI. The `<value>` is in the range of 1500–9600.

Linux Support for iface

The following keys apply to Linux hosts with support for iface files:

IPV4ADDR

Displays only iface files for IPv4 static configuration.

IPV4DHCP

Displays only iface files for IPv4 dynamic configuration.

IPV6ADDR

Displays only iface files for IPv6 static configuration.

IPV6DHCP

Displays only iface files for IPv6 dynamic configuration.

IPV4INAME

Sets the initiator name for IPv4 static configuration in an iface file.

IPV4VLANID

Sets the VLAN ID for IPv4 static configuration in an iface file. The `<value>` is the range of 0–4094.

IPV4IFNUM

Sets the iface number for IPv4 static configuration in an iface file. The `<value>` is a integer started from 0.

IPV6INAME

Sets the initiator name for IPv6 static configuration in an iface file.

IPV6VLANID

Sets the VLAN ID for IPv6 static configuration in an iface file. The `<value>` is the range of 0–4094.

IPV6IFNUM

Sets the iface number for IPv6 static configuration in an iface file. The `<value>` is a integer started from 0.

DHCP4INAME

Sets the initiator name for IPv4 dynamic configuration in an iface file.

DHCP4VLANID

Sets the VLAN ID for IPv4 dynamic configuration in an iface file. The `<value>` is the range of 0–4094.

DHCP4IFNUM

Sets the iface number for IPv4 dynamic configuration in an iface file. The `<value>` is a integer started from 0.

DHCP6INAME

Sets the initiator name for IPv6 dynamic configuration in an iface file.

DHCP6VLANID

Sets the VLAN ID for IPv6 dynamic configuration in an iface file. The `<value>` is the range of 0–4094.

DHCP6IFNUM

Sets the iface number for IPv6 dynamic configuration in the iface file. The `<value>` is a integer started from 0.

Notes

Issuing the `cfg iSCSIMgmt` command without options displays all iSCSI management keys and their current settings.

When the system is booted up through an iSCSI Host Bus Adapter:

- With an IPv4 address, the IPv4-related keywords are non-configurable, and not available in this command.
- With an IPv6 address, the IPv6-related keywords are non-configurable, and not available in this command.

Examples The following examples show the `cfg iSCSIMgmt` command:

```
qcscli -t iscsi -f MAC -i 000e1e5027b1 "cfg iSCSIMgmt"
```

```
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iSCSI Management  
-----
```

3-QCS CLI Commands

Configure iSCSI Management

```
Vlan ID (VLANID): 0
Jumbo Mtu (MTU): 1500
IPv4 Configurations:
IPv4 DHCP (IPV4DHCP): Enable
IPv6 Configurations:
IPv6 DHCP (IPV6DHCP): Disable
IPv6 Process Router Advertisements (IPV6PRA): Enable
IPv6 Address (IPV6ADDR):
Subnet Prefix Length (IPV6PL): 0

IPv6 Default Gateway (IPV6DG):
```

qcscli -t iscsi -f MAC -i 000e1e5027b1 "cfg iSCSIMgmt MTU=9000"

QLogic Control Suite Command Line Interface Utility for Windows v30.0.80.0
iSCSI Management

```
-----
Vlan ID (VLANID): 0
Jumbo Mtu (MTU): 9000
IPv4 Configurations:
IPv4 DHCP (IPV4DHCP): Enable
IPv6 Configurations:
IPv6 DHCP (IPV6DHCP): Disable
IPv6 Process Router Advertisements (IPV6PRA): Enable
IPv6 Address (IPV6ADDR):
Subnet Prefix Length (IPV6PL): 0

IPv6 Default Gateway (IPV6DG):
```

Configure iSCSI Management (on a Linux Host)

This variant of the `cfg iSCSIMgmt` command (see [Configure iSCSI Management](#)) is used to manage and edit iface files on a Linux host.

Both key and value (except `<Initiator name>`) are case insensitive and must be specified within the double quotes if they contain any spaces or special characters.

Syntax **cfg iSCSIMgmt**
 `{{-a | -e | -d} <iface file name>}`
 `[ [-dhcp4 | -dhcp6] | {{-ipv4 | -ipv6} <IP address>} ]`
 `[-m <Subnet mask>]`
 `[-p <Subnet Prefix Length>]`
 `[-i <Initiator name>]`
 `[-v <VLAN ID>]`
 `[-n <iface NUM> ]`
 `[-mtu < iface MTU> ]`

Keywords **-a <iface file name>**
 Adds a new iface file.

-e <iface file name>
 Edits an existing iface file.

-d <iface file name>
 Deletes an existing iface file.

The `add` and `edit` commands have the following options:

-dhcp4
 Enables DHCP for IPv4 configuration.

-dhcp6
 Enables DHCP for IPv6 configuration.

-ipv4
 Sets or modifies the IPv4 address.

-ipv6
 Sets or modifies the IPv6 address.

-m
 Sets or modifies the subnet mask for IPv4 configuration.

-p

Sets or modifies the subnet prefix length for IPv6 configuration.

-i

Sets or modifies the initiator name.

-v

Sets or modifies the VLAN ID in an iface file. The `<VLAN ID>` is the range of 0–4094.

-n

Sets or modifies the iface number in an iface file. The `<iface number>` is a integer started from 0.

-mtu

Sets or modifies the MTU for the iface file; the range is 1500–9600.

Examples The following examples show the `cfg iSCSIMgmt` command:

```
qcscli "cfg iSCSIMgmt -a new_dhcp_ipv4 -dhcp"
```

```
root@debian904:/opt/QLogic_Corporation/QCS# ./QCScli -t iscsi -f  
bdf -i 82:00.04 "cfg iscsimgmt -a new_iface -dhcp4"
```

```
QLogic Control Suite Command Line Interface Utility for Linux  
v40.0.19.0
```

```
....
```

List the iSCSI config(iface) using following command to verify.

```
root@debian904:/opt/QLogic_Corporation/QCS# ./QCScli -t iscsi -f  
bdf -i 82:00.04 "cfg iscsimgmt"
```

```
QLogic Control Suite Command Line Interface Utility for Linux  
v40.0.19.0
```

```
....
```

```
iSCSI Management
```

```
-----
```

```
IPv4 Configurations:
```

```
    IPv4 DHCP (IFIPV4DHCP)
```

IFace File Name:	new_iface
Initiator Name (DHCP4INAME0):	
IFace Vlan ID (DHCP4VLANID0):	-1
IFace Num (DHCP4IFNUM0):	-1
Jumbo Mtu:	-1

3-QCS CLI Commands

Configure iSCSI Management (on a Linux Host)

```
qcscli "cfg iSCSIMgmt -d bnx2i.00:10:18:a7:1b:11_2001::3742"
./QCScli -t iscsi -f bdf -i 82:00.04 "cfg iscsimgmt"
QLogic Control Suite Command Line Interface Utility for Linux
v40.0.19.0
```

....

iSCSI Management

IPv4 Configurations:

IPv4 Address (IFIPV4ADDR)

IFace File Name:

qedi.f4:e9:d4:73:af:ec

IPv4 Address (IPV4ADDR0): 192.168.100.51

IPv4 Subnet Mask (IPV4MASK0):

Initiator Name (IPV4INAME0):

IFace Vlan ID (IPV4VLANID0): -1

IFace Num (IPV4IFNUM0): -1

Jumbo Mtu: -1

IFace File Name:

sta_iface145

IPv4 Address (IPV4ADDR1):

192.168.100.20

IPv4 Subnet Mask (IPV4MASK1):

Initiator Name (IPV4INAME1):

IFace Vlan ID (IPV4VLANID1): -1

IFace Num (IPV4IFNUM1): -1

Jumbo Mtu: -1

IPv6 Configurations:

```
qcscli "cfg iSCSIMgmt -e dhcpv6 -v 10 -n 55"
```

```
root@debian904:/opt/QLogic_Corporation/QCS# ./QCScli -t iscsi -f
bdf -i 82:00.04 "cfg iscsimgmt -e new_iface dhcpv6 -v 10 -n 55"
```

```
QLogic Control Suite Command Line Interface Utility for Linux
v40.0.19.0
```

....

Configure iSCSI Secret

Sets the CHAP secret of the system iSCSI initiator.

Syntax **cfg iSCSI Secret**
 <value>

Keywords **value**
Specifies the CHAP secret.

Examples The following example shows the `cfg iSCSI Secret` command:

```
qcscli "cfg iSCSI Secret xxxxxxxxxxxx"
I:\QCS_cli Release\v40.0.19\Release_64\Release>QCScli.exe "cfg
iSCSI Secret Password1234"

QLogic Control Suite Command Line Interface Utility for Windows
v40.0.19.0
.....
```

Configure Licenses

Displays all license parameters and their current settings. This command is only available if the actively selected device is a virtual block device (VBD) and belongs to the Marvell Network Adapter family of devices.

Syntax **cfg Licenses**

Keywords None

Examples The following example shows the `cfg Licenses` command:

```
qcscli -r 172.28.62.78 -u winuser1\Administrator -p abc@USA -t VBD -f mac -I  
00101896DC9C "cfg Licenses"
```

```
QLogic Control Suite Command Line Interface Utility for Windows v17.0.7.0
```

```
.....
```

```
Licenses
```

```
-----
```

TCP Offload Engine (TOE) :	Maximum Connections
iSCSI Offload Engine:	Maximum Connections
FCoE Offload Engine:	Maximum Connections

Configure Multifunction

Configures a physical adapter with NIC partition settings. The parameters shown in the **Keywords** section are used in the NPAR configuration XML file (see [“NIC Partition Configuration XML Files” on page 188](#)).

Syntax **cfg Multi-Function**

["Multi-Function mode" | -s <filename> | -c <filename> | -p port# [-f function#]
key]

Additional syntax:

QCScli.exe -t phyadapters -f bdf -i <BDF values> "cfg multi-function -s
<filename>.xml" ' For saving multifunction configuration file

QCScli.exe -t phyadapters -f bdf -i <BDF values> "cfg multi-function -c
<filename>.xml" ' For upload the updated file

Keywords **-s**

Saves the current configuration to the specified XML file.

-c

Configures the NIC settings from the specified XML file.

See [“NIC Partition Configuration XML Files” on page 188](#) for an example of the multifunction (NPAR) and single function (SF) mode XML files, which you can edit and use the files to change the configuration of an adapter. Changes to maximum bandwidth and relative bandwidth weight (of an adapter in NPAR mode) can be applied while traffic is running (you need not reboot). All other changes require a system reboot.

-p

Specifies the port number.

-f

Specifies the function number. The default function number is the first function of the specified port. For example, the `cfg multi-function -p 0 -f 2 \ "FCoE\ "` command displays port0 function2 FCoE configuration.

key

Specifies the key type used in the XML file, see the example [page 63](#). All keys are case insensitive. Refer to [Table B-6 on page 195](#) for a list of valid keys.

-o

Overwrites the SR-IOV configuration if there is any conflict between multifunction configuration and SR-IOV configuration. For example, `cfg multi-function -o <filename>` configures the selected adapter settings with the specified XML file.

Notes To display the current configuration, issue the `cfg multi-function` command without options.

Examples The following examples show the `cfg multi-function` command.

To view the adapter's current multifunction mode, issue the following command:

```
qcscli -r 172.28.63.183 -u winuser1\Administrator -p abc@USA -t phyadapters -f BDF -i 42:00 "cfg multi-function"
```

QLogic Control Suite Command Line Interface Utility for Windows v17.0.7.0
Adapter is in Single-Function mode.

To write multifunction settings from a saved XML file back to the adapter, issue the following command:

```
qcscli -r 172.28.63.183 -u winuser1\Administrator -p abc@USA -t phyadapters -f BDF -i 42:00 "cfg multi-function -c MF.xml"
```

QLogic Control Suite Command Line Interface Utility for Windows v17.0.7.0
You have to reboot the system for the settings to take effect.

To save the adapter's current multifunction settings to an editable XML file, issue the following command:

```
qcscli -r 172.28.63.183 -u winuser1\Administrator -p abc@USA -t phyadapters -f BDF -i 42:00 "cfg multi-function -s MF1.xml"
```

QLogic Control Suite Command Line Interface Utility for Windows v17.0.7.0
Please wait while configurations are being saved...

To view a quad-port 10GbE 57840 adapter in NPAR mode, issue the following command, which shows a switch-independent NIC partitioning (NPAR) or Multi-Function configuration XML file:

```
qcscli -r 172.28.63.183 -u winuser1\Administrator -p abc@USA -t phyadapters -f BDF -i 42:00 "cfg multi-function"
```

QLogic Control Suite Command Line Interface Utility for Windows v17.0.7.0
Multi-Function Info

```
-----  
Multi-Function mode           Multi-Function  
Port 0  
Flow Control                  Rx & Tx Enabled  
Function                       0  
Ethernet/Ndis                 Enable  
iSCSI                         Disable  
FCoE                           Enable  
Relative Bandwidth Weight (%) 0  
Maximum Bandwidth (%)         10  
Function                       4
```

3-QCS CLI Commands

Configure Multifunction

Ethernet/Ndis	Enable
iSCSI	Enable
FCoE	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	20

Port 1

Flow Control	Tx Enabled
Function	1
Ethernet/Ndis	Enable
iSCSI	Enable
FCoE	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	30
Function	5
Ethernet/Ndis	Enable
iSCSI	Disable
FCoE	Enable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	40

Port 2

Flow Control	Rx Enabled
Function	2
Ethernet/Ndis	Enable
iSCSI	Disable
FCoE	Enable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	40
Function	6
Ethernet/Ndis	Enable
iSCSI	Enable
FCoE	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	10

Port 3

Flow Control	Auto
Function	3
Ethernet/Ndis	Enable
iSCSI	Enable
FCoE	Disable

```

Relative Bandwidth Weight (%)    0
Maximum Bandwidth (%)           10
Function                         7
Ethernet/Ndis                   Enable
iSCSI                           Disable
FCoE                            Enable
Relative Bandwidth Weight (%)    0
Maximum Bandwidth (%)           20

```

To configure RoCE and iWARP on 41000 and 45000 Series Adapters, issue the following command. For an example of the XML file, see [“RoCE and iWARP Configuration \(Windows and Linux\)” on page 194](#).

```
qcscli -r 172.28.63.183 -u winuser1\Administrator -p abc@USA -t phyadapters -f BDF -i 42:00 "cfg multi-function"
```

Multi-Function Info

```

-----
Multi-Function mode           Multi-Function
EP Mode                       Enabled

Port 0
  Flow Control                Auto
  Function                    0
  Ethernet/Ndis               Enable
  RoCE                        Disable
  iWARP                       Disable
  Relative Bandwidth Weight (%) 0
  Maximum Bandwidth (%)       100
  Function                    2
  Ethernet/Ndis               Disable
  RoCE                        Disable
  iWARP                       Disable
  FCoE                        Enable
  Relative Bandwidth Weight (%) 0
  Maximum Bandwidth (%)       100
  Function                    4
  Ethernet/Ndis               Enable
  RoCE                        Disable
  iWARP                       Disable
  iSCSI                       Disable
  Relative Bandwidth Weight (%) 0

```

Maximum Bandwidth (%)	100
Function	6
Ethernet/Ndis	Enable
RoCE	Disable
iWARP	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	100
Function	8
Ethernet/Ndis	Enable
RoCE	Disable
iWARP	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	100
Function	10
Ethernet/Ndis	Enable
RoCE	Disable
iWARP	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	100
Function	14
Ethernet/Ndis	Enable
RoCE	Disable
iWARP	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	100
Port 1	
Flow Control	Auto
Function	1
Ethernet/Ndis	Enable
RoCE	Enable
iWARP	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	100
Function	3
Ethernet/Ndis	Disable
RoCE	Disable
iWARP	Disable
FCoE	Enable
Relative Bandwidth Weight (%)	0

3-QCS CLI Commands

Configure Multifunction

Maximum Bandwidth (%)	100
Function	5
Ethernet/Ndis	Enable
RoCE	Enable
iWARP	Disable
iSCSI	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	100
Function	7
Ethernet/Ndis	Enable
RoCE	Enable
iWARP	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	100
Function	9
Ethernet/Ndis	Enable
RoCE	Disable
iWARP	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	100
Function	13
Ethernet/Ndis	Enable
RoCE	Disable
iWARP	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	100
Function	15
Ethernet/Ndis	Enable
RoCE	Disable
iWARP	Disable
Relative Bandwidth Weight (%)	0
Maximum Bandwidth (%)	100

```
C:\Automation_QCSCLI>QCScli.exe -t phyadapters -f bdf -i 01:00
"cfg multi-function -s multi.xml"
```

```
QLogic Control Suite Command Line Interface Utility for Windows
v30.0.103.0
```

```
.....
```

```
Please wait while configurations are being saved...
```

```
C:\Automation_QCSCLI>QCScli.exe -t phyadapters -f bdf -i 01:00
"cfg multi-function -c multi.xml"
QLogic Control Suite Command Line Interface Utility for Windows
v30.0.103.0
.....
You have to reboot the system for the settings to take effect.
```

Configure Resource

Gets and sets resource keys for the device, if the actively selected device is a VBD and belongs to the Marvell Network Adapter family of devices. This command is currently valid only for Windows OS.

Syntax **cfg Resource**

[[key] | [key=value]]

Keywords Both the key and value are case insensitive and must be specified within the double quotes if they contain any spaces or special characters. Do not use spaces around the = (equal sign) in key=value.

CONFIGURABLE

Displays all the configurable resource reservation parameters with their current setting and valid settings. It does not require any <rsc_value>.

CONFIG

Sets the “Offload Configuration” to either “First Come First Served” or “Reserved Resources”. It requires the <value> of either FCFS or RSVD.

NDIS

Sets the “Pre-Allocated Resources for NDIS” for the NDIS Ethernet instance on the physical function to enabled or disabled. It requires a <value> of either Enable or Disable.

TOE

Sets the “Pre-Allocated Resources for TOE” to enabled or disabled. It requires a <value> of either Enable or Disable.

iSCSI

Sets the “Pre-Allocated Resources for iSCSI” for the iSCSI-Offload instance on the physical function to enabled or disabled. It requires a <value> of either Enable or Disable.

FCOE

Sets the “Pre-Allocated Resources for FCOE” for the FCoE-Offload instance on the physical function to enabled or disabled. It requires a <value> of either Enable or Disable.

TOECONN

Sets the “TOE Connections” to the number allowed in the range displayed in the cfg Resource CONFIGURABLE command. It requires a numeric <value>.

iSCSICONN

Sets the “iSCSI Connections” to the number allowed in the range displayed in the `cfg Resource CONFIGURABLE` command. It requires a numeric `<value>`.

iSCSIPT

Sets the “iSCSI Pending Task per Connection” to valid numbers specified in the `cfg Resource CONFIGURABLE` command. It requires a numeric `<value>`.

TOEPROTOCOL

Sets the “TOE Protocol Type” to the valid settings specified in the `cfg Resource CONFIGURABLE` command. It requires a `<value>` of either IPv4 or IPv6.

iSCSIProtocol

Sets the “iSCSI Protocol Type” to the valid settings specified in the `cfg Resource CONFIGURABLE` command. It requires a `<value>` of either IPv4 or IPv6.

MEMCONSUMPTION

Sets the “iSCSI Memory Consumption” to the valid settings specified in the `cfg Resource CONFIGURABLE` command. This keyword applies only to 57xxx adapters. It requires the `<value>` to be a percentage from 25 percent to 100 percent, with increments of 25 percent.

TOERSS

Sets the “TOE RSS” to either `Enable` or `Disable`. This keyword applies only to 57xxx adapters. It requires a `<value>` that can be either `Enable` or `Disable`.

Notes

Issuing the `cfg Resource` command without options displays all resource reservation keys and their current settings.

Examples

The following examples show the `cfg Resource` command.

To view the currently enabled protocols (TOE, NDIS L2 Ethernet, iSCSI offload and FCoE offload) of a port (in Single Function mode) and other port settings, issue the following command. Note that TOE is not supported on the Marvell 3400/8400/41000/45000 Series Adapters. In addition, NPAR FCoE and iSCSI offload is controlled by the NPAR XML file settings (see [“NIC Partition Configuration XML Files” on page 188](#)).

```
qcscli -t VBD -f MAC -i 001018AD45E0 "cfg Resource"
```

```
QLogic Control Suite Command Line Interface Utility for Windows v17.0.7.0
```

```
.....
```

```
Resource Reservations
```

```
-----
```

3-QCS CLI Commands

Configure Resource

Pre-Allocated Resources:	NDIS iSCSI Offload Engine FCoE
Max TOE Connections:	0
Max iSCSI Offload Engine Conn:	64
Pending Tasks Per Connection:	512
Memory Consumption(%) :	100
TOE RSS:	Disabled

To view the currently enabled protocols in Single Function mode port settings, issue the following command, which is similar to the [Configure Multifunction](#) command.

```
qcscli -t VBD -f MAC -i 001018AD45E0 "cfg Resource Configurable"
QLogic Control Suite Command Line Interface Utility for Windows v17.0.7.0
.....
Resource Reservations
-----
Pre-Allocated Resources (NDIS, TOE, ISCSI, FCOE)
-----
Current setting:
NDIS:                Enable
TOE:                 Disable
ISCSI:               Enable
FCoE:                Enable
Valid setting:       Disable or Enable

TOE Connections (TOECONN)
-----
Current setting:      0
Valid settings:       Parameter is Read Only

TOE RSS (TOERSS)
-----
Current setting:      Disable
Valid settings:       Parameter is Read Only

iSCSI Connections (ISCSICONN)
-----
Current setting:      64
Valid settings:       from 1 to 128
```

Pending Tasks Per Connection (ISCSIPT)

```
-----
Current setting:                    512
Valid settings:                    Parameter is Read Only
```

Memory Consumption(%) (MEMCONSUMPTION)

```
-----
Current setting:                    100
Valid settings:                    25, 50, 75, or 100
```

The following command is similar to the [Configure Multifunction](#) command.

```
qcscli -t VBD -f MAC -i 001018AD45E0 "cfg Resource"
```

QLogic Control Suite Command Line Interface Utility for Windows v17.0.7.0

.....

Resource Reservations

```
-----
Pre-Allocated Resources:            NDIS
                                    TCP Offload Engine(TOE)
                                    iSCSI Offload Engine
                                    FCoE

Max TOE Connections:                8192
Max iSCSI Offload Engine Conn:      64
Pending Tasks Per Connection:       512
Memory Consumption(%):              75
TOE RSS:                            Disabled
```

To enable FCoE offload on the selected (through its MAC address) Single Function mode port, issue the following command:

```
qcscli -t VBD -f MAC -i 001018AD45E0 "cfg Resource \"fcoe\"=\"enable\""
```

To enable iSCSI offload on the selected (through its MAC address) Single Function mode port, issue the following command:

```
qcscli -t VBD -f MAC -i 001018AD45E0 "cfg Resource \"iscsi\"=\"enable\""
```

To enable TOE on the selected (through its MAC address) Single Function mode port, issue the following command:

```
qcscli -t VBD -f MAC -i 001018AD45E0 "cfg Resource \"toe\"=\"enable\""
```

Note that TOE is not supported on the Marvell 3400/8400/41000/45000 Series Adapters.

To enable TOE RSS on the selected (through its MAC address) Single Function mode port, issue the following command:

```
qcscli -t VBD -f MAC -i 001018AD45E0 "cfg Resource \"toerss\"=\"enable\""
```

Note that TOE is not supported on the Marvell 3400/8400/41000/45000 Series Adapters, and TOE must be enabled before enabling TOE RSS.

Configure SR-IOV

Configures and displays physical adapter single root input/output virtualization (SR-IOV) settings. The parameters shown in the **Keywords** section are used in the SR-IOV configuration XML file (see [“SR-IOV Configuration XML Files” on page 198](#)).

Syntax **cfg SRIOV**

-s <filename>
-c <filename>

Additional syntax:

QCScli.exe-tphyadapters-fbdf-i<BDF values>"cfgsriov-s<filename>.xml"
' For saving sriov configuration file
QCScli.exe-tphyadapters-fbdf-i<BDF values>"cfgsriov-c<filename>.xml"
' For upload the updated file

Keywords **-s**

Saves the current configuration to an XML file.

-c

Configures the SR-IOV settings from the specified XML file.

Valid keys include the following:

FlowControl

Ethernet/Ndis

iSCSI

FCoE

MaxBandwidth

RelativeBandwidth

Notes

Issue the `cfg SRIOV` command without keywords to display the current configuration.

All keys and operands are case insensitive. Multi-Function mode, `-p`, and `-f` options are not required.

Examples

The following example shows the `cfg SRIOV` command:

```
qcscli -t phyadapters -f BDF -i 04:00 "cfg SRIOV"
```

QLogic Control Suite Command Line Interface Utility for Windows
v17.0.7.0

```

.....
SRIOV Configurations
-----
SRIOV                               Enabled

Port 0
SRIOV VFs per PF                    32

Port 1
SRIOV VFs per PF                    32

Port 2
SRIOV VFs per PF                    0
SRIOV Max Chains per VF            16

Port 3
SRIOV VFs per PF                    0

```

QCScli.exe -t phyadapters -f bdf -i 01:00 "cfg multi-function -s multi.xml"

QLogic Control Suite Command Line Interface Utility for Windows v30.0.103.0

```

.....
Please wait while configurations are being saved...

```

QCScli.exe -t phyadapters -f bdf -i 01:00 "cfg multi-function -c multi.xml"

QLogic Control Suite Command Line Interface Utility for Windows v30.0.103.0

```

.....
You have to reboot the system for the settings to take effect.

```

QCScli.exe -t phyadapters -f bdf -i 01:00 "cfg sriov -s sriov.xml"

QLogic Control Suite Command Line Interface Utility for Windows v30.0.103.0

```

.....
Please wait while configurations are being saved...

```

C:\Automation_QCSCLI>**QCScli.exe -t phyadapters -f bdf -i 01:00 "cfg sriov -c sriov.xml"**

QLogic Control Suite Command Line Interface Utility for Windows v30.0.103.0

```

.....
You have to reboot the system for the settings to take effect.

```

cfg sriov -c save.xml

Note: Number of VFs cannot be set on this adapter in Single Function Mode.

Configure SRIOV on port 0 successfully. Please reboot the host for the configuration to take effect.

Configure SRIOV on port 1 successfully. Please reboot the host for the configuration to take effect.

Configure SRIOV on port 2 successfully. Please reboot the host for the configuration to take effect.

Configure SRIOV on port 3 successfully. Please reboot the host for the configuration to take effect.

Configure System TOE

NOTE

Chimney has been deprecated by Microsoft in the OS.

Enables, disables, or automates the TCP offload engine (TOE) in Chimney for the system. This command is only available to the system target and is valid only for Windows OS.

Syntax **cfg Systoe**
 [value]

Keywords **value**
Enables, disables, or automates TOE in Chimney for the system.

Notes To display the current TOE Chimney state, issue the `cfg Systoe` command without options.

Examples The following example shows the `cfg Systoe` command:

```
qcscli "cfg Systoe"
QLogic Control Suite Command Line Interface Utility for Windows
v17.0.7.0
.....
System TOE Settings
-----
Chimney Offload State      : Enabled
```

Configure vLAN

Displays, saves, and sets the virtual LAN (vLAN) configuration for the selected 3400 or 8400 Series Adapter.

Syntax `cfg vlan`
`{ -[s|c] <filename> }`

Keywords `-s`
 Saves the current configuration into an XML file with the specified file name.

`-c`
 Configure the selected vLAN file, where `VLAN_IDPool` is string of bits to be set. For example, `2 3 4-8 9` indicates the adapter settings in the specified XML file.

Notes Note the following regarding vLAN configuration:

- If you change the `VLAN_MODE` to `Normal` mode for any port, the values for the function configuration `[VLAN_ID/VLAN_PRIORITY/VLAN_ID_POOL]` are set to `0` (empty).
- For QinQ and filtering mode, `VLAN_ID` value `0` is not accepted.
- For QinQ mode, `VLAN_ID` must be unique across all PFs of the same port.
- `VLAN_ID_POOL` may contain bitmap values up to 256 at the adapter level. You can configure up to 256 vLANs with any combination on each port. However, if `VLAN_ID` is not inside the pool, the ID is counted separately and added into the final count.
- `VLAN_ID_POOL` must contain multiple `VLAN_IDS` separated by spaces. To add a range of values, use a hyphen (-). Valid examples of `VLAN_ID_POOL` values include:
 - ☐ `2 4 6 8-35`
Configures the vLAN on 2, 4, 6, and 8 to 35 IDs (total of 31)
 - ☐ `7 90 100-200`
Configures vLANs on 7, 90, and 100 to 200 (total of 103)
 - ☐ `0`
An invalid value.
- For the changes to take effect, you must reboot the system.

Examples To display the current vLAN configuration, issue the `cfg vlan` command as follows:

```
localhost.localdomain\devcon\0\adapter>cfg vlan
```

```
VLAN Table Info
```

```

-----
Port 0
VLAN_MODE                               QinQ
  Function                               0
  VLAN_ID                               4
  VLAN_PRIORITY                         0
  VLAN_ID_POOL
Port 1
VLAN_MODE                               Normal
  Function                               1
  VLAN_ID                               0
  VLAN_PRIORITY                         0
  VLAN_ID_POOL

```

To save the current vLAN configuration to an XML file, issue the `cfg vlan -s save.xml` command as follows:

```
localhost.localdomain\devcon\0\adapter>cfg vlan -s save.xml
```

Please wait while configurations are being saved...

For example, the specified XML file may appear as follows:

```

<?xml version="1.0" encoding="UTF-8"?>
<VLAN_Configuration><Version>1</Version><Port><PortNumber>0</PortNumber><VLAN_MODE>QinQ</VLAN_MODE><Function><FunctionNumber>0</FunctionNumber><VLAN_ID>4</VLAN_ID><VLAN_Priority>0</VLAN_Priority><VLAN_IDPool></VLAN_IDPool></Function></Port><Port><PortNumber>1</PortNumber><VLAN_MODE>Normal</VLAN_MODE><Function><FunctionNumber>1</FunctionNumber><VLAN_ID>0</VLAN_ID><VLAN_Priority>0</VLAN_Priority><VLAN_IDPool></VLAN_IDPool></Function></Port></VLAN_Configuration>

```

To edit the vLAN XML file, configure parameters in the appropriate fields. For example, to change the `VLAN_MODE` for Port-0 in the preceding example XML, change the value to `Normal`, as shown in the following.

```

<?xml version="1.0" encoding="UTF-8"?>
<VLAN_Configuration><Version>1</Version><Port><PortNumber>0</PortNumber><VLAN_MODE>Normal</VLAN_MODE><Function><FunctionNumber>0</FunctionNumber><VLAN_ID>4</VLAN_ID><VLAN_Priority>0</VLAN_Priority><VLAN_IDPool></VLAN_IDPool></Function></Port><Port><PortNumber>1</PortNumber><VLAN_MODE>Normal</VLAN_MODE><Function><FunctionNumber>1</FunctionNumber><VLAN_ID>0</VLAN_ID><VLAN_Priority>0</VLAN_Priority><VLAN_IDPool></VLAN_IDPool></Function></Port></VLAN_Configuration>

```

Then save the XML file and display the vLAN information again:

```
localhost.localdomain\devcon\0\adapter>cfg vlan -c save.xml
```

You have to reboot the system for the settings to take effect.

```
localhost.localdomain\devcon\0\adapter>cfg vlan
```

VLAN Table Info

```
-----  
Port 0  
VLAN_MODE                      Normal  
  Function                      0  
  VLAN_ID                      0  
  VLAN_PRIORITY                 0  
  VLAN_ID_POOL  
Port 1  
VLAN_MODE                      Normal  
  Function                      1  
  VLAN_ID                      0  
  VLAN_PRIORITY                 0  
  VLAN_ID_POOL
```

Create Multiple NPIV Ports

Creates multiple NPIV ports at the same time. This command is available in the context of an FCoE Host Bus Adapter.

Syntax **createmultipivport**
 {{-s <x:xx>}
 {-e <x:xx>
 {-n <num>}}

Keywords **-s**
Specifies the starting name range, where only three digits (1–3) are allowed to change; for example: 2x:xx:00:10:18:aa::bb::cc. The other digits are from the current WWPN or WWNN.

-e
Specifies the ending name range, where only three digits(1–3) are allowed to change; for example: 2x:xx:00:10:18:aa::bb::cc. The other digits are from the current WWPN or WWNN.

-n
Specifies the quantity of NPIV ports to be created.

Examples The following example shows the `createmultipivport` command:

```
qcscli -t FCoE -f MAC -i 000e1e5027b3 "createmultipivport -s 1:11
-e 9:99 -n 9"
:Creating NPIV ports
-----
21:11:00:0e:1e:50:27:b3      : Succeed
21:12:00:0e:1e:50:27:b3      : Succeed
21:13:00:0e:1e:50:27:b3      : Succeed
21:14:00:0e:1e:50:27:b3      : Succeed
21:15:00:0e:1e:50:27:b3      : Succeed
21:16:00:0e:1e:50:27:b3      : Succeed
21:17:00:0e:1e:50:27:b3      : Succeed
21:18:00:0e:1e:50:27:b3      : Succeed
21:19:00:0e:1e:50:27:b3      : Succeed
Result: 9 NPIV port(s) created.
```

Create NPIV Port

Creates an NPIV port. This command is available when an FCoE Host Bus Adapter is selected.

Syntax **createnpivport**
 {{-s <x:xx>} | {-p <WWPN>}}

Keywords **-s**
Specifies the save mode, where only three digits(1–3) are allowed to change; for example: 2x:xx:00:10:18:aa::bb::cc. Other digits are from the current WWPN or WWNN.

-p
(Windows only) Specifies an expert mode in which a complete WWPN value can be passed. For example: 21:34:00:10:18:aa::bb::cc.

Examples The following example shows the `createnpivport` command:

```
qcscli -t FCoE -f MAC -i 000e1e5027b3 "createmultinpivport -s 1:11  
-e 9:99 -n 9"
```

```
:Creating NPIV ports
```

```
-----
```

```
21:11:00:0e:1e:50:27:b3                : Succeed
```

```
21:12:00:0e:1e:50:27:b3                : Succeed
```

```
21:13:00:0e:1e:50:27:b3                : Succeed
```

```
21:14:00:0e:1e:50:27:b3                : Succeed
```

```
21:15:00:0e:1e:50:27:b3                : Succeed
```

```
21:16:00:0e:1e:50:27:b3                : Succeed
```

```
21:17:00:0e:1e:50:27:b3                : Succeed
```

```
21:18:00:0e:1e:50:27:b3                : Succeed
```

```
21:19:00:0e:1e:50:27:b3                : Succeed
```

```
Result: 9 NPIV port(s) created.
```

Diagnostics

Configures the parameters of a diagnostic test and runs the test. If no options are specified, displays all the tests available for the currently selected target. This command is available on Marvell 3400/8400/41000/45000 Series Adapters and adapters based on 57xxx controllers physical port devices.

Syntax

diag

```
{[-c REG ] [-c MII ] [-c EEP] [-c MEM] [-c CPU] [-c INT] [-c MACLB ] [-c PHYLB]
  [-c LED] | [-c ALL]}
```

```
[-l <cnt> ]
```

```
[-v <LEDIntv> ]
```

Additional syntax:

```
QCSccli.exe -t phyports -f bdf -i < BDF values> "diag -c all" ' Diag test
```

Keywords

-c

Specifies the individual test type to run, including:

REG

Runs the Control Register test.

MII

Runs the MII Register test.

EEP

Runs the EEPROM test.

MEM

Runs the Internal Memory test.

CPU

Runs the On-Chip CPU test.

INT

Runs the Interrupt test.

MACLB

Runs the MAC Loopback test.

PHYLB

Runs the PHY Loopback test.

LED

Runs the LED test with the `LEDIntv` value.

ALL

Runs all of the preceding tests.

-l

Specifies the iteration. If not specified, the test uses the default iteration of 1.

-v

Specifies the LED interval. If not specified, the test uses the default LED interval of 5.

Notes

To stop a running test, press the CTRL+BREAK keys.

Examples

The following examples show the `diag` command:

```
qcscli -t phyports -f BDF -i 04:00.00 "diag -c all -l 10 -v 5"
```

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```
.....
```

Test	Pass/Fail	Result

Control Registers	10 /0	Pass
EEPROM	10 /0	Pass
Internal Memory	10 /0	Pass
On Chip CPU	10 /0	Pass
Interrupt	10 /0	Pass
Loopback MAC	10 /0	Pass
Loopback PHY	10 /0	Pass
LED	10 /0	Pass

```
qcscli -t phyports -f BDF -i 04:00.00 "diag -c CPU -c LED -v 5"
```

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```
.....
```

Test	Pass/Fail	Result

On Chip CPU	1 /0	Pass
LED	1 /0	Pass

```
C:\Automation_QCSCLI>QCScli.exe -t phyports -f bdf -i 01:00.00
"diag -c all"
QLogic Control Suite Command Line Interface Utility for Windows
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.....
.....
Test                Pass/Fail    Result
-----
Control Registers   1    /0      Pass
EEPROM              1    /0      Pass
Internal Memory     1    /0      Pass
On Chip CPU         1    /0      Pass
Interrupt           1    /0      Pass
Loopback PHY        1    /0      Pass
LED                 1    /0      Pass
```

Discover Host

Discovers hosts from the specified IP range and adds them to the managed host list.

Syntax

discoverhost

-s <start IP address>
 -e <end IP address>
 -p <password>
 [-protocol <rpc | local>]
 [-persist]

Keywords

-s

Specifies the starting IP address of the searching range.

-e

Specifies the ending IP address of the searching range.

-p

Specifies the password.

-protocol

Specifies one of the following connection protocols:

rpc

Uses a remote procedure call protocol to connect to QLogic Control Suite.

local

Uses the local protocol to connect to QLogic Control Suite on the local system. The `local` option does not require an agent.

-persist

Specifies that the host information is saved to the persistent hosts in the saved file when you close QCS CLI by issuing the `q` command. All host files are automatically connected when you restart QCS CLI.

Notes

To stop discovering hosts, press the CTRL+BREAK keys.

Examples

The following example shows the `discoverhost` command:

```
QCScli.exe "discoverhost -s 175.28.63.12 -e 175.28.63.13 -u Administrator -p Qlogic01"
```

```
QLogic Control Suite Command Line Interface Utility for Windows v30.0.14.0
```

```
.....
```

```
Please add a host that you want to manage.
```

3-QCS CLI Commands

Discover Host

C	Status	Name	OS	Platform
-	-----	-----	-----	-----
0	connected	DL380G7-3809	Windows Server 2012	64 bit (x64)
1	connected	720-4883-10022	N/A	64 bit (x64)

DL380G7-3809\host\0>

Fallback

Falls back to primary adapters from standby. This command is only applicable for SLF-AFD team type and in the `teamview` view listing.

This command is valid only for Windows OS, but is not supported on Windows Server 2012 R2 or later.

Syntax **fallback**

Keywords None.

Examples The following example shows the `fallback` command:

```
qcscli -t TEAM "fallback"
```

```
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```

```
.....
```

Help

Lists available commands.

Syntax `help`

Keywords None

Examples The following example shows the `help` command.

`qcscli help`

Following is the list of available commands.

For detailed information of each command, please type 'command ?'

All commands are case insensitive.

```
q           : quit the program
log         : log all command input and output to a file
version     : display version of this program
list        : list target items in different views
select      : select a target item from the list of available targets
showsel     : show the selected target item
help        : list of available commands
addhost     : add a host to managed host list
removehost  : remove a host from managed host list
removeallhosts: remove all hosts from managed host list
discoverhost: discover hosts from a range of IP address and add them to host
list
info        : display host information
help        : display help for host context
refresh     : refresh the specified system for hardware/configuration changes
refreshall  : refresh all systems for hardware/configuration changes
cfg         : configure the host
adddiscoveryportal: add discovery portal to the host
removediscoveryportal: remove discovery portal from the host
refreshdiscoveryportal: refresh discovery portal on the host
listdiscoveryportals: list discovery portal on the host
addisnssserver: add iSNS server address to the host
removeisnssserver: remove iSNS server address from the host
refreshisnssserver: refresh iSNS server address on the host
listisnssservers: list iSNS server address on the host
addtarget   : add static target to the host system
removetarget: remove static target from host system
removepersistenttarget: remove persistency for the target
```

3-QCS CLI Commands

Help

login : establishes a login session to the target
logout : close the indicated login session

Information (Adapter)

Displays adapter information for the selected NIC. This command is available if the actively selected device is an NDIS device.

Syntax info

[all | vitalsigns | driver | eee | hardware | resource | os | initiator | dcbx |
nicpartition | system | hwinventory | swinventory | switch | vf]
[-a]

Keywords Available information display options include:

all

Displays information for all categories.

vitalsigns

Displays vital signs information.

driver

Displays driver information.

eee

Displays EEE information.

hardware

Displays hardware information.

resource

Displays resource information.

os

Displays host OS information.

initiator

Displays iSCSI initiator information.

dcbx

Displays DCBX information.

nicpartition

Displays NIC partition information.

-a

Displays more DCBX advanced information for a physical port device; this option applies only if the *all* or *dcbx* value is specified.

Examples The following examples show the `info` command:

QCScli.exe "info hardware"

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...

Host Information

Host Name: R720-E3E4
OS Version Info: Windows Server 2012 R2
Platform: 64 bit (x64)
Connection Details:
 Protocol Local
 IP localhost
QLASP Driver: Installed

iSCSI Initiator

Name:

Portal List

Portal 192.168.86.235
Portal fe80::210:18ff:fee3:a5c1

qcscli -t VBD "info"

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..

Driver Information

Driver Version: : 7.12.1.0
Driver Date: : 11/5/2014
Driver Name: : evbda.sys
Driver Status: : Loaded

Vital Signs

Link Speed(in Mbps) : 10000
Ndis MAC Address : 00101896DC9C
iSCSI MAC Address : 00101896DC9D
FCoE MAC Address : 00101896DC9D

FCoE Node WWN : 10:00:00:10:18:96:dc:9d
FCoE Port WWN : 20:00:00:10:18:96:dc:9d

SRIOV Switch Information

Number of HW Available VFs : 0
Number of Available VFs : 0
Max VF Chains Per VFs : 0
VF Chains Pool Size : 0
Switch Friendly Name :

SRIOV VF Information

No hardware is available for SRIOV.

qcscli -t NDIS "info"

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..

Vital Signs

NDIS MAC Address : 00101896DC9C
Permanent MAC Address : 00101896DC9C
IPV4 Address : 152.30.4.173
IPV6 Address : 3001:3001:3001:3001:e129:df2b:a5b3:2fb6
Offload Capabilities : TOE, LSO, CO, RSS
MTU : 1500
Flow Control : Rx Enabled

Driver Information

Driver Version: : 7.12.1.0
Driver Date: : 11/5/2014
Driver Name: : bxnd60a.sys
Driver Status: : Loaded

qcscli -t iSCSI "info"

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..

Vital Signs

```
-----
iSCSI MAC Address:           : 00101896DC9D

*** IPv4 Configuration 152.30.4.197 ***
IPv4 DHCP                   : Enable
IPv4 Address                 : 152.30.4.197
IPv4 Subnet Mask             : 255.255.0.0
IPv4 Default Gateway         :

*** IPv6 Configuration fe80::210:18ff:fe96:dc9d ***
IPv6 DHCP                   : Disable
IPv6 Address                 : fe80::210:18ff:fe96:dc9d
IPv6 Scope Id                : 2
IPv6 Flow Info               : 0
IPv6 Default Gateway         :

Driver Information
-----
Driver Name:                 : bxois.sys
Driver Status:               : Loaded
Driver Version:              : 7.12.0.0
Driver Date:                 : 11/11/2014

Driver Parameters
-----
Initial R2T                  : yes
Immediate Data               : yes
Data Sequence in Order      : yes
Data PDU in Order           : yes
Max Outstanding R2T          : 1
First Data Burst Length      : 65536
Max Data Burst Length        : 262144
Error Recovery Level         : 0
Default Time to Wait         : 2
Default Time to Retain       : 20
Max Connections              : 1
    Portal 152.30.4.197
    Portal fe80::210:18ff:fe96:dc9d

qcscli -t devcon -f BDF -i 05:00 "info"
QLogic Control Suite Command Line Interface Utility for Windows
v30.0.54.0
```

3-QCS CLI Commands

Information (Adapter)

```
.....
Resource
-----
Bus Type                PCI-E (8x)
Slot Number             5

Hardware
-----
ASIC Version            xxxxxxxxxxxx
Bootcode Version        8.18.2.0
PXE Boot Version        PCI30 MBA 2.0.5;EFIx64 2.1.0.1

Vendor Product Description :
MN                        : 1028
Product Number           : QLE3442-CU
Product Name              : QLogic QLE3442 Adapter
Serial Number           : AFE1645F13737
V0                        : FFv8.14.11
V1                        : DSV1028vPDF.VER2.1
V2                        : NPY2
V3                        : PMTD
V4                        : NMVQLogic
V5                        : DTINIC
V6                        : DCM1001FFFFFFF1202FFFFFFF1403...
Temperature in Celsius    : 31
```

For 3400/8400 Series Adapters, DCBX information is shown at the port level. For example:

```
Vital Signs
-----
Link State                N/A

Resource
-----
Bus Number                32
Device Number             0
MSI Version               2
Function Per Port         4
Memory Address            0x0080
Physical Type             Optical
```

DCBX

DCB enable	Enabled							
DCBX version	CEE							
Priority Tagging	Operational							
Networking PRI	0							
FCoE PRI	3							
iSCSI HBA PRI	0							
Priority Flow Control (PFC)	Operational							
PFC Enabled on Priorities	3							
PFC Disabled on Priorities	0 1 2 4 5 6 7							
Enhanced Transmission Selection (ETS)	Operational							
Priority to Priority Group Mapping								
Priorities in Priority Group 0	1 2 4 5 6 7							
Priorities in Priority Group 1	0							
Priorities in Priority Group 2	3							
Priority Group Bandwidth								
Priority Group	0	1	2	3	4	5	6	7
Bandwidth %	0	50	50	0	0	0	0	0

Information (Team)

Displays adapter information for the team selected in the `teamview` view.

Syntax **info**
 [all | members | livelink | general]

Keywords **all**
Displays all the information for the selected team.

members
Displays member devices for the selected team.

livelink
Displays LiveLink™ information for the selected team.

general
Displays general information for the selected team.

Examples The following example shows the `info` command for viewing team information.

```
qcscli -t TEAM "info"
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.....
Team Information
-----
Team Name           : Team 1
Team Type           : Smart Load Balancing(TM) and Failover
Team Offload Capabilities : LSO, CO, RSS
Team MTU            : 1500
Driver Name         : QLasp.sys
Driver Version      : 2.0.2
Driver Date         : 12/3/2014
```

List (Targets)

Lists the target items in different views. This command is available in all views.

Syntax **list**
 [-f MAC | BDF]
 [-b]
 [-r]
 [-h <host>]
 [<view>]

Keywords -f

Identifies the device. If the `-f` option is not specified, the default behavior is to identify devices using the MAC address. Values include:

MAC

Identifies devices using the MAC address. The MAC option is not appropriate for host, physical adapter, and physical port devices because a single MAC address does not apply to them.

BDF

Identifies devices using the bus, device, or function (BDF) numbers, which are in hexadecimal format.

-b

Lists team names for the selected device. This option is useful only in the `teamview` and is valid only for Windows OS. If specified, the view lists only the team names for the selected device. If not specified, the default behavior is to list teams and the (physical and virtual) members that constitute the team.

-h

Lists targets corresponding to the specified host in the view. This option is useful in most of the commands.

-r

Forces the host to reconnect. This option valid only when the `-h` option is used. This option is useful when you want to reconnect to a host that has recovered from a reboot or link-down.

view

Specifies the view. Available views are defined as follows:

devcon

Lists devices by connection.

teamview

Lists teams and the devices that participate in a team.

ndis

Lists NDIS devices.

vbd

Lists VBD devices.

iscsi

Lists iSCSI devices.

fcoe

Lists FCoE devices.

iscsitargets

Lists iSCSI targets.

fcoetargets

Lists FCoE targets.

phyadapters

Lists physical adapters on the system.

phyports

Lists physical ports on the system.

hosts

Lists the connected hosts.

luns

Lists the LUNs.

vnic

Lists virtual adapters in the system.

iscsiportal

Lists iSCSI portal devices.

fcport

Lists Fibre Channel port devices.

Notes

To stop the list target operation, press the CTRL+BREAK keys.

3-QCS CLI Commands

List (Targets)

Examples The following example shows the `list` command:

```
qcscli "list -f MAC devcon"
```

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.....

C	ID	DevType	Name
-	-----	-----	-----
		WIN-D0ICI9TJD00 Host	WIN-D0ICI9TJD00
0	05:00	Adapter	[VS2019] QLogic FastLinQ QL45000 Series 25GbE Adapter
1	05:00.00	Port	-Port0
2	000E1E080520	VBD	--[0059] [VS2019] QLogic FastLinQ QL45000 Series 25GbE Adapter #59
3	000E1E080520	NDIS	---[0018] QLogic FastLinQ QL45000 Series 25GbE Adapter (VBD Client) #4
4	05:00.01	Port	-Port1
5	000E1E080520	VBD	--[0060] [VS2019] QLogic FastLinQ QL45000 Series 25GbE Adapter #60
6	000E1E080520	NDIS	---[0009] QLogic FastLinQ QL45000 Series 25GbE Adapter (VBD Client)
7	05:00.02	Port	-Port2
8	000E1E080522	VBD	--[0061] [VS2019] QLogic FastLinQ QL45000 Series 25GbE Adapter #61
9	000E1E080522	NDIS	---[0011] QLogic FastLinQ QL45000 Series 25GbE Adapter (VBD Client) #3
10	05:00.03	Port	-Port3
11	000E1E080522	VBD	--[0062] [VS2019] QLogic FastLinQ QL45000 Series 25GbE Adapter #62
12	000E1E080522	NDIS	---[0010] QLogic FastLinQ QL45000 Series 25GbE Adapter (VBD Client) #2
13	01:00	Adapter	QLogic BCM57800 10 Gigabit Ethernet
14	01:00.00	Port	-Port0
15	B82A72E1C447	VBD	--[0261] QLogic BCM57800 10 Gigabit Ethernet #261
16	B82A72E1C447	NDIS	---[0030] QLogic BCM57800 10 Gigabit Ethernet (NDIS VBD Client) #5
17	01:00.01	Port	-Port1
18	B82A72E1C449	VBD	--[0262] QLogic BCM57800 10 Gigabit Ethernet #262
19	B82A72E1C449	NDIS	---[0031] QLogic BCM57800 10 Gigabit Ethernet (NDIS VBD Client) #6
20	01:00.02	Port	-Port2
21	B82A72E1C44B	VBD	--[0055] QLogic BCM57800 Gigabit Ethernet #55
22	B82A72E1C44B	NDIS	---[0014] QLogic BCM57800 Gigabit Ethernet (NDIS VBD Client) #2
23	01:00.03	Port	-Port3
24	B82A72E1C44D	VBD	--[0056] QLogic BCM57800 Gigabit Ethernet #56
25	B82A72E1C44D	NDIS	---[0007] QLogic BCM57800 Gigabit Ethernet (NDIS VBD Client)
26	00:00	Adapter	Microsoft KM-TEST Loopback Adapter
27		Port	-Port0
28	02004C4F4F50	VBD	--[0024] Npcap Loopback Adapter
29	02004C4F4F50	NDIS	---[0024] Npcap Loopback Adapter
30	04:00	Adapter	[VS2019] QLogic 2x25Gb QL41262HLCU Adapter

3-QCS CLI Commands

List (Targets)

```
31  04:00.00      Port      -Port0
32  F4E9D4EDA7DD VBD        --[0040] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter #40
33  F4E9D4EDA7DD NDIS        ---[0020] QLogic 2x25Gb QL41262HLCU Adapter (NDIS) #9
34  F4E9D4EDA7DE VBD        --[0042] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter (FCoE) #42
35  F4E9D4EDA7DE FCoE        ---[0002] QLogic 2x25Gb QL41262HLCU FCoE Device
36  F4E9D4EDA7DF VBD        --[0044] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter #44
37  F4E9D4EDA7DF NDIS        ---[0021] QLogic 2x25Gb QL41262HLCU Adapter (NDIS) #10
38  F4E9D4EDA7E0 VBD        --[0046] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter #46
39  F4E9D4EDA7E0 NDIS        ---[0019] QLogic 2x25Gb QL41262HLCU Adapter (NDIS) #8
40  F4E9D4EDA7E1 VBD        --[0048] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter #48
41  F4E9D4EDA7E1 NDIS        ---[0008] QLogic 2x25Gb QL41262HLCU Adapter (NDIS) #5
42  F4E9D4EDA7E2 VBD        --[0050] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter #50
43  F4E9D4EDA7E2 NDIS        ---[0002] QLogic 2x25Gb QL41262HLCU Adapter (NDIS) #2
44  F4E9D4EDA7E3 VBD        --[0051] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter #51
45  F4E9D4EDA7E3 NDIS        ---[0015] QLogic 2x25Gb QL41262HLCU Adapter (NDIS) #7
46  F4E9D4EDA7E4 VBD        --[0052] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter #52
47  F4E9D4EDA7E4 NDIS        ---[0013] QLogic 2x25Gb QL41262HLCU Adapter (NDIS) #6
48  04:00.01      Port      -Port1
49  F4E9D4EDA7E5 VBD        --[0041] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter #41
50  F4E9D4EDA7E5 NDIS        ---[0006] QLogic 2x25Gb QL41262HLCU Adapter (NDIS) #4
51  F4E9D4EDA7E6 VBD        --[0043] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter #43
52  F4E9D4EDA7E6 NDIS        ---[0017] QLogic 2x25Gb QL41262HLCU Adapter (NDIS) #11
53  F4E9D4EDA7E8 VBD        --[0047] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter #47
54  F4E9D4EDA7E8 NDIS        ---[0001] QLogic 2x25Gb QL41262HLCU Adapter (NDIS)
55  F4E9D4EDA7E9 VBD        --[0049] [VS2019] QLogic 2x25Gb QL41262HLCU Adapter #49
56  F4E9D4EDA7E9 NDIS        ---[0005] QLogic 2x25Gb QL41262HLCU Adapter (NDIS) #3
```

List Discovery Portals

Displays the list of persisted target portals that the iSCSI initiator service will use for discovery for the specified host. This command is available in the context of a host.

Syntax **listdiscoveryportals**
 {-h <host>}

Keywords **-h**
 Specifies the host.

Examples The following example shows the `listdiscoveryportals` command:

```
qcscli "listdiscoveryportals -h 172.28.40.40"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.10.0.
```

C	MAC	Address	Port
-	-----	-----	-----
0	000E1E5027B3	172.16.249.230	3260

List iSNS Servers

Lists iSNS servers. This command is available in the context of a host.

Syntax **listisnsservers**

Keywords None

Examples The following examples show the `listisnsservers` command:

```
qcscli "list -f MAC fcoetargets/fcoe/luns"
```

C	MAC	Dev	Type	Name
-	-----	-----	-----	-----
		Host		WIN-8D9UAMDVMLJ
0	000E1E5027B3	FCoE.T		50:06:01:69:3C:E0:65:61
1	000E1E5027B3	FCoE.T		50:06:01:61:3C:E0:65:61

```
QCScli.exe "listisnsservers"
```

```
QLogic Control Suite Command Line Interface Utility for Windows
v40.0.18.0
```

```
.....
C   iSNS Server Address
-   -----
0   192.168.100.3
```

Log

Saves input and output to a log file. This command is available in all views.

Syntax **log**
 [<file>]

Keywords **file**
Specifies the name of the log file in which to save all input and output. If a file name is not specified, the current log file is closed.

Notes In non-interactive mode, this command creates a log file but does not capture information.

Examples The following example shows the `log` command issued to turn on log capturing to the `ndis.txt` file:

```
qccli "log ndis.txt"
```

C	Status	Name	OS	Platform
0	connected	DL380G7-3809	Windows Server 2012	64 bit (x64)

To turn off log capturing, issue the following command:

```
qccli "log"
```

Login

Logs in to a host. This command is available in the context of a host, iSCSI device, or iSCSI target.

Syntax

login

```

{-m <iSCSI HBA MAC Address>}
{-t <TargetName>}
[-h <hostname>]
[-i <TargetPortalAddress> ]
[-n <TargetPortalSocket>]
[-u <CHAP name>]
[-p <CHAP secret>]
[-if <iface file name>]
[-iu <Initiator CHAP name>]
[-ip <Initiator CHAP secret>]
[-b]
[-e]
[-d]
[-o]
[-mu <CHAP authentication>]
[-ir <iSER target type>]

```

Keywords

-m

Specifies the MAC address of an iSCSI Host Bus Adapter.

-t

Specifies the target name.

-h

Specifies the host name. Required when this command is used in iSCSI target context, because a target might connect to a Host Bus Adapter on different host.

-i

(Optional) Specifies the target IP address. If you specify the `-i` option (target portal IP address), you must also specify the `-n` option (target port number).

-n

Specifies that the login operation is performed using the Host Bus Adapter identified by the `-m` parameter, the target identified by the `-t` parameter, and the target portal address.

-u

Specifies the CHAP name. If the `-u` option (CHAP name) and `-u` option (CHAP secret) are both specified, CHAP authentication is used for login.

-p

Specifies the CHAP secret. If the `-u` option (CHAP name) and `-u` option (CHAP secret) are both specified, CHAP authentication is used for login.

-if

Specifies the iface file name for a Linux host.

-iu

Specifies the initiator CHAP name for a Linux host.

-ip

Specifies the initiator CHAP secret for a Linux host.

-b

Specifies whether the connection should persist across reboot sessions.

-e

Specifies that the header digest is used for login.

-d

Specifies that the data digest is used for login.

-o

Enables MPIO.

-mu

Specifies the mutual CHAP authentication type.

-ir

Specifies the iSCSI Extensions for RDMA (iSER) target type.

Examples The following example shows the `login` command:

```
qcscli "login -m 000e1e5027b1 -t iqn.1986-03.com.hp:storage.p2000g3.1206144efa
-h 172.28.40.40 -i 172.16.249.230"
```

QLogic Control Suite Command Line Interface Utility for Windows v17.0.10.0

.

```
qcscli -t iscsi -f MAC -i 000e1e5027b1 "list iscsitargets"
```

QLogic Control Suite Command Line Interface Utility for Windows v17.0.10.0

```
C    Status      DevType Name
```

3-QCS CLI Commands

Login

```
- -----  
0    connected      iSCSI.T iqn.1986-03.com.hp:storage.p2000g3.1206144efa
```

Logout

Logs out of the specified session. This command is available in the context of a host, iSCSI device, or iSCSI target.

Syntax **logout**
 {-s <SessionID>}
 [-h <host>]

Keywords **-s**
 Specifies the session ID to log out of.

-h
 Required only when the command is used on an iSCSI target, because a target might connect to a Host Bus Adapter on a different host.

Examples The following example shows the `logout` command:

```
qcscli "logout -s FFFFE00005FF8010-FFFFE0001CBB6540 -h 172.28.40.40"
QLogic Control Suite Command Line Interface Utility for Windows v17.0.10.0
.
qcscli -t iscsi -f MAC -i 000ele5027b1 "list iscsitargets"
QLogic Control Suite Command Line Interface Utility for Windows v17.0.10.0
C   Status           DevType Name
-   -----
0   disconnected iSCSI.T iqn.1986-03.com.hp:storage.p2000g3.1206144efa
```

Network Diagnostics

Runs a network diagnostic test on the selected device. This command is available on all NDIS device selections.

Syntax **networkdiag**
 [-p <ipaddr>]

Keywords **-p**
Specifies the IP address used for the test. If this option is not specified, the test uses the IP address found in the gateway IP addresses list.

Notes To stop the test, press the CTRL+BREAK keys.

Examples The following example shows the `networkdiag` command:

```
qcscli -t NDIS -f MAC -i 00101896DC9C "networkdiag -p 151.32.2.63"
QLogic Control Suite Command Line Interface Utility for Windows
v17.0.0.0
..
.....
Link: 10 Gbps Full Duplex
Network test completed successfully.
```

Ping Test

Runs an iSCSI ping test. This command is currently valid only for Windows OS.

Syntax **pingtest**
 <ipaddr>

Keywords None

Examples The following example shows the `pingtest` command:

```
qcscli -t iscsi -f MAC -i 000e1e5027b3 "pingtest 172.16.249.230"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.10.0
```

```
Test completed with no error.
```

Quit

Exits the program.

Syntax **q**

Keywords None

Refresh

Scans the system for hardware and configuration changes. This command is only available in the host view.

Syntax **refresh**

Keywords None

Examples The following example shows the `refresh` command:

```
qcscli refresh
```

```
Refresh host started
```

```
Refresh host finished
```

Refresh All

Scans all systems for hardware and configuration changes. This command is only available in the host view.

Syntax **refreshall**

Keywords None

Notes To stop the refresh operation, press the CTRL+BREAK keys.

Examples The following examples show the `refreshall` command:

```
qcscli refreshall
```

```
Refresh host WIN-OQ5P8PAQ7IP started
```

```
Refresh host WIN-OQ5P8PAQ7IP finished
```

```
# ./QCScli refreshall
```

```
QLogic Control Suite Command Line Interface Utility for Linux  
v30.0.10.0
```

```
semaphore_create: semaphore exists, getting semaphore!
```

```
...
```

```
Refresh host rh67-721-nabaria-d started
```

```
Refresh host rh67-721-nabaria-d finished
```

Refresh Discovery Portal

Performs a SendTargets operation to the target portal and adds the discovered targets to the list of targets maintained by the service. This command is available in the context of a host.

Syntax **refreshdiscoveryportal**
 {-m <iSCSI HBA MAC Address>}
 {-i <TargetPortalAddress>}
 [-n <TargetPortalSocket>]

Keywords **-m**
 Specifies the iSCSI Host Bus Adapter MAC address.

-i
 Specifies the target portal address.

-n
 Specifies the target portal socket. If a value is not specified, the default 3260 port number is used.

Examples The following example shows the `refreshdiscoveryportal` command:

```
qcscli "refreshdiscoveryportal -m 000E1E502683 -i 193.169.101.17"
```

```
C   Status      DevType Name
-   -
0   connected   iSCSI.T
iqn.2001-05.com.equallogic:0-8a0906-07f4b7e04-8496c0a767e54930
1   disconnected iSCSI.T
iqn.2001-05.com.equallogic:0-8a0906-4734b7e04-df70000002051ffa
2   disconnected iSCSI.T
iqn.2001-05.com.equallogic:0-8a0906-5844b7e04-a2b0000002852021
3   disconnected iSCSI.T
iqn.2001-05.com.equallogic:0-8a0906-9db4b7e04-b3a0000005c5231b
4   disconnected iSCSI.T
iqn.2001-05.com.equallogic:0-8a0906-0bc4b7e04-b020000007452692
5   disconnected iSCSI.T
iqn.2001-05.com.equallogic:0-8a0906-7c94b7e04-1596c0a617d52d65
6   disconnected iSCSI.T
iqn.2001-05.com.equallogic:0-8a0906-80c4b7e04-f9e6c0a618952e77
7   disconnected iSCSI.T
iqn.2001-05.com.equallogic:0-8a0906-9364b7e04-5716c0a619652f4f
```

Refresh iSNS Server

Refreshes the list of targets discovered from the specified iSNS server. This command is available in the context of a host.

Syntax **refreshisnsserver**
 {-i <iSNS Server Address>}

Keywords **-i**
 Specifies the address of the iSNS server.

Examples The following example shows the `refreshisnsserver` command:

```
qcscli "refreshisnsserver -i 192.168.30.51"
QCScli.exe "refreshisnsserver -i 192.168.100.3"
C:\Users\Administrator\Desktop\Bod\QCS\40.0.18>QCScli.exe
"refreshisnsserver -i 192.168.100.3"
QLogic Control Suite Command Line Interface Utility for Windows
v40.0.18.0
.....
```

Remove (Team)

Removes the specified team or all teams. This command is only available if the active selection is a team or in the `teamview` view.

This command is valid only for Windows OS.

Syntax **remove**
 {-c all | <team name>}

Keywords **-c**
Specifies the team or teams to remove:

all

Specifies that all teams in the system are removed.

team name

Specifies that the team with this name is removed.

Examples The following examples show the `remove` command:

```
qcscli -t TEAM "remove -c all"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.7.0
```

```
.....  
Please wait while executing Teaming operation...  
Remove Team succeeded.
```

```
qcscli -t TEAM "remove ""Team 2"""
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.7.0
```

```
.....  
Please wait while executing Teaming operation...  
Remove Team succeeded.
```

Remove All Hosts

Removes all remote hosts from the host management list.

This command is valid only for Windows OS.

Syntax **removeallhosts**

Keywords None

Examples The following example shows the `removeallhosts` command:

```
qcscli "removeallhosts"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.7.0
```

```
.....
```

Remove Discovery Portal

Removes a portal from the list of portals to which the iSCSI initiator service sends SendTargets request to discover targets. This command is available in the context of a host.

Syntax **removediscoveryportal**
 {-m <iSCSI HBA MAC Address>}
 {-i <TargetPortalAddress>}
 [-n <TargetPortalSocket>]

Keywords **-m**
 Specifies the MAC address of the iSCSI Host Bus Adapter.

-i
 Specifies the target portal address.

-n
 Specifies the target portal socket. If a value is not specified, the default port number of 3260 is used.

Examples The following example shows the `removediscoveryportal` command:

```
qcscli "removediscoveryportal -m 000E1E502683 -i 193.169.101.17"
```

```
C    Status            DevType Name
-    -
0    connected        iSCSI.T
ign.2001-05.com.equallogic:0-8a0906-07f4b7e04-8496c0a767e54930
```

Remove Host

Removes (disconnects) the specified host.

Syntax **removehost**
 <host name | IP address>

Keywords **host name**
 Specifies the name of the host to be disconnected.

IP address
 Specifies the IP address of the host to be disconnected.

Examples The following examples show the `removehost` command:

```
qcscli "removehost 172.28.63.183"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.0.0
```

```
.....
```

```
qcscli "removehost 720-4911-10022"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.0.0
```

```
.....
```

Remove iSNS Server

Removes the specified iSNS server. This command is available when a host is selected.

Syntax **removeisnsserver**
 {-i <iSNS Server Address>}

Keywords **-i**
Identifies the iSNS server to be removed.

Examples The following example shows the `removeisnsserver` command:

```
qcscli "removeisnsserver -i 192.168.55.02"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v40.0.18.0
```

```
.....
```

Remove Multiple NPIV Ports

Removes multiple NPIV ports at the same time. This command is available in the context of an FCoE Host Bus Adapter.

Syntax **removemultipivport**
 {{-s <x:xx>} {-e <x:xx>}}

Keywords **-s**
Specifies the starting name range, where only three digits (1–3) are allowed to change; for example: 2x:xx:00:10:18:aa::bb::cc. The other digits are from the current WWPN or WWNN.

-e
Specifies the ending name range, where only three digits (1–3) are allowed to change; for example: 2x:xx:00:10:18:aa::bb::cc. The other digits are from the current WWPN or WWNN.

Examples The following example shows the `removemultipivport` command:

```
qcscli -t FCoE -f MAC -i 000e1e5027b3 "removemultipivport -s 1:11  
-e 1:19"
```

```
Removing NPIV ports
```

```
-----
```

```
21:11:00:0e:1e:50:27:b3      : Succeed  
21:12:00:0e:1e:50:27:b3      : Succeed  
21:13:00:0e:1e:50:27:b3      : Succeed  
21:14:00:0e:1e:50:27:b3      : Succeed  
21:15:00:0e:1e:50:27:b3      : Succeed  
21:16:00:0e:1e:50:27:b3      : Succeed  
21:17:00:0e:1e:50:27:b3      : Succeed  
21:18:00:0e:1e:50:27:b3      : Succeed  
21:19:00:0e:1e:50:27:b3      : Succeed
```

```
Result: 9 NPIV port(s) removed.
```

Remove NPIV Port

Removes the NPIV port from the adapter. This command is available when an FCoE Host Bus Adapter is selected.

Syntax **removenpivport**
 {-p <WWPN>}

Keywords **-p**
 Specifies the port by WWPN.

Examples The following example shows the `removenpivport` command:

```
qcscli -t FCoE -f MAC -i 000e1e5027b3 "removenpivport -p
21:11:00:0e:1e:50:27:b3
```

```
QLogic Control Suite Command Line Interface Utility for Windows
v17.0.10.0
```

```
.....
```

Remove Persistent Target

Removes a target from the list of persistent targets. This command is available in the context of a host or in the context of an iSCSI target.

Syntax **removepersistenttarget**
 {-t <TargetName>}
 [-h <host>]

Keywords **-t**
Required when using this command on a host. Do not specify this keyword when using this command on an iSCSI target.

-h
Required when using this command on an iSCSI target, because a target might connect to the Host Bus Adapter on a different host.

Examples The following examples show the `removepersistenttarget` command:

```
qcscli "removepersistenttarget -t
iqn.2001-05.com.equallogic:0-8a0906-07f4b7e04-8496c0a767e54930 -h
173.29.42.254"

C   Status      DevType      Name
-   -
0   disconnected iSCSI.T iqn.2001-05.com.equallogic:0-8a0906-07f4b7e04-8496c0a767e54930
```

Remove Target

Removes the specified static target. This command is available in the context of a host.

Syntax **removetarget**
 {-t <TargetName>}

Keywords **-t**
 Identifies the static target.

Examples The following example shows the `removetarget` command:

```
qcscli "removetarget -t  
iqn.2001-05.com.equallogic:0-8a0906-07f4b7e04-8496c0a767e54930"  
No devices to list.
```

Reset Session Statistics

Resets statistical information for either all sessions or for a specified session. This command is only available if the active selection is an iSCSI device.

Syntax **resetsessionstats**
 [**<sessionid>**]

Keywords **sessionid**
Identifies a single session for which to reset statistics.

Examples The following example shows the `resetsessionstats` command:

```
qcscli resetsessionstats  
Session Statistics  
-----  
Session Name           :  
iqn.2001-05.com.equallogic:0-8a0906-07f4b7e04-8496c0a767e54930  
Session Id             : FFFFE000095EC010-FFFFE0003634C7C0  
Bytes sent              : 0  
Bytes received         : 119964  
PDU sent               : 104  
PDU received           : 104  
Digest errors          : 0  
Connection Timeout errors: 0  
Format errors          : 0
```

Reset Statistics

Resets the statistics for the selected device. This command is available everywhere that the `stats` command (see [Statistics](#)) is applicable.

Syntax `resetstats`

Keywords None

Examples The following example shows the `resetstats` command:

```
qcsccli -t NDIS -f MAC -i 001018AD45E0 "resetstats"
QLogic Control Suite Command Line Interface Utility for Windows
v17.0.7.0
.....
```

Restore (Team)

Restores team configuration from the specified file and deletes existing team configurations. This command is only available if the active selection is a team or in the `teamview` view.

This command is valid only for Windows OS.

Syntax **restore**
 [-h <host>] <file>

Keywords **-h**
Specifies the host. Required if no team is selected and QCS CLI needs to know to which host the teams will be restored.

Examples The following example shows the `restore` command:

```
qcscli -t TEAM "restore c:\team_bdf.txt"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.7.0
```

```
.....
```

```
Please wait while executing Teaming operation...
```

```
Restore Team succeeded.
```

Save (Team)

Saves teaming configuration to a specified file and format. This command is only available if the active selection is a team or in the `teamview` view.

This command is valid only for Windows OS.

Syntax **save**
 {-f MAC | BDF} <file>

Keywords **-f MAC**

Saves the team configuration to the specified <file> in MAC format.

-f BDF

Saves the team configuration to the specified <file> in bus-device-function (BDF) hexadecimal format.

Examples The following examples show the `save` command:

```
qcscli -t TEAM "save -f BDF c:\team_bdf.txt"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.7.0
```

```
.....
```

```
Please wait while executing Teaming operation...
```

```
Save Team succeeded.
```

```
qcscli -t TEAM "save -f MAC c:\team_mac.bcg"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.7.0
```

```
.....
```

```
Please wait while executing Teaming operation...
```

```
Save Team succeeded.
```

Select (Target)

Selects a target from the current view or displays the selected target in the current view if `<index>` is not specified. This command is available in all views.

Syntax `select`
 `[<index>]`

Keywords None

Examples The following example shows the `select` command:

```
qcscli "Select 2"
```

```
QLogic Control Suite Command Line Interface Utility for Windows v17.0.0.0
```

```
.....
```

C	Status	Name	OS	Platform
0	connected	DL380G7-3809	Windows Server 2012	64 bit (x64)
1	connected	720-4883-10022	N/A	64 bit (x64)
2	connected	720-4911-10022	N/A	64 bit (x64)

Session Statistics

Displays statistical information for either all sessions or for a specified session. This command is only available if the active selection is an iSCSI device.

Syntax **sessionstats**
 [<session_id>]

Keywords **sessionid**
Displays statistics for the session specified with this ID.

Examples The following example shows the `sessionstats` command:

```
qcscli -t FCoE -f MAC -i 000e1e5027b1 "sessionstats <session_id>"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.10.0
```

```
Session Statistics
```

```
-----
```

```
Session Name           :  
iqn.1986-03.com.hp:storage.p2000g3.1206144efa_FFFFFFFA8  
03E622240  
Session Id              : FFFFFFFA806EFA6008-FFFFFFA803E622240  
Bytes sent               : 0  
Bytes received           : 3398667  
PDU sent                 : 3031  
PDU received             : 3101  
Digest errors            : 0  
Connection Timeout errors: 0  
Format errors            : 0
```

Sessions

Lists iSCSI sessions on the selected iSCSI adapter. This command is only available if the actively selected device is an iSCSI device.

Syntax **sessions**

Keywords None

Examples The following example shows the `sessions` command:

```
qcscli -t iscsi -f MAC -i 000e1e5027b1 "sessions"
QLogic Control Suite Command Line Interface Utility for Windows v17.0.10.0
iSCSI Session Id                               Session Name
-----
FFFFFFFFA806EFA6008-FFFFFFFFA803E622240:
iqn.1986-03.com.hp:storage.p2000g3.1206144efa
_FFFFFFFA803E622240
```

Show Selected (Target)

Displays the selected target item.

Syntax `showsel`

Keywords None

Examples The following example shows the `showsel` command:

`qcscli showsel`

C	Status	Name	OS	Platform
-	-----	-----	-----	-----
0	connected	WIN-OQ5P8PAQ7IP	Windows Server 2012 R2	64 bit (x64)

Statistics

Displays statistical information for the selected NDIS device. This command is available on all NDIS, iSCSI, and VBD device selections.

Syntax

For NDIS:

stats

[all | general | IEEE | qlasp]

For iSCSI:

stats

[all | login | instance | custom]

For VBD:

stats

[all | custom | switch | vf]

Keywords

all

Displays all the statistical information for the selected iSCSI adapter.

general

Displays general statistics for the selected iSCSI adapter.

IEEE

Displays IEEE statistics for the selected iSCSI adapter.

qlasp

Displays QLASP statistics for the selected iSCSI adapter.

login

Displays login statistics for the selected iSCSI adapter.

instance

Displays instance statistics for the selected iSCSI adapter.

custom

Displays custom statistics for the selected iSCSI adapter.

switch

Displays SR-IOV switch statistics.

vf

Displays SR-IOV virtual function statistics.

Notes If no keyword is specified, the `stats` command displays all of the statistical information for the selected iSCSI adapter.

Examples The following examples show the `stats` command:

```
qcscli -t NDIS -f MAC -i 001018AD45E0 "stats"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.7.0
```

```
.....
```

```
General Statistics
```

```
-----
```

```
Frames Tx. OK                : 2480622  
Frames Rx. OK                : 72108500  
Directed Frames Tx.         : 2150747  
Multicast Frames Tx.        : 301584  
Broadcast Frames Tx.        : 28291  
Directed Frames Rx.         : 2119082  
Multicast Frames Rx.        : 62955498  
Broadcast Frames Rx.        : 7033925  
Frames Rx. with CRC Error   : 0
```

```
IEEE 802.3 Statistics
```

```
-----
```

```
Frames Rx. with Alignment Error      : 0  
Frames Tx. with one Collision        : 0  
Frames Tx. with more than one Collision : 0  
Frames Tx. after Deferral            : 0
```

```
qcscli -t VBD -f MAC -i 001018AD45E0 "stats"
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.7.0
```

```
.....
```

```
Custom Statistics
```

```
-----
```

```
Large Send Offload Transmit Requests      : 6  
Frames Discard Due to Lack of On-chip Buffer Space : 0  
Frames Discard Due to Lack of Host Rx Buffers   : 0  
Total Offload TCP Connections             : 0  
Total Offload iSCSI Connections           : 0
```

```
SRIOV Switch Statistics
```

```
-----
          Num of Active VFs : 0
```

```
SRIOV VF Statistics
```

```
-----
No hardware is available for SRIOV.
```

```
qcscli -t iSCSI -f MAC -i 001018AD45E1 "stats all"
```

```
QLogic Control Suite Command Line Interface Utility for Windows
v17.0.7.0
```

```
.....
Initiator Login Statistics
```

```
-----
Login Accept Responses                      : 0
Login other failed Responses                : 0
Login Redirect Responses                   : 0
Login Authentication Failed Responses      : 0
Login target authentication failure        : 0
Login target negotiation failure           : 0
Normal logout command PDU                  : 0
Other logout command PDU                   : 0
Local initiator login failures             : 0
```

```
Custom Statistics
```

```
-----
Total Offload iSCSI Connections            : 0
```

```
Initiator Instance Statistics
```

```
-----
Session digest errors                      : 0
Session connection timeout error           : 0
Session format error                      : 0
Sessions failed                           : 0
```

```
qcscli -t NDIS -f MAC -i 001018AD45E2 "stats QLASP"
```

```
QLogic Control Suite Command Line Interface Utility for Windows
v17.0.7.0
```

```
.....
QLASP Statistics
```

```
-----
Tx. Packet                                2564477
```

```

Tx. Packet Discarded      0
Rx. Packet                 72748822
Rx. Packet Discarded      0

```

qcscli -t iscsi -f MAC -i 000e1e5027b1 "stats"

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Initiator Login Statistics

```

-----
Login Accept Responses           : 0
Login other failed Responses     : 0
Login Redirect Responses        : 0
Login Authentication Failed Responses : 0
Login target authentication failure : 0
Login target negotiation failure  : 0
Normal logout command PDU       : 0
Other logout command PDU       : 0
Local initiator login failures   : 0
Custom Statistics
-----

```

```

Total Offload iSCSI Connections : 0
Initiator Instance Statistics
-----

```

```

Session digest errors           : 0
Session connection timeout error : 0
Session format error            : 0
Sessions failed                  : 0

```

qcscli -t fcoe -f MAC -i 000e1e5027b1 "stats"

QLogic Control Suite Command Line Interface Utility for Windows
v17.0.10

Statistics

```

-----
Input requests                   0
Output requests                  0
Control requests                 0
Input megabytes                  0
Output megabytes                 0
FCoE transmitted frames         0
FCoE transmitted bytes          0

```

Transmitted FCP frames	0
FCoE received frames	0
FCoE received bytes	0
Received FCP frames	0
FIP VLAN negotiations performed	0
FIP fabric discoveries performed	0
FLOGIs performed	0
FDISCs performed	0
Packets received with wrong FC-CRC	0
Wrong FCoE version count	0
Wrong delimiter count	0
Missing frame count	0
Receive sequence timeout count	0
ULP_TOV expiration count	0
REC expiration count	0
ABTS count	0
SRR count	0
Reset LUN count	0
Reset target count	0
Session recovery count	0
FCoE dropped frames	0
Dropped sequences	0

The following example shows the `stats` command with LinkDump support for 41000 and 45000 Series Adapters:

```
localhost.localdomain\devcon\11\L2NIC>stats
Statistics
-----
Single Collision Frames      0
Multiple Collision Frames   0
Deferred Transmissions      0
Late Collisions             0
Excessive Collisions        0
Carrier Sense Errors        0
Packets Transmitted         249
Packets Received            1077
Total Transmit Error        0
Total Receive Error         0
Directed Frames Transmitted 0
```

3-QCS CLI Commands

Statistics

Multicast Frames Transmitted	187
Broadcast Frames Transmitted	62
Directed Frames Received	0
Multicast Frames Received	58
Broadcast Frames Received	1019
RDMA Inbound Bytes	0
RDMA Outbound Bytes	0
RDMA Inbound Frames	0
RDMA Outbound Frames	0
RDMA Icrc Errors	0
RDMA Retransmit Events	0
RDMA Silent Drops	0
RDMA Rnr Nacks Sent	0
RDMA Ecn Pkt Rcv	0
RDMA Cnp Pkt Rcv	0
RDMA Cnp Pkt Sent	0
RDMA Cnp Pkt Reject	0
RDMA Implied Nak Seq Err	0
RDMA Duplicate Request	0
RDMA Local Ack Timeout Err	0
RDMA Out Of Sequence	0
RDMA Packet Seq Err	0
RDMA Rnr Nak Retry Err	0
RDMA Req Cqe Error	0
RDMA Req Remote Access Errors	0
RDMA Req Remote Invalid Rqst	0
RDMA Resp Cqe Error	0
RDMA Resp Local Length Error	0
RDMA Resp Remote Access Errors	0
Transmit OK	0
Receive OK	0
Transmit Error	0
Receive Error	0
Receive No Buffer	0
Directed Bytes Transmit	0
Directed Frames Transmit	0
Multicast Bytes Transmit	0
Multicast Frames Transmit	0
Broadcast Bytes Transmit	0

3-QCS CLI Commands

Statistics

Broadcast Frames Transmit	0
Directed Bytes Receive	0
Directed Frames Receive	0
Multicast Bytes Receive	0
Multicast Frames Receive	0
Broadcast Bytes Receive	0
Broadcast Frames Receive	0
Receive CRC Error	0
Transmit Queue Length	0
Receive Discards	0
Transmit Discards	0
Bytes Receive	0
Bytes Transmit	0
Receive Error Alignment	0
Transmit One Collision	0
Transmit Deferred	0
Transmit More Collisions	0
Transmit Max Collisions	0
Receive Overrun	0
Transmit Underrun	0
Transmit Heartbeat Failure	0
Transmit Times CRS Lost	0
Transmit Late Collisions	0

LDUMP Info

FecMode	FEC_MODE_NONE
FecCoreErrCnt	0
FecUnCoreErrCnt	0
FecRsvd	0
ChipMode	CHIP_MODE_2X25G
MfwLinkState	LINK_UP
Pcs Link	0
MacFault	NONE
RxSignalDetect	PRESENT
LinkSpeed	SPEED_10GBPS
Module	10G_SR
FlowControl	UNKNOWN
Autoneg	DISABLED
AnLpAdvSpeeds	UNKNOWN

3-QCS CLI Commands

Statistics

AnLpAdvFlowControl	UNKNOWN
TxPreFir	0
TxMainFir	0
TxPostFir	0
RxDfe	RX_DFE_OFF
LinkChangeCount	2
LfaCount	3
LfaState	0
MfwRawLinkState	14680327

Unassigned (Adapters)

Lists adapters that are not yet part of a team. This command is only available in `teamview view` listing.

This command is valid only for Windows OS.

Syntax **unassigned**
 [-h <host>]

Keywords **-h**
Required if no team is selected and QCS CLI needs to know the host to which the unassigned NICs belong.

Examples The following shows an example of the `unassigned` command:

```
qcscli -t TEAM "unassigned"
QLogic Control Suite Command Line Interface Utility for Windows v17.0.7.0
.....
MAC                Unassigned Adapter Name
-----
D4AE52897638 [0007] QLogic BCM57800 10 Gigabit Ethernet (NDIS VBD Client) #84
D4AE52897632 [0004] QLogic BCM57800 10 Gigabit Ethernet (NDIS VBD Client) #85
D4AE52897634 [0002] QLogic BCM57800 Gigabit Ethernet (NDIS VBD Client) #86
D4AE52897636 [0003] QLogic BCM57800 Gigabit Ethernet (NDIS VBD Client) #87
```

Upgrade (Firmware)

Upgrades the firmware with the Multiboot Image (MBI) file at the adapter level. An upgrade or downgrade depends on the image version currently in use. This command forces a firmware upgrade to the version present in the MBI file.

Syntax **upgrade**
 -mbi <filepath and name>

Keywords **filepath and name**
Directory path to the location of the MBI file and the file name.

Notes After the firmware upgrade, issue the `refresh` command (see [“Refresh” on page 112](#)) at the host level to confirm the updated MBI version shown in the output of the `info` command (see [“Information \(Adapter\)” on page 91](#)) at the adapter level.

Examples The following examples show the `upgrade` command:

```
upgrade -mbi C:\ql_mbi_81521.bin
```

```
WIN-OPRU38IV51J\devcon\13\adapter>upgrade -mbi  
C:\Users\Administrator\Desktop\Bod\fwupg\2.10.42\ql_hp_bb_mbi_8.34  
.05.bin  
Starting FW-Upgrade Operation.. Please Wait.. It may take some  
time !!  
Upgrade Operation is Successful! Please Reboot the server to take  
the changes in effect!!
```

Version

Displays the version of QLogic Control Suite CLI installed.

Syntax `version`

Keywords None

Examples The following shows an example of the `version` command:

```
qcscli version
```

```
QLogic Control Suite Command Line Interface Utility for Windows  
v17.0.7.0
```

Legacy Commands

Legacy commands include commands used by QLogic QCS CLI v1.0.10 or earlier. These commands are deprecated, and only supported to provide backward compatibility. Marvell recommends that you use the POSIX-compliant commands described in [“Command List” on page 19](#).

QCSCLI [-mac MAC | -bdf BDF] [Command]

[Table 3-3](#) lists the available legacy commands.

Table 3-3. Legacy Commands

Command	Description
help	Lists available commands
q	Exits QCS CLI
dev	Selects an adapter or lists available adapters
bdf	Displays the PCI bus, device, or function number of each device
log	Logs all input and output to a file
version	Displays the version of QCS CLI
info	Displays adapter information for the selected NIC
stats	Displays statistical information for the selected NIC
cfg	Configures parameters for the selected device
diag	Configures and runs a diagnostic test

4 QCS CLI Usage

This chapter provides the following information about using QLogic Control Suite CLI to perform tasks:

- [Using Interactive Mode](#)
- [Performing Management Tasks](#)

Using Interactive Mode

To start QLogic Control Suite CLI in interactive mode, issue the `QCSCLI` command without additional parameters.

For example:

```
QCSCLI
```

To exit from QLogic Control Suite CLI interactive mode, issue the `q` command.

For example:

```
q
```

Performing Management Tasks

This section provides methods for performing some common management tasks using QCS CLI. The methods for performing the task in the CLI mode (noninteractive), interactive mode, or both, are included as appropriate.

- [Listing Target Adapters of Different Views in CLI Mode](#)
- [Obtaining Context Help for each Command](#)
- [Switching Between Different Views of Target Adapters](#)
- [Selecting a Target for the Command to Operate On](#)
- [Selecting a Target in Interactive Mode](#)
- [Getting Information for a Selected Target](#)
- [Configuring Advanced Parameters for a Selected Target](#)
- [Displaying Licenses for a Selected Target](#)

- [Configuring the iSCSI Parameter for a Selected Target](#)
- [Configuring the Resource of a Selected Target](#)
- [Configuring OoB Management for a Selected Target](#)
- [Configuring Systoe for the System](#)
- [Getting Statistics for a Selected Target](#)
- [Resetting Statistics for a Selected Target](#)
- [Running Diagnostic Tests for a Selected Target](#)
- [Running a Cable Diagnostic Test for a Selected Target](#)
- [Running a Network Diagnostic Test for a Selected Target](#)
- [Connecting to an FCoE Target](#)
- [Connecting to an iSCSI Target](#)
- [Creating a Team Configuration](#)
- [Restoring a Team Configuration](#)
- [Saving a Team Configuration](#)
- [Removing One or More Teams](#)
- [Displaying Unassigned Adapters](#)
- [Switching Between Primary and Standby Adapters in a Team](#)
- [Connecting an iSCSI Target](#)
- [Performing Other Common Tasks](#)

Listing Target Adapters of Different Views in CLI Mode

CLI Mode

Issue the `QCSCLI list <view>` command to list all the target adapters of the <view>. For example:

`QCSCLI list NDIS` lists all NDIS adapters in the system.

`QCSCLI list devcon` lists all adapters by connection.

Obtaining Context Help for each Command

CLI Mode

Use the following syntax to obtain the help text for a specific operation. For example:

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "info help"` displays all the help text for the information of the selected NDIS adapter.

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "cfg help"` displays help for the configuration of the selected NDIS adapter and their current settings.

Interactive Mode

In interactive mode, use the keywords `help` or `?` to show help for a specific command. For example:

`cfg Advanced help` displays help text for the advanced configuration for the selected device.

`info ?` displays help text for the information of the selected adapter

Switching Between Different Views of Target Adapters

CLI Mode

In the CLI mode, use the following syntax to list devices in different views:

```
QCSCLI "list [-f MAC | BDF] [<view>]"
```

For example:

`QCSCLI "list -f MAC NDIS"` lists all NDIS devices that are present in the system with the MAC address as identifiers.

`QCSCLI "list -f BDF NDIS"` lists all NDIS devices that are present in the system with bus-device-function (BDF) hexadecimal numbers as identifiers.

`QCSCLI "list -f MAC devcon"` lists all adapters by the connection identifying devices using their MAC address.

`QCSCLI "list -f BDF devcon"` lists all adapters by the connection identifying devices using their BDF address.

Interactive Mode

Issue the `list [-f MAC|BDF] [<view>]` command to switch between different views of target adapters. The default view of QCS CLI is `devtype` in interactive mode. All target adapters for the selected view are listed and the selected default adapter is highlighted. For example:

`list -f MAC NDIS` lists all NDIS devices that are present in the system with MAC address as identifiers.

`list -f BDF NDIS` lists all NDIS devices that are present in the system with BDF hexadecimal numbers as identifiers.

`list -f MAC devcon` lists all adapters by the connection identifying devices using their MAC address.

`list -f BDF devcon` lists all adapters by the connection identifying devices using their BDF hexadecimal address.

Selecting a Target for the Command to Operate On

CLI Mode

Use the `-t`, `-f`, and `-i` options to uniquely select the target for the following `<command string>` to operate on it. For example:

`QCSCLI -t NDIS -f mac -i 0010181a1b1c info` displays the adapter information of the selected NDIS adapter whose current MAC address is 00:10:18:1a:1b:1c.

`QCSCLI -t VBD -f mac -i 0010181a1b1d info` displays the adapter information of the selected VBD adapter whose current MAC address is 00:10:18:1a:1b:1d.

`QCSCLI -t iSCSI -f mac -i 0010181a1b1e info` displays the adapter information of the selected iSCSI adapter whose current MAC address is 00:10:18:1a:1b:1e.

`QCSCLI -t TEAM -f name -i Team1 "info"` displays the team information of the selected (Team1) team.

`QCSCLI -t VNIC -f name -i VLAN1 "info"` displays the virtual adapter information which is identified by a VLAN name (VLAN1).

Interactive Mode

`select [<index>]` selects the target from the target adapters of the current view.

`showsel` displays the selected target adapter. Any command entered after this command applies to the selected target adapter.

`list NDIS` and `select 3` selects the third adapter from the list of all target adapters in the `NDIS` view.

`list devcon` and `select 5` selects the fifth adapter from the list of all target adapters of `devcon` view.

Selecting a Target in Interactive Mode

Interactive Mode

Issue the `select [<index>]` command to select the target from the `target0` adapters of the current view.

Issue the `showsel` command to display the selected target adapter. Any command entered after this command applies to the selected target adapter.

Examples:

`list NDIS` and `select 3` selects the third adapter from the list of all target adapters in the `NDIS` view.

`list devcon` and `select 5` selects the fifth adapter from the list of all target adapters in the `devcon` view.

Getting Information for a Selected Target

No information is available for NIC partition or virtual function targets on Linux systems. Issue the following command:

```
info {all | [ vitalsigns | driver | nicpartition | switch | vf]}
```

CLI Mode

To get information about the selected target, issue the following command, which is available for NDIS, VBD and team:

```
QCSCLI -t <target type> -f <format> -i <target ID> info [ all |  
vitalsigns | driver | hardware | resource | members | livelink |  
general ]
```

Where the `resource` keyword applies only on Windows systems.

Examples:

QCSCLI -t NDIS -f mac -i 0010181a1b1c "info" displays all the information for the selected NDIS adapter.

QCSCLI -t NDIS -f mac -i 0010181a1b1c "info vitalsigns" displays the vital signs information for the selected NDIS adapter.

QCSCLI -t NDIS -f mac -i 0010181a1b1c "info resource" displays the resources information for the selected NDIS adapter.

QCSCLI -t VBD -f mac -i 0010181a1b1d "info driver" displays the driver information for the selected VBD adapter.

QCSCLI -t NDIS -f mac -i 0010181a1b1c "info hardware" displays the hardware information for the selected NDIS adapter.

QCSCLI -t NDIS -f mac -i 0010181a1b1c "info all" displays all the information for the selected NDIS adapter.

QCSCLI -t TEAM -f name -i team1 "info all" displays all the information for the selected team.

QCSCLI -t TEAM -f name -i team1 "info members" displays all the information for the selected team.

QCSCLI -t TEAM -f name -i team1 "info livelink" displays LiveLink settings for the selected team.

QCSCLI -t TEAM -f name -i team1 "info general" displays the general information for the selected team.

QCSCLI -t TEAM -f name -i team1 "info" displays all the information for the selected team.

`QCSCLI -t TEAM -f name -i "^Team1 "info"` displays the team information for the selected (^Team1) team. If the team name contains a special character, it must be enclosed within double quotes.

`QCSCLI -t VNIC -f name -i "^VLAN1 "info"` displays the virtual adapter information that is identified by a VLAN name (^VLAN1). If the VLAN name contains a special character, it must be enclosed within double quotes.

Interactive Mode

Use the `list <view>` and `select <idx>` commands to select the target device.

Use the `info [ all | vitalsigns | driver | hardware | resource ]` command to get information about the selected target.

Examples:

`info` or `info all` displays all the information for the selected target.

`info vitalsigns` displays vital signs information for the selected target.

`info resource` displays resource information for the selected target. (The 41000 Series Adapters do not support the `info resource` command.)

`info driver` displays driver information for the selected target.

`info hardware` displays hardware information for the selected target. (The 41000 Series Adapters do not support the `info hardware` command.)

`info members` displays all the information for the selected team.

`info livelink` displays LiveLink settings for the selected team.

`info general` displays the general information for the selected team.

Configuring Advanced Parameters for a Selected Target

CLI Mode

Issue the following command to display and configure the advanced parameters for the selected target:

```
QCSCLI -t <target type> -f <target format> -i <target ID> cfg  
Advanced [param | param=value]
```

Examples:

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "cfg Advanced"` displays all advanced parameters of the selected NDIS adapter and their current settings.

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "cfg Advanced \"802.1p QOS\""` displays the current setting and all valid settings of the specified advanced parameter 802.1p QOS.

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "cfg Advanced \"802.1p QOS\"=\"Enable\""` sets the advanced parameter, 802.1p QOS, to Enable.

```
QCSCLI -t NDIS -f mac -i 0010181a1b1c "cfg advanced \"Flow Control\"=\"Rx \"&\" Tx enabled\"" sets the advanced parameter Flow Control to Rx & Tx enabled.
```

Interactive Mode

Use the `list <view>` and `select <idx>` commands to select the target device.

Use the `cfg Advanced [param | param=value]` command to display and configure the advanced parameter of the selected target.

Examples:

`cfg Advanced` displays all advanced parameters of the selected adapter and their current settings.

`cfg Advanced "802.1p QOS"` displays the current setting and all valid settings of the 802.1p QOS advanced parameter.

`cfg Advanced "802.1p QOS"="Disable"` sets the 802.1p QoS parameter to Disable.

Displaying Licenses for a Selected Target

CLI Mode

Issue the `QCSCLI -t <target type> -f <target format> -i <target ID> cfg licenses` command to display the licenses of the selected target.

Example:

`QCSCLI -t VBD -f mac -i 0010181a1b1d "cfg Licenses"` displays all licenses of the selected VBD adapter and their current settings.

Interactive Mode

Issue the `list <view>` and `select <idx>` commands to select a target device. Then issue the `cfg Licenses` command to display the licenses of the selected target.

Example:

`cfg Licenses` displays all license parameters for the selected adapter and their current settings.

Configuring the iSCSI Parameter for a Selected Target

CLI Mode

Issue the following command to display and configure the iSCSI parameter for the selected target:

```
QCSCLI -t <target type> -f <target format> -i <target ID> cfg iSCSIMgmt [key | key=value]
```

Examples:

`QCSCLI -t iSCSI -f mac -i 0010181a1b1e "cfg iSCSIMgmt"` displays all iSCSI management keys for the selected iSCSI adapter and their current settings.

`QCSCLI -t iSCSI -f mac -i 0010181a1b1e "cfg iSCSIMgmt \IPV4DHCP\""` displays the current setting and all valid settings for the IPV4DHCP key.

`QCSCLI -t iSCSI -f mac -i 0010181a1b1e "cfg iSCSIMgmt \IPV4DHCP\"="Enable\""` sets the IPV4DHCP key to Enable.

Interactive Mode

Issue the `list <view>` and `select <idx>` commands to select the target device. Then issue the `cfg iSCSIMgmt [key | key=value]` command to display and configure the iSCSI parameter for the selected target.

Examples:

`cfg iSCSIMgmt` displays the current settings of all iSCSI management parameters for the selected adapter and their current settings.

`cfg iSCSIMgmt "IPV4DHCP"` displays the current setting and all valid settings of the IPV4DHCP key.

`cfg iSCSIMgmt "IPV4DHCP"="Enable"` enables the IPV4DHCP setting.

Configuring the Resource of a Selected Target

CLI Mode

Issue the following command to display and configure the resource parameter of the selected target:

```
QCSCLI -t <target type> -f <target format> -i <target ID> cfg
Resource [key | key=value]
```

Examples:

`QCSCLI -t VBD -f mac -i 0010181a1b1d "cfg Resource"` displays all resource reservation keys and their current settings.

`QCSCLI -t VBD -f mac -i 0010181a1b1d "cfg Resource \CONFIGURABLE\""` displays all configurable Resource Reservation keys with their current settings and valid settings.

`QCSCLI -t VBD -f mac -i 0010181a1b1d "cfg Resource \CONFIG\"="RSVD\""` sets the "Offload Configuration" to "Reserved Resources".

`QCSCLI -t VBD -f mac -i 0010181a1b1d "cfg Resource \TOECONN\"="50\""` sets the "TOE Connections" to 50. The command is valid only when the "Offload Configuration" is in "Reserved Resources".

`QCSCLI -t VBD -f mac -i 0010181a1b1d "cfg Resource \\"iSCSI\\"=\"Disable\""` sets the "Pre-Allocated Resources for iSCSI" to Disable. All reserved resources for iSCSI is released. The command is valid only when the "Offload Configuration" is in "Reserved Resources".

Interactive Mode

Issue the `list <view>` and `select <idx>` commands to select the target device. Then issue the `cfg Resource [key|key=value]` command to display and configure the resource parameter for the selected target.

Examples:

`cfg Resource` displays all resource reservation parameters for the selected adapter and their current settings.

`cfg Resource "CONFIGURABLE"` displays all configurable resource reservation parameters with their current settings and valid settings.

`cfg Resource "CONFIG"="FCFS"` sets the "Offload Configuration" to "First Come First Served".

`cfg Resource "iSCSI_CONN"="20"` sets the "iSCSI Connections" to 20. The command is valid only when the "Offload Configuration" is in "Reserved Resources".

`cfg Resource "TOE"="Disable"` sets the "Pre-Allocated Resources for TOE" to Disable. All reserved resources for TOE are released. The command is valid only when the "Offload Configuration" is in "Reserved Resources".

Configuring OoB Management for a Selected Target

CLI Mode

Issue the following command to display and configure the iSCSI parameter of the selected target:

```
QCSCLI -t <target type> -f <target format> -i <target ID> cfg
iSCSIMgmt [key|key=value]
```

Examples:

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "cfg OOBMgmt"` displays all OOB Management parameters and their current settings or information.

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "cfg OOBMgmt \\"Heartbeat Transmit Interval\\""` displays the Heartbeat Transmit Interval current settings and its valid input.

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "cfg OOBMgmt \\"Heartbeat Transmit Interval\\"=\"50\""` sets the Heartbeat Transmit Interval to 50 seconds.

Interactive Mode

Issue the `list <view>` and `select <idx>` commands to select the target device.

Issue the `cfg OOBMgmt [key|key=value]` command to display and configure the OOB management parameter for the selected target.

Examples:

`cfg OOBMgmt` displays all OOB management parameters of the selected adapter and their current settings.

`cfg OOBMgmt "Heartbeat Transmit Interval"` displays the Heartbeat Transmit Interval current settings and its valid input.

`cfg OOBMgmt "Heartbeat Transmit Interval"="50"` sets the Heartbeat Transmit Interval to 50 seconds

Configuring Systoe for the System

NOTE

Chimney has been deprecated by Microsoft in the OS.

CLI Mode

Issue the following command to display and configure the TCP offload engine (TOE) of the system Chimney:

```
QCSCLI -t System "cfg Systoe [value]"
```

Examples:

`QCSCLI -t system "cfg Systoe"` displays the current Chimney Offload State.

`QCSCLI -t system "cfg Systoe Enabled"` enables the current Chimney Offload State.

`QCSCLI -t system "cfg Systoe Disabled"` disables the current Chimney Offload State.

`QCSCLI -t system "cfg Systoe Automatic"` sets the current Chimney Offload State to automatic (Windows 7 kernel and later).

Interactive Mode

Issue the `list System` command; by default, the system is selected because there is only one system.

Issue the `cfg Systoe [value]` command to display and configure the TCP Offload of the system Chimney.

Examples:

`cfg Systoe` displays current Chimney Offload State.

`cfg Systoe Enabled` enables current Chimney Offload State.

`cfg Systoe Disabled` disables current Chimney Offload State.

`cfg Systoe Automatic` sets the current Chimney Offload State to automatic (Windows 7 kernel and later).

Getting Statistics for a Selected Target

CLI Mode

Issue the following command:

```
QCSCLI -t <target type> -f <target format> -i <target ID> "stats  
<options>"
```

Examples:

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "stats"` displays all statistics for the selected NDIS adapter.

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "stats general"` displays general statistics for the selected NDIS adapter.

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "stats IEEE"` displays IEEE statistics for the selected NDIS adapter.

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "stats QLASP"` displays QLASP statistics for the selected NDIS adapter.

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "stats all"` displays all the statistics information for the selected NDIS adapter.

`QCSCLI -t iSCSI -f mac -i 0010181a1b1e "stats login"` displays login statistics information for the selected iSCSI device.

`QCSCLI -t iSCSI -f mac -i 0010181a1b1e "stats instance"` displays instance statistics information for the selected iSCSI device.

`QCSCLI -t iSCSI -f mac -i 0010181a1b1e "stats custom"` displays custom statistics information for the selected iSCSI device.

`QCSCLI -t iSCSI -f mac -i 0010181a1b1e "stats all"` displays all the statistics information for the selected iSCSI device.

`QCSCLI -t iSCSI -f mac -i 0010181a1b1e "stats"` displays all the statistics information for the selected iSCSI device.

`QCSCLI -t TEAM -f name -i asdf "stats"` displays all the statistics information for the `asdf` team.

`QCSCLI -t VNIC -f name -i asdf "stats"` displays all the statistics information for the virtual adapter whose VLAN name is `asdf`.

Interactive Mode

The target type and target identifiers need not be specified. The `stats` command is based on the active target selection.

Examples:

```
stats displays all statistics for the selected NDIS adapter.  
stats general displays general statistics for the selected NDIS adapter.  
stats IEEE displays IEEE statistics for the selected NDIS adapter.  
stats QLASP displays QLASP statistics for the selected NDIS adapter.  
stats all displays all the statistics information for the selected NDIS adapter.  
stats login displays login statistics information for the selected iSCSI device.  
stats instance displays instance statistics information for the selected iSCSI device.  
stats custom displays custom statistics information for the selected iSCSI device.  
stats all displays all the statistics information for the selected iSCSI device.  
stats displays all the statistics information for the selected iSCSI device.  
stats displays all the statistics information for the selected team.  
stats displays all the statistics information for the selected virtual adapter.
```

Resetting Statistics for a Selected Target

CLI Mode

The `resetstats` command is not applicable in the CLI mode.

Interactive Mode

Issue the `resetstats` command to reset all statistical information for a selected or active NDIS, VNIC, or iSCSI team.

Running Diagnostic Tests for a Selected Target

CLI Mode

Issue the following command to run NIC diagnostics tests for the selected target. This command is available for NDIS and VBD adapters:

```
QCSCLI -t <target type> -f <target format> -i <target ID> diag {[-c  
REG ] [-c MII ] [-c EEP] [-c MEM] [-c CPU] [-c INT] [-c MACLB ] [-c  
PHYLB] [-c LED] | [-c ALL]} [-l <cnt> ] [ -v <LEDIntv> ]
```

Examples:

```
QCSCLI -t NDIS -f mac -i 0010181a1b1c "diag" displays all the  
diagnostics tests available for the currently selected target.
```

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "diag -c MII -c LED"`
runs the MII and LED test for the selected NDIS adapter.

`QCSCLI -t VBD -f mac -i 0010181a1b1d "diag -c all -l 5 -v 8"`
runs all the tests for 5 times with LED test interval 8 milliseconds for the selected VBD adapter.

Interactive Mode

Issue the `list <view>` and `select <idx>` commands to select the target device.

Issue the following command to run diagnostics tests for the selected target:

```
diag {[-c REG ] [-c MII ] [-c EEP] [-c MEM] [-c CPU] [-c INT] [-c  
MACLB ] [-c PHYLB] [-c LED] | [-c ALL]} [-l <cnt> ] [ -v <LEDIntv>  
]
```

Examples:

`diag` displays all the diagnostics tests available for the currently selected target.

`diag -c MII -c LED` runs the MII and LED test for the selected NDIS adapter.

`diag -c all -l 5 -v 8` runs all the tests for 5 times with LED test interval of 8 milliseconds for the selected VBD adapter.

Running a Cable Diagnostic Test for a Selected Target

CLI Mode

Issue the following command to run the cable diagnostics test for the selected target. This command is available for NDIS and VBD adapters:

```
QCSCLI -t <target type> -f <target format> -i <target ID> cablediag
```

Examples:

`QCSCLI -t NDIS -f mac -i 0010181a1b1c "cablediag"` runs the cable diagnostics test for the currently selected NDIS adapter.

`QCSCLI -t VBD -f mac -i 0010181a1b1d "cablediag"` runs the cable diagnostics test for the currently selected VBD adapter.

Interactive Mode

Issue the `list <view>` and `select <idx>` commands to select the target device. Then issue the `cablediag` command to run the cable diagnostics test for the selected target.

Example:

`cablediag` runs the cable diagnostics test for the currently selected NDIS adapter.

Running a Network Diagnostic Test for a Selected Target

CLI Mode

Issue the following command to run a cable diagnostics test for the selected target. This command is available for NDIS and virtual adapters:

```
QCSCLI -t <target type> -f <target format> -i <target ID>  
networkdiag [-p <IP address>]
```

Examples:

```
QCSCLI -t NDIS -f mac -i 0010181a1b1c "networkdiag -p  
192.168.1.5" runs the network test for the currently selected NDIS adapter.
```

```
QCSCLI -t VNIC -f mac -i 0010181a1b1c "networkdiag" runs the  
network test for the currently selected virtual adapter. Because no IP address is  
specified, QCS CLI uses the gateway address for the test.
```

Interactive Mode

Issue the `list <view>` and `select <idx>` commands to select the target device. Then issue `networkdiag [-p <IP address>]` to run cable diagnostics test for the selected target.

Examples:

```
networkdiag -p 192.168.1.5 runs the network test for the currently  
selected NDIS adapter.
```

```
networkdiag runs the network test for the currently selected virtual adapter.
```

Connecting to an FCoE Target

CLI Mode

N/A

Interactive Mode

Content unavailable at the time of publication.

Connecting to an iSCSI Target

CLI Mode

N/A

Interactive Mode

Issue the `list devcon`, `adddiscoveryportal`, `listdiscoveryportals`, and `list iscsitargets` commands. Then `select` an iSCSI target, `login` to the target, and `list iscsitargets` again to ensure that it is connected. Lastly, check the iSCSI target and session `info`.

Example:

list devcon

C	ID	DevType	Name
-	-	-	-
	R740-3-iDRAC-IP-40221	Host	R740-3-iDRAC-IP-40221
0	c8:00	Adapter	QLogic QL41262 25 Gigabit Ethernet
1	c8:00.00	Port	-Port0
2	000E1EF1FEA8	L2NIC	---QLogic QL41262 25 Gigabit Ethernet rev 02 (p1p1)
3	000E1EF1FEA8	iSCSI	---iSER over p1p1
.	.	.	.
29	F4E9D4E86E99	iSCSI	---QLogic 2x1GE+2x10GE QL41162HMRJ CNA (iSCSI) rev 02 (host16)
30	17:00.02	Port	-Port2

adddiscoveryportal -m F4E9D4E86E99 -i 100.69.4.26 -n 3260 port-0

listdiscoveryportals

C	MAC	Address	Port
-	-	-	-
0	F4E9D4E86E99	100.69.4.26	3260

list iscsitargets

C	Status	DevType	Name
-	-	-	-
0	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-506bac223-b7c0000002d58fa4-mell-cx3-2k16-uefi
1	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-e22f5bf23-94c40f33a0f5afc5-kik-r940xa

select 1

C	Status	DevType	Name
-	-	-	-
0	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-506bac223-b7c0000002d58fa4-mell-cx3-2k16-uefi
1	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-e22f5bf23-94c40f33a0f5afc5-kik-r940xa

login -m F4E9D4E86E99 -i 100.69.4.26 -n 3260 -if port-0

list iscsitargets

C	Status	DevType	Name
-	-	-	-
0	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-506bac223-b7c0000002d58fa4-mell-cx3-2k16-uefi
1	connected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-e22f5bf23-94c40f33a0f5afc5-kik-r940xa

```
info
iSCSI Target Information
-----
Target Type           Member of Discovery Group
Portal List           100.69.4.26
Target IQN Name
iqn.2001-05.com.equallogic:0-1cb196-e22f5bf23-94c40f33a0f5afc5-kik-r940xa
Persistency Flag      No
LUN List              LUN0 - EQLOGIC - 100E-00

iSCSI Session Information
-----
3
  Target
iqn.2001-05.com.equallogic:0-1cb196-e22f5bf23-94c40f33a0f5afc5-kik-r940xa
  Session State        Connected
  Target Portal        100.69.4.26
  Initiator Portal     100.69.41.213
  Session Type         Normal
  InitialR2T           False
  Immediate Data       True
  Max Outstanding R2T  1
  Max Unsolicited -
    First Data Burst Length 65536
  Data Sequence In Order True
  Data PDU In Order    True
  Error Recovery Level 120
  Max Connection Per Session 1

  Connection
    Local TCP Port      0
    Remote TCP Port     3260
    Authentication Type  None
    Header Digest       None
    Data Digest         None
```

Creating a Team Configuration

CLI Mode

Issue the following command to add a team using a configuration file:

```
QCSCLI -t TEAM "add <config file name>"
```

Examples:

```
QCSCLI -t team "add o:\temp\asdf1.bcg"
```

Because no host is specified, a host that has teaming is selected to add the team file. This command might fail if the team file is not for the selected host.

```
QCScli -r host -u user -p pass -t team "add
o:\temp\asdf1.bcg" specifies the host to which to add a team.
```

Interactive Mode

Issue the `add <config file name>` command to add a team using a configuration file.

Example:

```
add o:\temp\asdf1.bcg
```

Restoring a Team Configuration

CLI Mode

Issue the following command to restore teams using a configuration file:

```
QCSCLI -t TEAM "restore <config file name>"
```

Examples:

```
QCSCLI -t team "restore o:\temp\asdf.bcg"
```

Because no host is specified, a host that has teaming is selected to restore the team file. This command might fail if the team file is not for the selected host.

```
QCScli -r host -u user -p pass -t team "restore  
o:\temp\asdf1.bcg" specifies a host to restore a team.
```

Interactive Mode

Issue the `restore <config file name>` to restore teams using a configuration file.

Example:

```
restore o:\temp\asdf.bcg
```

Saving a Team Configuration

CLI Mode

Issue the following command to save the current teaming configuration to a file:

```
QCSCLI -t TEAM "save {-f MAC|BDF} <file_path_name>"
```

Examples:

```
QCSCLI -t team "save -f MAC o:\temp\asdf.bcg" saves the current  
teaming configuration to a file named asdf.bcg with adapters identified using  
the MAC address.
```

```
QCSCLI -t team "save -f BDF o:\temp\asdf.bcg" saves the current  
teaming configuration to a file named asdf.bcg with adapters identified using  
the BDF hexadecimal numbers.
```

```
QCScli -r host -u user -p pass -t team "save  
o:\temp\asdf1.bcg" specifies a host to save a team.
```

Interactive Mode

Issue the `save {-f MAC|BDF} <file_path_name>` to save the current teaming configuration to a file from within the `teamview` view listings.

Examples:

`save -f MAC o:\temp\asdf.bcg` saves the current teaming configuration to a file named `asdf.bcg` with adapters identified using the MAC address.

`save -f BDF o:\temp\asdf.bcg` saves the current teaming configuration to a file named `asdf.bcg` with adapters identified using the BDF hexadecimal number.

Removing One or More Teams

CLI Mode

Issue the following command to remove one or more teams from the system:

```
QCSCLI -t team -f name -i <team name> "remove"
```

Examples:

`QCSCLI -t team -f name -i asdf "remove"` removes a team named `asdf` from the system.

`QCSCLI -t team "remove -c all"` removes all teams from the system.

`QCScli -r host -u user -p pass -t team "remove -c all"` specifies a host from which to remove team.

Interactive Mode

Issue the `remove {-c all | <team_name>|}` command to remove one or more teams from the system.

Examples:

`remove asdf` removes a team named `asdf` from the system.

`remove -c all` removes all teams from the system.

`remove` removes the active and selected team from the system.

Displaying Unassigned Adapters

CLI Mode

Issue the `QCSCLI -t TEAM unassigned` command to display a list of adapters that are not assigned to a team.

Interactive Mode

Issue the `unassigned` command in the `teamview` to display a list of adapters that are not assigned to a team.

Switching Between Primary and Standby Adapters in a Team

CLI Mode

Issue the `QCSCLI -t TEAM "fallback"` command in a Smart Load Balancing™ and Auto-Fallback Disable (SLB-AFD) team to switch adapters from standby to primary.

Interactive Mode

Issue the `fallback` command in an SLB-AFD team to switch adapters from standby to primary.

Connecting an iSCSI Target

Interactive Mode

1. Launch QCS CLI from `/opt/QLogic_Corporation/QCS/`.

```
R740-3-iDRAC-IP-40221:/opt/QLogic_Corporation/QCS # ./QCScli
QLogic Control Suite Command Line Interface Utility for Linux v40.0.19.0
.....
Please add a host that you want to manage.
```

C	Status	Name	OS	Platform
-	-----	-----	-----	-----
0	connected	R740-3-iDRAC-IP-40221	Linux SuSE 15.0	64 bit (x64)

No entry for terminal type "xterm-256color";
using dumb terminal settings.

2. Get the iSCSI MAC address.

In this example, the address is F4E9D4E86E99.

```
R740-3-iDRAC-IP-40221\host\0>list devcon
C  ID          DevType Name
-  -
R740-3-iDRAC-IP-40221 Host      R740-3-iDRAC-IP-40221
0  c8:00        Adapter QLogic QL41262 25 Gigabit Ethernet
1  c8:00.00     Port    -Port0
2  000E1EF1FEA8 L2NIC    ---QLogic QL41262 25 Gigabit Ethernet rev 02 (plp1)
3  000E1EF1FEA8 iSCSI    ---iSER over plp1
4  c8:00.01     Port    -Port1
5  000E1EF1FEA9 L2NIC    ---QLogic QL41262 25 Gigabit Ethernet rev 02 (plp2)
6  000E1EF1FEA9 iSCSI    ---iSER over plp2
7  c9:00        Adapter QLogic QL41162 10 Gigabit Ethernet
8  c9:00.00     Port    -Port0
9  F4E9D4EFDD98 L2NIC    ---QLogic QL41162 10 Gigabit Ethernet rev 02 (p2p1)
```

4-QCS CLI Usage

Performing Management Tasks

```
10 F4E9D4EFDD98 iSCSI ---iSER over p2p1
11 c9:00.01 Port -Port1
12 F4E9D4EFDD99 L2NIC ---QLogic QL41162 10 Gigabit Ethernet rev 02 (p2p2)
13 F4E9D4EFDD99 iSCSI ---iSER over p2p2
14 25:00 Adapter FastLinQ QL45212-DE 25GbE Adapter
15 25:00.00 Port -Port0
16 000E1EC50772 L2NIC ---FastLinQ QL45212-DE 25GbE Adapter rev 10 (p3p1)
17 000E1EC50772 iSCSI ---iSER over p3p1
18 25:00.01 Port -Port1
19 000E1EC50773 L2NIC ---FastLinQ QL45212-DE 25GbE Adapter rev 10 (p3p2)
20 17:00 Adapter QLogic 2x1GE+2x10GE QL41162HMRJ CNA
21 17:00.00 Port -Port0
22 F4E9D4E86E94 L2NIC ---QLogic 2x1GE+2x10GE QL41162HMRJ CNA rev 02 (em1_1)
23 F4E9D4E86E94 iSCSI ---iSER over em1_1
24 F4E9D4E86E98 L2NIC ---QLogic 2x1GE+2x10GE QL41162HMRJ CNA rev 02 (em1_2)
25 F4E9D4E86E98 iSCSI ---iSER over em1_2
26 17:00.01 Port -Port1
27 F4E9D4E86E95 L2NIC ---QLogic 2x1GE+2x10GE QL41162HMRJ CNA rev 02 (em2_1)
28 F4E9D4E86E95 iSCSI ---iSER over em2_1
29 F4E9D4E86E99 iSCSI ---QLogic 2x1GE+2x10GE QL41162HMRJ CNA (iSCSI) rev 02 (host16)
30 17:00.02 Port -Port2
31 F4E9D4E86E96 L2NIC ---QLogic 2x1GE+2x10GE QL41162HMRJ CNA rev 02 (em3)
32 F4E9D4E86E96 iSCSI ---iSER over em3
33 17:00.03 Port -Port3
34 F4E9D4E86E97 L2NIC ---QLogic 2x1GE+2x10GE QL41162HMRJ CNA rev 02 (em4)
35 F4E9D4E86E97 iSCSI ---iSER over em4
```

3. Return to the host.

```
R740-3-iDRAC-IP-40221\devcon\0\adapter> >list hosts
```

4. Add the discovery portal using the iSCSI MAC address.

```
R740-3-iDRAC-IP-40221\host\0>adddiscoveryportal -m F4E9D4E86E99 -i 100.69.4.26 -n 3260 port-0
```

5. Check the iSCSI portal.

```
R740-3-iDRAC-IP-40221\host\0>listdiscoveryportals
```

C	MAC	Address	Port
-	-----	-----	-----
0	F4E9D4E86E99	100.69.4.26	3260

6. Check the iSCSI target.

```
R740-3-iDRAC-IP-40221\host\0>list iscsitargets
```

C	Status	DevType	Name
0	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-506bac223-b7c0000002d58fa4-mell-cx3-2k16-uefi
1	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-e22f5bf23-94c40f33a0f5afc5-kik-r940xa

7. Select the iSCSI target.

```
R740-3-iDRAC-IP-40221\iSCSITarget\0>select 1
```

C	Status	DevType	Name
0	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-506bac223-b7c0000002d58fa4-mell-cx3-2k16-uefi
1	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-e22f5bf23-94c40f33a0f5afc5-kik-r940xa

8. Login to the iSCSI target.

```
R740-3-iDRAC-IP-40221\iSCSITarget\0>login -m F4E9D4E86E99 -i 100.69.4.26 -n 3260 -if port-0
```

9. Check the iSCSI target.

```
R740-3-iDRAC-IP-40221\iSCSITarget\1>list iscsitargets
```

C	Status	DevType	Name
0	disconnected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-506bac223-b7c0000002d58fa4-mell-cx3-2k16-uefi
1	connected	iSCSI.T	iqn.2001-05.com.equallogic:0-1cb196-e22f5bf23-94c40f33a0f5afc5-kik-r940xa

10. Check the iSCSI target information.

```
R740-3-iDRAC-IP-40221\iSCSITarget\1>info
```

iSCSI Target Information

Target Type	Member of Discovery Group
Portal List	100.69.4.26
Target IQN Name	iqn.2001-05.com.equallogic:0-1cb196-e22f5bf23-94c40f33a0f5afc5-kik-r940xa
Persistence Flag	No
LUN List	LUN0 - <name> - 100E-00

iSCSI Session Information

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Target

iqn.2001-05.com.equallogic:0-1cb196-e22f5bf23-94c40f33a0f5afc5-kik-r940xa

Session State	Connected
Target Portal	100.69.4.26
Initiator Portal	100.69.41.213
Session Type	Normal
InitialR2T	False
Immediate Data	True
Max Outstanding R2T	1
Max Unsolicited -	
First Data Burst Length	65536
Data Sequence In Order	True
Data PDU In Order	True
Error Recovery Level	120
Max Connection Per Session	1

Connection

Local TCP Port	0
Remote TCP Port	3260
Authentication Type	None
Header Digest	None
Data Digest	None

Refreshing a Host

CLI Mode

Issue the `QCScli -t hosts "refresh"` command to get the latest host status and statistics.

Interactive Mode

Issue the `refresh` command under a host node to get the latest host status and statistics.

Performing Other Common Tasks

- `help` displays a list of available commands.
- `version` displays the version of QCS CLI.
- `log <file>` turns on the log and logs all the information to `<file>`.
- `log` turns off the log.
- `showsel` shows the currently selected target.

A Security Considerations

This appendix provides the following security information for QLogic Control Suite CLI:

- [Management Connectivity Model](#)
- [“Managing Security” on page 168](#)

Management Connectivity Model

[Figure A-1](#) shows how QLogic Control Suite CLI communicates in typical deployments.

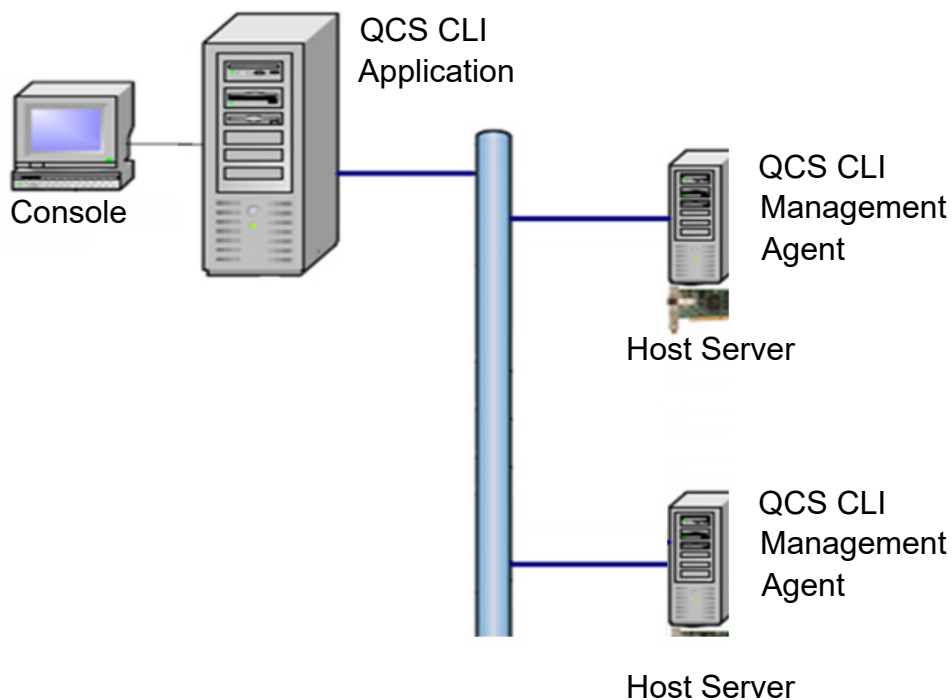


Figure A-1. QCS CLI Management Connectivity

Managing Security

QLogic Control Suite CLI security includes local connectivity, password and access control, Windows firewall configuration, and secure sockets layer (SSL) data encryption.

Connectivity Options

As shown in [Figure A-1 on page 167](#), you can connect to QLogic Control Suite CLI using:

- A localhost connection (on the Web application server)
- A remote connection (from another computer)

For information on how to connect to QCS CLI from the local host server or from a remote connection, see [“Adding a Host” on page 16](#).

Password Protection for Access Control

You can password-protect a host server to prevent unauthorized changes to that host's installed adapters. When you do this, authorization is required to make changes to the following on the host or any of its adapters:

- Adapter parameters
- Flash BIOS
- Adapter driver
- Port configuration
- Password (updates to the password itself)

To set up password protection on a host, you must have administrator or root privileges on that host.

Windows Firewall Configuration

When Windows Firewall is enabled with default settings, connections between QLogic Control Suite and remote hosts are blocked. To use Windows Firewall but allow connections between QCS CLI and remote hosts, add the following programs to the exceptions list in the firewall settings on both client and server machines:

- For x64 running native x64 CLI and agents:

```
C:\Program Files\QLogic Corporation\portmap\portmap.exe  
C:\Program Files\QLogic Corporation\QLRemote\qlnxremote.exe
```

- For x86 running native x86 CLI and agents:

```
C:\Program Files\QLogic Corporation\portmap\portmap.exe  
C:\Program Files\QLogic Corporation\QLRemote\qlnxremote.exe
```

Next, configure the Windows Firewall as described in the following procedure.

To configure the firewall on a host or client using Server Manager:

1. Do *one* of the following in Windows Server:
 - ☐ Click **Start**, click **Administrative Tools**, and then click **Windows Firewall with Advanced Security**.
 - ☐ Run the Server Manager application, click **Tools**, and then click **Windows Firewall with Advanced Security**.
2. Refer to the list of programs in [“Windows Firewall Configuration” on page 168](#).
For the first program in the list:
 - a. Right-click **Inbound Rules**, and then click **New Rule**.
 - b. In the New Inbound Rule wizard, select **Program**, and then click **Next**.
 - c. Click **Browse** to specify the path to the `.exe` file, and then click **Next**.
 - d. Select **Allow the connection**, and then click **Next**.
 - e. Select **Domain** and **Public**. Then click **Next**.
 - f. Enter a name and description for the rule, and then click **Finish**.
 - g. Server Manager displays the new rule under **Inbound Rules**.
3. Repeat [Step 2](#) for each program in the list, except in [Step 2a](#), click **Outbound Rules** and **New Rules** to open the New Outbound Rule wizard.
4. If any agent was previously running, restart it.

Figure A-2 shows how the default Windows firewall settings prevent QCS CLI from connecting to remote hosts, and Figure A-3 shows how modifying the settings lets QCS CLI connect to remote hosts.

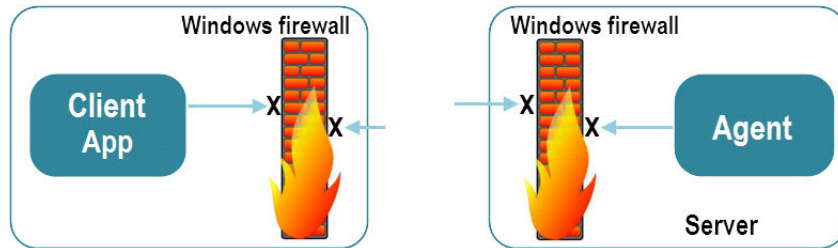


Figure A-2. Default Windows Firewall Settings: QCS CLI Blocked from Remote Hosts

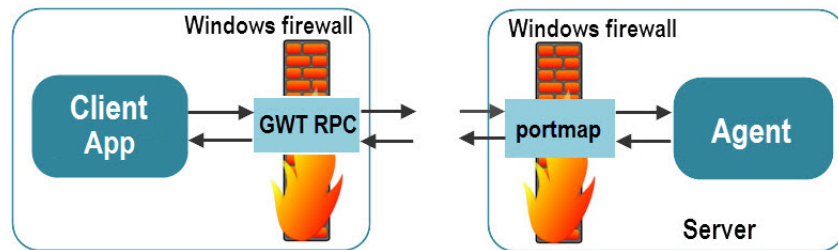


Figure A-3. Modified Firewall Settings: QCS CLI Able to Connect to Remote Hosts

Linux Firewall Configuration

If the Linux firewall (iptables¹ rules) is enabled with default settings, connections between QCS CLI and remote hosts are not allowed. To use Linux firewall but allow connections between QCS CLI and remote hosts, set up the rules for your Linux distribution as described in the following sections.

RHEL 7.x/8.x and Later; and SLES 12.x and Later

To configure the firewall:

1. Check the Fedora® FirewallD status (enabled or disabled):

```
# systemctl status firewalld
```

2. If FirewallD is enabled:

- a. Identify the current active zone and related interface:

```
# firewall-cmd --get-active-zones
public interfaces: em1
```

¹ `iptables` is the userspace command line program used to configure the Linux 2.4.x and later packet filtering ruleset.

- b. Set a permanent port range of 1–1024 for TCP and UDP protocols (to allow the qlnxremote management agent to communicate with the outside world):

```
# firewall-cmd --permanent --zone=public  
--add-port=1-1024/tcp  
# firewall-cmd --permanent --zone=public  
--add-port=1-1024/udp
```

- c. Allow QCS CLI servers to connect to client machines:

- Identify the current active zone and related interface (displayed by the command in [Step a](#) for the server machine).
- Set options permanently to allow port 8080 for HTTP and 8443 for HTTPS:

```
# firewall-cmd --permanent --zone=public --add-port=8080/tcp  
# firewall-cmd --permanent --zone=public --add-port=8080/udp  
# firewall-cmd --permanent --zone=public --add-port=8443/tcp  
# firewall-cmd --permanent --zone=public --add-port=8443/udp
```

NOTE

The preceding commands reflect the default port numbers of 8080 for HTTP and 8443 for HTTPS. If different port numbers were specified during QCS CLI installation, replace the port numbers in the preceding commands with the port numbers specified during installation.

3. To reload the firewall settings to make the current permanent settings the new runtime settings, issue the following command:

```
# firewall-cmd --reload
```

NOTE

To ensure the `rpcbind` (universal addresses to RPC program number mapper) service is running, issue the following commands:

```
# service rpcbind start  
# systemctl enable rpcbind.service
```

B Configuration File Examples

This appendix provides examples of the following configuration files used with QLogic Control Suite CLI to save and restore iSCSI, FCoE, Multi-Boot Agent (MBA) boot configuration, and SR-IOV and NPAR configuration:

- [Understanding Team Configuration Files](#)
- [“Multi-Boot Agent Configuration XML File” on page 175](#)
- [“iSCSI Boot Configuration XML File” on page 177](#)
- [“FCoE Boot Configuration XML File” on page 183](#)
- [“NIC Partition Configuration XML Files” on page 188](#)
- [“SR-IOV Configuration XML Files” on page 198](#)

Understanding Team Configuration Files

This section provides the following information about team configuration files:

- [Syntax Plain Text File](#)
- [Team Configuration File Requirements](#)
- [Team Configuration File Examples](#)

Syntax Plain Text File

The syntax plain text file is defined as follows, where the team parameters can be specified either using `NO_LIVELINK_PARAMS` or using `LIVELINK_PARAMS`.

```
TEAM_CFG
[TEAM_CFG]
...
```

Without LiveLink Parameters

Each `TEAM_CFG` with `NO_LIVELINK_PARAMS` is defined as follows:

```
name: team_name
[type: team_type]
pnic: MAC_address
[pnic: MAC_address]
```

```
...]  
[snic: MAC_address]  
[[ip: IP_address  
smask: subnetmask]  
|[vname: VLAN_name v  
id: VLAN_ID  
[ip: IP_address  
smask: subnetmask]]  
...]
```

With LiveLink Parameters

Each `TEAM_CFG` with `LIVELINK_PARAMS` is defined as follows:

```
name: livelinkteam_name  
[type: livelink_team_type]  
target_ip: ip1  
[target_ip: ip2  
target_ip: ip3  
target_ip: ip4]  
[retry: 3]  
[freq: 2000]  
[retry_freq: 2000]  
[livelink_vid: 1234]  
[pnic: MAC_address|PCIINFO  
livelink_ip: ll_ip  
livelink_ipv6: ll_ipv6  
pnic: MAC_address|PCIINFO  
livelink_ip: ll_ip  
pnic:...]  
[snic: MAC_address|PCIINFO  
livelink_ip: ll_ip]  
[[ip: IP_address  
smask: subnetmask]  
|[vname: VLAN_name  
vid: VLAN_ID  
[ip: IP_address  
smask: subnetmask]]  
...]
```

Team Configuration File Requirements

Requirements for the team configuration file include the following:

- A team configuration file must contain at least one team configuration. `QCSCLI.exe` recognizes `name` as the starting point of a team configuration section. All lines after `name` apply to the `team_name` until another `name` or end-of-file is encountered.
- Each team configuration must contain at least one physical network adapter. Otherwise the configuration of the team is ignored.
- If `type` is missing, the default is set to Smart Load Balancing (SLB). The four team types include: SLB = 0, GEC = 1, LACP = 2 and SLB-AFD = 4.

NOTE

In NPAR mode, the team configuration type GEC is not supported.

- If `ip` is set, `smask` must also be set or `ip` is ignored. If `ip` is not set, DHCP is used.
- The team IP can be set if no VLANs are configured. If any VLAN is configured, the team IP is ignored. Multiple VLAN configurations are allowed. Each VLAN configuration allows an optional static IP information. Each IP must be coupled with `smask`, otherwise it is ignored.
- There must at least one white space or tab between the tag (`name`, `pnice`, and so on) and the value pair (`team_name`, `MAC_address`, and so on) in each line.
- Keywords before semicolons are case sensitive.

Team Configuration File Examples

The following show examples of a team configuration XML file. For details about the command, see [“Save \(Team\)” on page 128](#).

Example Without LiveLink Parameters

The following shows an example of the `TeamConfig.txt` configuration file with `NO_LIVELINK_PARAMS`:

```
name: BCMTeam type: 1
pnice: 00101801794D
pnice: 00:0B.2 vname: VLAN2 vid: 2
vname: VLAN3 vid: 3
ip: 172.16.8.3
smask: 255.255.255.0
vname: VLAN4 vid: 4
```

```
ip: 172.16.8.4
smask: 255.255.255.0
vname: VLAN5 vid: 5
ip: 172.16.8.5
smask: 255.255.255.0
```

Example with LiveLink Parameters

The following shows an example of the `TeamConfig.txt` configuration file with `LIVELINK_PARAMS`:

```
name: LiveLinkTeam type: 0
target_ip: 172.16.8.66
target_ip: 172.16.8.77
target_ip: 172.16.8.88
target_ip: fc01::101
livelink_vid: 1234 pnic: 00101801794D
livelink_ip: 172.16.8.10
livelink_ipv6: fc01::218
pnic: 01:0D.0 livelink_ip: 172.16.8.11
snic: 02:03.00
livelink_ip: 172.16.8.20
vname: VLAN2
vid: 2 vname: VLAN3 vid: 3
ip: 172.16.8.3
smask: 255.255.255.0
vname: VLAN4 vid: 4
ip: 172.16.8.4
smask: 255.255.255.0
vname: VLAN5 vid: 5
ip: 172.16.8.5
smask: 255.255.255.0
```

Multi-Boot Agent Configuration XML File

The following shows an example of the Multi-Boot Agent (MBA) configuration XML file. For details about the command, see [“Boot Configuration” on page 33](#).

```
<?xml version="1.0" encoding="UTF-8"?>
<MBAConfiguration>
  <QCScli>
    <Version>30.0.44.0</Version>
```

```

</QCScli>
<OptionROM>Enabled</OptionROM>
<BootProtocol>None</BootProtocol>
<BootStrapType>Auto</BootStrapType>
<HideSetupPrompt>Disabled</HideSetupPrompt>
<SetupKeyStroke>Ctrl-S</SetupKeyStroke>
<BannerMessageTimeout>5</BannerMessageTimeout>
<LinkSpeed>AutoNeg</LinkSpeed>
<Pre-bootWakeonLAN>disable</Pre-bootWakeonLAN>
<VLANmode>disable</VLANmode>
<VLANID>1</VLANID>
<BootRetryCount>0</BootRetryCount>
</MBAConfiguration>

```

Table B-1 lists the range of values for the MBA configuration parameters used in the MBA configuration XML file.

Table B-1. MBA Configuration Parameters

Parameter	Value Range
OptionROM	Enabled Disabled
BootProtocol ^a	None PXE RPL iSCSI Boot ^b FCoE Boot (for adapters based on 57xxx controllers) None UNDI (PXE) iBFT ^c (for 3400/8400/41000/45000 Series Adapters)
BootStrapType	Auto BBS INT 18h INT 19h
HideSetupPrompt	Enabled Disabled
SetupKeyStroke	Ctrl-S Ctrl-B
BannerMessageTimeout	0-14
LinkSpeed ^d	AutoNeg 1 Gbps 10 Gbps 10M Half 100M Half 100M Full 1000M Half 1000M Full 2.5G 10G 20G 25G 40G 50G 100G SmartAN(TM) ^e
Pre-bootWakeonLAN	Enabled Disabled
VLANmode	Enabled Disabled
VLANID	1-4094

Table B-1. MBA Configuration Parameters (Continued)

Parameter	Value Range
BootRetryCount	0–7

^a For 3400/8400/41000/45000 Series Adapters, only the last saved `BootProtocol` setting is used.

If the `BootProtocol` is set to `iBFT`, set the iSCSI Boot Mode to Non-Offload and the iSCSI `BootProtocol` to `iSCSI` (see [Table B-3 on page 182](#)).

^b The iSCSI Boot protocol applies to boot or iBFT.

^c For software iSCSI boot, set the `BootProtocol` to `iBFT` and the iSCSI boot configuration `BootMode` to Non-Offload (see [Table B-3 on page 182](#)).

For hardware iSCSI boot, set the `BootProtocol` to `iSCSI Boot` and the iSCSI boot configuration `BootMode` to Offload (see [Table B-3 on page 182](#)).

^d For PXE boot, ensure that the `LinkSpeed` is the same value on both ports.

^e SmartAN (TM) is supported as a `LinkSpeed` value on some FastLinQ 45000 Series Adapters.

iSCSI Boot Configuration XML File

The following sections show examples of iSCSI boot configuration XML files. For details about the command, see [“Boot Configuration” on page 33](#).

iSCSI Example Configuration File for Adapters Based on 57xxx Controllers

```
<?xml version="1.0" encoding="UTF-8"?>
  <iSCSIBootConfiguration>
    <QCScli><Version>30.0.56.0</Version></QCScli>
    <iSCSIGeneral>
      <TCPIPviaDHCP>Disabled</TCPIPviaDHCP>
      <iSCSIIPviaDHCP>Disabled</iSCSIIPviaDHCP>
      <CHAPAuthentication>Disabled</CHAPAuthentication>
      <BoottoiSCSITarget>Enabled</BoottoiSCSITarget>
      <DHCPVendorID>QLGCISAN</DHCPVendorID>
      <LinkUpDelayTime>0</LinkUpDelayTime>
      <UseTCPTimestamp>Disabled</UseTCPTimestamp>
      <TargetasFirstHDD>Disabled</TargetasFirstHDD>
      <LUNBusyRetryCount>0</LUNBusyRetryCount>
      <IPVersion>IPv4</IPVersion>
```

```
<SubnetMask>0.0.0.0</SubnetMask>
<WindowsHBABootMode>Disabled</WindowsHBABootMode>
</iSCSIGeneral>
<IscsiInitiator>
  <IPAddress>192.168.100.24</IPAddress>
  <SubnetMask>255.255.255.0</SubnetMask>
  <DefaultGateWay>192.168.100.1</DefaultGateWay>
  <PrimaryDNS>192.168.100.2</PrimaryDNS>
  <SecondaryDNS>192.168.100.3</SecondaryDNS>
  <iSCSIName>iqn</iSCSIName>
  <CHAPID>user</CHAPID>
  <CHAPSecret>password</CHAPSecret>
</IscsiInitiator>
<IscsiTarget>
  <Connect>Enabled</Connect>
  <IPAddress>192.168.100.10</IPAddress>
  <TCPPort>3260</TCPPort>
  <BootLun>1</BootLun>
  <iSCSIName>targetiqn</iSCSIName>
  <CHAPID>admin</CHAPID>
  <CHAPSecret>password1</CHAPSecret>
</IscsiTarget>
<IscsiSecTarget>
  <Connect>Disabled</Connect>
  <IPAddress></IPAddress>
  <TCPPort>0</TCPPort>
  <BootLun>0</BootLun>
  <iSCSIName></iSCSIName>
  <CHAPID></CHAPID>
  <CHAPSecret></CHAPSecret>
</IscsiSecTarget>
<IscsiMPIO>
  <EnableMPIO>Enabled</EnableMPIO>
  <SecondaryDevice ></SecondaryDevice>
  <UseIndependentTargetPortal>Disabled</UseIndependentTargetPortal>
  <UseIndependentTargetName>Disabled</UseIndependentTargetName>
</IscsiMPIO>
</iSCSIBootConfiguration>
```

Table B-2 lists the range of values for the iSCSI boot configuration parameters for adapters based on 57xxx controllers used in the iSCSI boot configuration XML file.

Table B-2. iSCSI Boot Configuration Parameters for Adapters Based on 57xxx Controllers

Parameter	Value Range or Description
iSCSI General Parameters	
TCPIPviaDHCP	Enabled Disabled
iSCSIviaDHCP	Enabled Disabled
CHAPAuthentication	Enabled Disabled
BoottoiSCSITarget	Enabled Disabled One Time Disabled
DHCPVendorID	Vendor ID string
LinkUpDelayTime	0-255
UseTCPTimestamp	Enabled Disabled
TargetasFirstHDD	Enabled Disabled
LUNBusyRetryCount	0-60
IPVersion	IPv4 IPv6
WindowsHBABootMode	Enabled Disabled
iSCSI Initiator Parameters	
IPAddress	IPv4 or IPv6 format IP address
SubnetMask	IPv4 or IPv6 format IP address
DefaultGateWay	IPv4 or IPv6 format IP address
PrimaryDNS	IPv4 or IPv6 format IP address
SecondaryDNS	IPv4 or IPv6 format IP address
iSCSIName	Initiator iSCSI qualified name (IQN)
CHAPID	CHAP ID
CHAPSecret	CHAP password

Table B-2. iSCSI Boot Configuration Parameters for Adapters Based on 57xxx Controllers (Continued)

Parameter	Value Range or Description
iSCSI Target and iSCSI Second Target Parameters	
Connect	Enabled Disabled
IPAddress	IPv4 or IPv6 format IP address
TCPPort	TCP port, default value is 3260
BootLun	0–65535
iSCSIName	Target IQN name
CHAPID	CHAP ID
CHAPSecret	CHAP password
EnableMPIO	Enabled Disabled
iSCSI MPIO Parameters	
SecondaryDevice	Secondary device name
UseIndependentTargetPort al	Enabled Disabled
UseIndependentTargetName	Enabled Disabled

iSCSI Example Configuration File for 8400/41000/45000 Series Adapters

```
<?xml version="1.0" encoding="UTF-8"?>
  <iSCSIBootConfiguration>
    <QCScli><Version>30.0.56.0</Version></QCScli>
    <iSCSIGeneral>
      <BootProtocol>iSCSI</BootProtocol>
      <BootMode>Non-Offload</BootMode>
      <TCPIPviaDHCP>Disabled</TCPIPviaDHCP>
      <iSCSIPviaDHCP>Disabled</iSCSIPviaDHCP>
      <CHAPAuthentication>Disabled</CHAPAuthentication>
      <DHCPVendorID>QLGC ISAN</DHCPVendorID>
      <DHCPRequestTimeout>60</DHCPRequestTimeout>
      <TargetLoginTimeout>60</TargetLoginTimeout>
    </iSCSIGeneral>
  </iSCSIBootConfiguration>
```

```
<IPVersion>IPv4</IPVersion>
<IPv4Fallback>Disabled</IPv4Fallback>
<VLANID>0</VLANID>
<AddressRedirect>Disabled</AddressRedirect>
</iSCSIGeneral>
<IscsiInitiator>
  <IPAddress>192.168.100.24</IPAddress>
  <SubnetMask>255.255.255.0</SubnetMask>
  <DefaultGateWay>192.168.100.1</DefaultGateWay>
  <PrimaryDNS>192.168.100.2</PrimaryDNS>
  <SecondaryDNS>192.168.100.3</SecondaryDNS>
  <iSCSIName>iqn</iSCSIName>
  <CHAPID>user</CHAPID>
  <CHAPSecret>password</CHAPSecret>
</IscsiInitiator>
<IscsiTarget>
  <Connect>Enabled</Connect>
  <IPAddress>192.168.100.10</IPAddress>
  <TCPPort>3260</TCPPort>
  <BootLun>1</BootLun>
  <iSCSIName>targetiqn</iSCSIName>
  <CHAPID>admin</CHAPID>
  <CHAPSecret>password1</CHAPSecret>
</IscsiTarget>
<IscsiSecTarget>
  <Connect>Disabled</Connect>
  <IPAddress></IPAddress>
  <TCPPort>0</TCPPort>
  <BootLun>0</BootLun>
  <iSCSIName></iSCSIName>
  <CHAPID></CHAPID>
  <CHAPSecret></CHAPSecret>
</IscsiSecTarget>
</iSCSIBootConfiguration>
```

Table B-3 lists the range of values for the iSCSI boot configuration parameters for 8400/41000/45000 Series Adapters used in the iSCSI boot configuration XML file.

Table B-3. iSCSI Boot Configuration Parameters for 8400/41000/45000 Series Adapters

Parameter	Value Range or Description
iSCSI General Parameters	
BootProtocol ^a	None iSCSI
BootMode	Offload Non-Offload
TCPIPviaDHCP	Enabled Disabled
iSCSIviaDHCP	Enabled Disabled
CHAPAuthentication	Enabled Disabled
DHCPVendorID	Vendor ID string
DHCPRequestTimeout	0-65535
TargetLoginTimeout	0-65535
IPVersion	IPv4 IPv6
IPv4Fallback	Enabled Disabled
VLANID	0-4094
Address Redirect	Enabled Disabled
iSCSI Initiator Parameters	
IPAddress	IPv4 or IPv6 format IP address
SubnetMask	IPv4 or IPv6 format IP address
DefaultGateWay	IPv4 or IPv6 format IP address
PrimaryDNS	IPv4 or IPv6 format IP address
SecondaryDNS	IPv4 or IPv6 format IP address
iSCSIName	Initiator iSCSI qualified name (IQN)
CHAPID	CHAP ID
CHAPSecret	CHAP password

Table B-3. iSCSI Boot Configuration Parameters for 8400/41000/45000 Series Adapters (Continued)

Parameter	Value Range or Description
iSCSI Target and iSCSI Second Target Parameters	
Connect	Enabled Disabled
IPAddress	IPv4 or IPv6 format IP address
TCPPort	TCP port; default value is 3260
BootLun	0–65535
iSCSIName	Target IQN name
CHAPID	CHAP ID
CHAPSecret	CHAP password

^a For 8400/41000/45000 Series Adapters, only the last saved `BootProtocol` setting is used. If the `BootProtocol` is set to `iSCSI` and the `BootMode` is set to `Non-Offload`, set the `MBA BootProtocol` to `iBFT`. (see [Table B-1 on page 176](#)).

FCoE Boot Configuration XML File

The following sections show examples of FCoE boot configuration XML files. For details about the command, see [“Boot Configuration” on page 33](#).

- [FCoE Example Configuration File for 8400 Series Adapters and Adapters Based on 57xxx Controllers](#)
- [FCoE Example Configuration File for 41000/45000 Series Adapters](#)

FCoE Example Configuration File for 8400 Series Adapters and Adapters Based on 57xxx Controllers

```
<?xml version="1.0" encoding="UTF-8"?>
  <FCoEBootConfiguration>
    <QCScli><Version>30.0.56.0</Version></QCScli>
    <FCoEGeneral>
      <BoottoFCoETarget>Enabled</BoottoFCoETarget>
      <TargetasFirstHDD>Disabled</TargetasFirstHDD>
      <LinkUpDelayTime>0</LinkUpDelayTime>
      <LUNBusyRetryCount>0</LUNBusyRetryCount>
      <FabricDiscoveryTimeout>4</FabricDiscoveryTimeout>
```

```
<FCoEHBABootMode>Enabled</FCoEHBABootMode>
</FCoEGeneral>
<FCoETarget>
  <Target>
    <Index>0</Index>
    <Connect>Enabled</Connect>
    <PortWWN>207000c0001e47fb</PortWWN>
    <BootLun>1</BootLun>
  </Target>
  <Target>
    <Index>1</Index>
    <Connect>Disabled</Connect>
    <PortWWN>0000000000000000</PortWWN>
    <BootLun>0</BootLun>
  </Target>
  <Target>
    <Index>2</Index>
    <Connect>Disabled</Connect>
    <PortWWN>0000000000000000</PortWWN>
    <BootLun>0</BootLun>
  </Target>
  <Target>
    <Index>3</Index>
    <Connect>Disabled</Connect>
    <PortWWN>0000000000000000</PortWWN>
    <BootLun>0</BootLun>
  </Target>
  <Target>
    <Index>4</Index>
    <Connect>Disabled</Connect>
    <PortWWN>0000000000000000</PortWWN>
    <BootLun>0</BootLun>
  </Target>
  <Target>
    <Index>5</Index>
    <Connect>Disabled</Connect>
    <PortWWN>0000000000000000</PortWWN>
    <BootLun>0</BootLun>
  </Target>
</FCoETarget>
```

```

    <Target>
      <Index>6</Index>
      <Connect>Disabled</Connect>
      <PortWWN>0000000000000000</PortWWN>
      <BootLun>0</BootLun>
    </Target>
    <Target>
      <Index>7</Index>
      <Connect>Disabled</Connect>
      <PortWWN>0000000000000000</PortWWN>
      <BootLun>0</BootLun>
    </Target>
  </FCoETarget>
</FCoEBootConfiguration>

```

Table B-4 lists the range of values for the FCoE boot configuration parameters for 8400 Series Adapters and adapters based on 57xxx controllers used in the FCoE boot configuration XML file.

Table B-4. FCoE Boot Configuration Parameters for 8400 Series Adapters and Adapters Based on 57xxx Controllers

Parameter	Value Range or Description
FCoE General Parameters	
BoottoFCoETarget	Enabled Disabled One Time Disabled
TargetasFirstHDD	Enabled Disabled
LinkUpDelayTime	0-255
LUNBusyRetryCount	0-60
FabricDiscoveryTimeout	0-8
FCoEHBABootMode	Enabled ^a

Table B-4. FCoE Boot Configuration Parameters for 8400 Series Adapters and Adapters Based on 57xxx Controllers (Continued)

Parameter	Value Range or Description
FCoE Target (0–7) Parameters	
Connect	Enabled Disabled
PortWWN	64-bit port WWN address
BootLun	0–65535

^a You cannot disable the `FCoEHBABootMode` parameter; it is always enabled for an FCoE remote boot.

FCoE Example Configuration File for 41000/45000 Series Adapters

```
<?xml version="1.0" encoding="UTF-8"?>
  <FCoEBootConfiguration>
    <QCScli><Version>30.0.56.0</Version></QCScli>
    <FCoEGeneral>
      <BootProtocol>FCoE</BootProtocol>
      <FIPVLAN>0</FIPVLAN>
      <FabricLoginRetryCount>5</FabricLoginRetryCount>
      <TargetLoginRetryCount>5</TargetLoginRetryCount>
    </FCoEGeneral>
    <FCoETarget>
      <Target>
        <Index>0</Index>
        <Connect>Enabled</Connect>
        <PortWWN>207000c0001e47fb</PortWWN>
        <BootLun>1</BootLun>
      </Target>
      <Target>
        <Index>1</Index>
        <Connect>Disabled</Connect>
        <PortWWN>0000000000000000</PortWWN>
        <BootLun>0</BootLun>
      </Target>
      <Target>
        <Index>2</Index>
```

```
        <Connect>Disabled</Connect>
        <PortWWN>0000000000000000</PortWWN>
        <BootLun>0</BootLun>
    </Target>
    <Target>
        <Index>3</Index>
        <Connect>Disabled</Connect>
        <PortWWN>0000000000000000</PortWWN>
        <BootLun>0</BootLun>
    </Target>
    <Target>
        <Index>4</Index>
        <Connect>Disabled</Connect>
        <PortWWN>0000000000000000</PortWWN>
        <BootLun>0</BootLun>
    </Target>
    <Target>
        <Index>5</Index>
        <Connect>Disabled</Connect>
        <PortWWN>0000000000000000</PortWWN>
        <BootLun>0</BootLun>
    </Target>
    <Target>
        <Index>6</Index>
        <Connect>Disabled</Connect>
        <PortWWN>0000000000000000</PortWWN>
        <BootLun>0</BootLun>
    </Target>
    <Target>
        <Index>7</Index>
        <Connect>Disabled</Connect>
        <PortWWN>0000000000000000</PortWWN>
        <BootLun>0</BootLun>
    </Target>
</FCoETarget>
</FCoEBootConfiguration>
```

Table B-5 lists the range of values for the FCoE boot configuration parameters for 41000/45000 Series Adapters used in the FCoE boot configuration XML file.

Table B-5. FCoE Boot Configuration Parameters for 41000/45000 Series Adapters

Parameter	Value Range or Description
FCoE General Parameters	
BootProtocol ^a	None FCoE
FIPVLAN	0-4094
FabricLoginRetryCount	0-60
TargetLoginRetryCount	0-60
FCoE Target (0-7) Parameters	
Connect	Enabled Disabled
PortWWN	64-bit port WWN address
BootLun	0-65535

^a For 41000/45000 Series Adapters, only the last saved `BootProtocol` setting is used.

NIC Partition Configuration XML Files

This section provides examples of:

- [NIC Partition Configuration \(Windows and Linux\)](#)
- [RoCE and iWARP Configuration \(Windows and Linux\)](#)
- [NPAR, RoCE, and iWARP Configuration Parameters](#)

NOTE

To view and configure NPAR for the 57xxx family of adapters, ensure that the virtual bus device (VBD) instances (in the Windows Device Manager under **System devices**) corresponding to PCI functions 0 and 1 are enabled.

NIC Partition Configuration (Windows and Linux)

The following show examples of NIC partition (NPAR) configuration XML files for Windows and Linux. For details about the command, see [“Configure Multifunction” on page 62](#).

Example 1: Configuring NIC Partitioning (3400 and 8400 Series Adapters Only)

```
<?xml version="1.0" encoding="UTF-8"?>
<MultiFunctionConfiguration>
  <Version>7</Version>
  <MultiFunctionMode>Multi-Function</MultiFunctionMode>
  <MF_PortConfig>
    <Port>0</Port>
    <FlowControl>Rx & Tx Enabled</FlowControl>
    <FunctionConfig>
      <Function>0</Function>
      <EthernetNdis>Enable</EthernetNdis>
      <iSCSI>Disable</iSCSI>
      <FCoE>Disable</FCoE>
      <RelativeBandwidth>50</RelativeBandwidth>
      <MaxBandwidth>50</MaxBandwidth>
    </FunctionConfig>
    <FunctionConfig>
      <Function>2</Function>
      <EthernetNdis>Enable</EthernetNdis>
      <iSCSI>Disable</iSCSI>
      <FCoE>Disable</FCoE>
      <RelativeBandwidth>25</RelativeBandwidth>
      <MaxBandwidth>100</MaxBandwidth>
    </FunctionConfig>
    <FunctionConfig>
      <Function>4</Function>
      <EthernetNdis>Enable</EthernetNdis>
      <iSCSI>Disable</iSCSI>
      <FCoE>Disable</FCoE>
      <RelativeBandwidth>25</RelativeBandwidth>
      <MaxBandwidth>100</MaxBandwidth>
    </FunctionConfig>
    .
    .
    .
  </MF_PortConfig>
</MultiFunctionConfiguration>
```

```
<FunctionConfig>
  <Function>6</Function>
  <EthernetNdis>Enable</EthernetNdis>
  <iSCSI>Disable</iSCSI>
  <FCoE>Disable</FCoE>
  <RelativeBandwidth>0</RelativeBandwidth>
  <MaxBandwidth>100</MaxBandwidth>
</FunctionConfig>
</MF_PortConfig>
<MF_PortConfig>
  <Port>1</Port>
  <FlowControl>Rx & Tx Enabled</FlowControl>
  <FunctionConfig>
    <Function>1</Function>
    <EthernetNdis>Enable</EthernetNdis>
    <iSCSI>Disable</iSCSI>
    <FCoE>Disable</FCoE>
    <RelativeBandwidth>0</RelativeBandwidth>
    <MaxBandwidth>100</MaxBandwidth>
  </FunctionConfig>
  <FunctionConfig>
    <Function>3</Function>
    <EthernetNdis>Enable</EthernetNdis>
    <iSCSI>Disable</iSCSI>
    <FCoE>Disable</FCoE>
    <RelativeBandwidth>0</RelativeBandwidth>
    <MaxBandwidth>100</MaxBandwidth>
  </FunctionConfig>
  <FunctionConfig>
    <Function>5</Function>
    <EthernetNdis>Enable</EthernetNdis>
    <iSCSI>Disable</iSCSI>
    <FCoE>Disable</FCoE>
    <RelativeBandwidth>0</RelativeBandwidth>
    <MaxBandwidth>100</MaxBandwidth>
  </FunctionConfig>
  <FunctionConfig>
    <Function>7</Function>
    <EthernetNdis>Enable</EthernetNdis>
```

```
        <iSCSI>Disable</iSCSI>
        <FCoE>Disable</FCoE>
        <RelativeBandwidth>0</RelativeBandwidth>
        <MaxBandwidth>100</MaxBandwidth>
    </FunctionConfig>
</MF_PortConfig>
</MultiFunctionConfiguration>
```

Example 2: Configuring NIC Partitioning Mode (41000 and 45000 Series Adapters Only)

```
<?xml version="1.0" encoding="UTF-8"?>
<MultiFunctionConfiguration>
    <Version>8</Version>
    <MultiFunctionMode>Single-Function</MultiFunctionMode>
    <DefaultMode_PortConfig>
        <Port>0</Port>
        <EthernetNdis>Enable</EthernetNdis>
        <iSCSI>Disable</iSCSI>
        <FCoE>Enable</FCoE>
        <RoCE>Enable</RoCE>
        <iWARP>Enable</iWARP>
    </DefaultMode_PortConfig>
    <DefaultMode_PortConfig>
        <Port>1</Port>
        <EthernetNdis>Enable</EthernetNdis>
        <iSCSI>Disable</iSCSI>
        <FCoE>Disable</FCoE>
        <RoCE>Enable</RoCE>
        <iWARP>Enable</iWARP>
    </DefaultMode_PortConfig>
    <MF_PortConfig>
        <Port>0</Port>
        <FlowControl>Auto</FlowControl>
        <FunctionConfig>
            <Function>0</Function>
            <EthernetNdis>Enable</EthernetNdis>
            <RoCE>Enable</RoCE>
            <iWARP>Enable</iWARP>
            <RelativeBandwidth>0</RelativeBandwidth>
            <MaxBandwidth>100</MaxBandwidth>
```

```
</FunctionConfig>
<FunctionConfig>
  <Function>2</Function>
  <EthernetNdis>Disable</EthernetNdis>
  <RoCE>Disable</RoCE>
  <iWARP>Disable</iWARP>
  <FCoE>Enable</FCoE>
  <RelativeBandwidth>0</RelativeBandwidth>
  <MaxBandwidth>100</MaxBandwidth>
</FunctionConfig>
<FunctionConfig>
  <Function>4</Function>
  <EthernetNdis>Enable</EthernetNdis>
  <RoCE>Disable</RoCE>
  <iWARP>Disable</iWARP>
  <iSCSI>Disable</iSCSI>
  <RelativeBandwidth>0</RelativeBandwidth>
  <MaxBandwidth>100</MaxBandwidth>
</FunctionConfig>
.
.
.
<FunctionConfig>
  <Function>14</Function>
  <EthernetNdis>Enable</EthernetNdis>
  <RoCE>Disable</RoCE>
  <iWARP>Disable</iWARP>
  <RelativeBandwidth>0</RelativeBandwidth>
  <MaxBandwidth>100</MaxBandwidth>
</FunctionConfig>
</MF_PortConfig>
<MF_PortConfig>
  <Port>1</Port>
  <FlowControl>Auto</FlowControl>
  <FunctionConfig>
    <Function>1</Function>
    <EthernetNdis>Enable</EthernetNdis>
    <RoCE>Enable</RoCE>
    <iWARP>Enable</iWARP>
```

```
        <RelativeBandwidth>0</RelativeBandwidth>
        <MaxBandwidth>100</MaxBandwidth>
    </FunctionConfig>
    <FunctionConfig>
        <Function>3</Function>
        <EthernetNdis>Disable</EthernetNdis>
        <RoCE>Disable</RoCE>
        <iWARP>Disable</iWARP>
        <FCoE>Enable</FCoE>
        <RelativeBandwidth>0</RelativeBandwidth>
        <MaxBandwidth>100</MaxBandwidth>
    </FunctionConfig>
    <FunctionConfig>
        <Function>5</Function>
        <EthernetNdis>Enable</EthernetNdis>
        <RoCE>Disable</RoCE>
        <iWARP>Disable</iWARP>
        <iSCSI>Disable</iSCSI>
        <RelativeBandwidth>0</RelativeBandwidth>
        <MaxBandwidth>100</MaxBandwidth>
    </FunctionConfig>
    .
    .
    .
    <FunctionConfig>
        <Function>15</Function>
        <EthernetNdis>Enable</EthernetNdis>
        <RoCE>Disable</RoCE>
        <iWARP>Disable</iWARP>
        <RelativeBandwidth>0</RelativeBandwidth>
        <MaxBandwidth>100</MaxBandwidth>
    </FunctionConfig>
</MF_PortConfig>
</MultiFunctionConfiguration>
```

Example 3: Disabling NIC Partitioning Mode

```
<?xml version="1.0" encoding="UTF-8" ?>
<MultiFunctionConfiguration>
    <Version>2</Version>
```

```
        <MultiFunctionMode>SingleFunction</MultiFunctionMode>
    </MultiFunctionConfiguration>
```

RoCE and iWARP Configuration (Windows and Linux)

```
<?xml version="1.0" encoding="UTF-8"?>
  <MultiFunctionConfiguration>
    <Version>8</Version>
    <MultiFunctionMode>Multi-Function</MultiFunctionMode>
    <EPMode>Disabled</EPMode>
    <PortConfig><Port>0</Port>
    <FlowControl>Auto</FlowControl>
    <FunctionConfig>
      <Function>0</Function>
      <EthernetNdis>Enable</EthernetNdis>
      <RoCE>Disable</RoCE>
      <RelativeBandwidth>0</RelativeBandwidth>
      <MaxBandwidth>100</MaxBandwidth>
    </FunctionConfig>
    <FunctionConfig>
      <Function>2</Function>
      <EthernetNdis>Enable</EthernetNdis>
      <RelativeBandwidth>0</RelativeBandwidth>
      <MaxBandwidth>100</MaxBandwidth>
    </FunctionConfig>
    <FunctionConfig>
      <Function>4</Function>
      <EthernetNdis>Enable</EthernetNdis>
      <RelativeBandwidth>0</RelativeBandwidth>
      <MaxBandwidth>100</MaxBandwidth>
    </FunctionConfig>
    .
    .
    .
  </PortConfig>
  <PortConfig><Port>1</Port>
  <FlowControl>Auto</FlowControl>
  <FunctionConfig>
    <Function>1</Function>
    <EthernetNdis>Enable</EthernetNdis>
    <RoCE>Disable</RoCE>
```

```
<RelativeBandwidth>0</RelativeBandwidth>
<MaxBandwidth>100</MaxBandwidth>
</FunctionConfig>
<FunctionConfig>
<Function>3</Function>
  <EthernetNdis>Enable</EthernetNdis>
  <RelativeBandwidth>0</RelativeBandwidth>
  <MaxBandwidth>100</MaxBandwidth>
</FunctionConfig>
.
.
.
<FunctionConfig>
<Function>15</Function>
  <EthernetNdis>Disable</EthernetNdis>
  <RelativeBandwidth>0</RelativeBandwidth>
  <MaxBandwidth>100</MaxBandwidth>
</FunctionConfig>
</PortConfig>
</MultiFunctionConfiguration>
```

NPAR, RoCE, and iWARP Configuration Parameters

[Table B-6](#) lists the range of values for the NPAR configuration parameters used in the NIC Partition configuration XML file. Not all parameters are available for all adapter models; refer to the table footnotes for details.

CAUTION

Marvell recommends that you create the XML file by issuing the `cfg multi-function -s "filename"` command. Creating the XML file on your own may result in improper formatting and result in errors.

Table B-6. NPAR Configuration Parameters

Parameter	Value Range
MultiFunctionMode	Multi-Function SingleFunction UFP
EP Mode ^a	Enabled Disabled

Table B-6. NPAR Configuration Parameters (Continued)

Parameter	Value Range
FlowControl	Auto Rx & Tx Enabled Tx Enabled Rx Enabled Disabled
EthernetNdis ^b	Enable Disable
RoCE ^c	Enable Disable
iWARP ^d	Enable Disable
iSCSI ^e	Enable Disable
FCoE ^f	Enable Disable
RelativeBandwidth ^g	0-100
MaxBandwidth	1-100
DefaultMode_PortConfig ^h	N/A
MF_PortConfig ⁱ	N/A
Version ^j	Read-only NPAR data version; do not modify this value.

Table B-6. NPAR Configuration Parameters (Continued)

Parameter	Value Range
Port ^j	Read-only port number; do not modify this value.
Function ^j	Read-only physical function (PF) number; do not modify this value.

^a Extended Partition mode (EP mode) is available only on the 41000 and 45000 Series Adapters.

^b On 57xxx (344x/744x/844x/34xxx) Adapters, the EthernetNdis parameter applies to Windows only. On Linux and VMware, this value is always enabled (that is, the system ignores this setting) even if iSCSI and FCoE offload are also enabled. On 41000 and 45000 Series Adapters, the first partition of each port always has Ethernet enabled (on all OSs), while any of the remaining partitions can be disabled (that is, hidden by not enabling any protocol: Ethernet/RDMA, iSCSI, or FCoE).

^c RoCE (v1/v2) is available on the RDMA-capable 41000 and 45000 Series Adapters and only on Ethernet-enabled partitions.

^d iWARP is available on the RDMA-capable 41000 and 45000 Series Adapters and only on Ethernet-enabled partitions.

^e On 57xxx (744x/844x) Adapters, the iSCSI parameter can be on any partition. However, if you are doing an iSCSI-offloaded remote boot, this parameter must be enabled on the first partition of the booting port. On 41000 and 45000 Series Adapters, set this parameter only on the third partition.

^f On 57xxx (844x) Adapters, the FCoE parameter can be on any partition. However, if you are doing FCoE-offloaded remote boot, this parameter must be enabled on the first partition of the booting port. On 41000 and 45000 Series Adapters, set this parameter only on the second partition.

^g RelativeBandwidth must be less than the MaxBandwidth value and collectively must be less than 100 for all NPAR functions on the same port.

^h DefaultMode_PortConfig is available only on 41000 and 45000 Series Adapters. This parameter represents the single function mode port configuration. All nodes under this parent node represent the default mode single function configuration.

ⁱ MF_PortConfig represents the NPAR mode port configuration. All nodes under this parent node represent the NPAR configuration.

^j When you create the XML file by issuing the `cfg multi-function -s "filename"` command, the generated XML file creates this node per the adapter configuration and in the proper format. Do not modify the Version, Port, or Function values.

SR-IOV Configuration XML Files

The following sections show examples of SR-IOV configuration XML files in single function mode (for models BCM57xxx, QLE344x, QLE844x, 41000 Series, and 45000 Series) and in NPAR mode. For details about the command, see [“Configure SR-IOV” on page 74](#).

Single Function Mode

```
<?xml version="1.0" encoding="UTF-8"?>
  <SriovConfigurations>
    <Mode>SF Mode</Mode>
    <SriovEnabled>Enabled</SriovEnabled>
    <PortCfg>
      <Port>0</Port>
      <NumOfVF>64</NumOfVF>
    </PortCfg>
    <PortCfg>
      <Port>1</Port>
      <NumOfVF>64</NumOfVF>
    </PortCfg>
  </SriovConfigurations>
```

NPAR Mode

For the following SR-IOV configuration example XML file in NPAR mode:

- SR-IOV is not allowed on NPAR'd partitions (functions) that have FCoE-Offload or iSCSI-Offload enabled on them.
- The SR-IOV configuration XML file does *not* take precedence in the current NPAR configuration. However, the NPAR configuration XML file *does* take precedence in the current SR-IOV configuration. Thus, you must set the NPAR configuration first, and then set the SR-IOV configuration second.
- SR-IOV is not enumerated on the following:
 - ☐ Port 0 Function 0 (with FCoE-Offload enabled)
 - ☐ Port 0 Function 2 (with iSCSI-Offload enabled)
 - ☐ Port 1 Function 5 (with FCoE-Offload enabled)
 - ☐ Port 1 Function 7 (with iSCSI-Offload enabled)

```
<?xml version="1.0" encoding="UTF-8"?>
  <SriovConfigurations>
    <Mode>MF Mode</Mode>
    <SriovEnabled>Disabled</SriovEnabled>
```

```
<PortCfg>
  <Port>0</Port>
  <FunctionCfg>
    <Function>4</Function>
    <NumOfVF>0</NumOfVF>
    <MaxChains>8</MaxChains>
  </FunctionCfg>
  <FunctionCfg>
    <Function>6</Function>
    <NumOfVF>0</NumOfVF>
    <MaxChains>56</MaxChains>
  </FunctionCfg>
</PortCfg>
<PortCfg>
  <Port>1</Port>
  <FunctionCfg>
    <Function>1</Function>
    <NumOfVF>48</NumOfVF>
    <MaxChains>16</MaxChains>
  </FunctionCfg>
  <FunctionCfg>
    <Function>3</Function>
    <NumOfVF>0</NumOfVF>
    <MaxChains>16</MaxChains>
  </FunctionCfg>
</PortCfg>
</SriovConfigurations>
```

Table B-7 lists the range of values for the SR-IOV configuration parameters used in the SR-IOV configuration XML file.

Table B-7. SR-IOV Configuration Parameters

Parameter	Value Range
Mode	MF Mode SF Mode
SriovEnabled	Enabled Disabled
Port	0–3 (Generally, these values are pre-enumerated)

Table B-7. SR-IOV Configuration Parameters (Continued)

Parameter	Value Range
Function	0–7 (BCM57xxx, single-port QL45xxx, QLE344x, and QLE844x adapters) 0–15 (dual- and quad-port QL45xxx adapters and single-, dual-, quad-port QL41xxx adapters) (Generally, these values are pre-enumerated.)
NumOfVF	0–<max allowed> all in groups of 8 total sum per port, where: ■ 32 (quad-port BCM57840 adapters) ■ 64 (single- and dual-port BCM57xxx, QLE344x, and QLE844x adapters) ■ 120 (single-, dual-, and quad-port QL45xxx adapters) ■ 192 (up to 192 VFs assigned to any PF on any port in single-, dual-, or quad-port modes for QL41xxx adapters.)
MaxChains	0–16 (the maximum quantity of RSS or TSS queues that could be allocated to an SR-IOV VF)

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- [“License of libxml2” on page 201](#)
- [“License of libedit \(Linux Only\)” on page 202](#)
- [“License of libcurl” on page 203](#)

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```

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*/
```

D Exit Codes

Table D-1 lists the exit codes that indicate the level of success for a command.

Table D-1. QLogic Control Suite CLI Exit Codes

Code Text	Code Number	Meaning
QCSCLI_OK	0	Operation successful
QCSCLI_QUIT	1	Quit program
QCSCLI_PARAM_ERROR	2	Not correct parameters
QCSCLI_ADAPTER_NOT_FOUND	3	Adapter not found
QCSCLI_CANNOT_LOCK_ADAPTER	4	Cannot lock adapter
QCSCLI_GET_CLOSE_EVENT	5	Get close event
QCSCLI_INIT_FAILED	6	Initialization failed
QCSCLI_UNSUPPORTED_BMAPI_VER	7	QLMAPI is too old
QCSCLI_UNKNOWN_COMMAND	8	Unknown command
QCSCLI_MALLOC_ERROR	9	Memory allocation error
QCSCLI_BMAPI_ERROR	10	QLMAPI call returns error

Table D-1. QLogic Control Suite CLI Exit Codes (Continued)

Code Text	Code Number	Meaning
QCSCLI_OS_NOT_SUPPORTED	11	OS is not supported
QCSCLI_NO_ADVANCED_PARAMS_NIC	12	No advanced parameter for the NIC
QCSCLI_INVALID_ADVANCED_PARAM_DETECTED	13	Invalid advanced parameter detected
QCSCLI_INVALID_ADVANCED_PARAM_SPECIFIED	14	Invalid advanced parameter specified
QCSCLI_INVALID_ADVANCED_VALUE_SPECIFIED	15	Invalid advanced value specified
QCSCLI_FEATURE_NOT_SUPPORTED_FOR_NIC	16	Feature not supported for the NIC
QCSCLI_SET_ADVANCED_PARAM_ERROR	17	Failed to set the advanced parameter with new value
QCSCLI_SYSTEM_REBOOT	18	System reboot required
QCSCLI_UNSUPPORT_PLATFORM	19	System platform is not supported
QCSCLI_NOT_ENOUGH_PRIVILEGE	20	Current user does not have enough privilege
QCSCLI_READ_LICENSE_FILE_ERROR	21	Error in reading license file
QCSCLI_INVALID_LICENSE_KEY	22	Invalid license key
QCSCLI_INVALID_ISCSI_PARAM_SPECIFIED	23	Invalid iSCSI management parameter specified
QCSCLI_INVALID_ISCSI_VALUE_SPECIFIED	24	Invalid iSCSI management value specified
QCSCLI_INVALID_RSC_PARAM_SPECIFIED	25	Invalid resource parameter specified
QCSCLI_INVALID_RSC_VALUE_SPECIFIED	26	Invalid resource value specified
QCSCLI_FEATURE_NOT_SUPPORTED_IN_FCFS	27	Feature not supported in FCFS mode
QCSCLI_PARAM_IS_READ_ONLY	28	This parameter cannot be modified; administrator authority is required

Table D-1. QLogic Control Suite CLI Exit Codes (Continued)

Code Text	Code Number	Meaning
QCCLI_NULL_IP_ADDRESS	29	The current IP address is NULL
QCCLI_CANNOT_UNLOCK_ADAPTER	30	Failed to unlock adapter
QCCLI_INVALID_VALUE_SPECIFIED	31	Invalid value specified
QCCLI_NIC_IS_PART_OF_GEC_LACP_TEAM	32	NIC is part of a GEC or LACP team
QCCLI_REGISTRY_ACCESS_ERROR	33	Error in accessing registry
QCCLI_NOT_AN_ISCSI_BOOT_DEVICE	34	This is not an iSCSI boot device
QCCLI_INVALID_IP_ADDRESS	35	Invalid IP address
QCCLI_DUPLICATE_IP_ADDRESS	36	Duplicate IP address
QCCLI_TEAM_DRIVER_NOT_LOAD	37	NIC(%s) driver must be loaded to make it a member of a team
QCCLI_NDIS6_DRIVER_REQUIRED	38	NDIS 6 driver is required for the NIC to join the team in Windows Vista and later
QCCLI_TEAM_UNKNOW_NIC	39	Unknown NIC (%s)
QCCLI_INVALID_SUBNET_MASK	40	Invalid subnet mask
QCCLI_INVALID_CMD	41	Invalid command
QCCLI_INVALID_FCOE_PARAM_SPECIFIED	42	Invalid FCoE management parameter specified
QCCLI_INVALID_FCOE_VALUE_SPECIFIED	43	Invalid FCoE management value specified
QCCLI_DEFAULT_FAILED	44	Failed to set the NDIS advanced settings as default values
QCCLI_DATA_OBJECT_IS_NULL	45	Failed to get the data object

Table D-1. QLogic Control Suite CLI Exit Codes (Continued)

Code Text	Code Number	Meaning
QCCLI_CLIID_IS_NULL	46	CLI ID is NULL
QCCLI_EXCEED_MAX_PWD_LENGTH	47	Exceed the maximum length of password
QCCLI_GET_BREAK_EVENT	48	Get break event
QCCLI_INVALID__FILE_NAME	49	Invalid iSCSI interface () file name specified
QCCLI_DUPLICATED__FILE_NAME	50	Duplicated file name specified
QCCLI_INCOMPATIBLE_ISCSI_PARAM_SPECIFIED	51	Incompatible iSCSI management parameter specified
QCCLI_ISCSI_PARAM_MISSING	52	iSCSI management parameter is missing
QCCLI_NOT_YET_IMPL	1000	This functionality is not yet implemented
QCCLI_UNWIND	1001	Unwind to the parent processor
QCCLI_NOT_APPLICABLE	1002	Command not applicable
QCCLI_NO_TARGET_SEL	1003	No active target selection
QCCLI_INVALID_CONTEXT	1004	Not a valid context
QCCLI_INVALID_FORMAT	1005	Invalid format selection
QCCLI_INVALID_TARGET_ID	1006	Invalid target identifier
QCCLI_FILE_DOES_NOT_EXIST	1007	File does not exist
QCCLI_INVALID_TEAM_NAME	1008	Supplied team name is invalid
QCCLI_TEAM_COMMIT_FAILED	1009	Failed to commit the team operation
QCCLI_TEAM_REMOVE_FAILED	1010	Failed to remove the team

Table D-1. QLogic Control Suite CLI Exit Codes (Continued)

Code Text	Code Number	Meaning
QCSCLI_CANNOT_OPEN_FILE	1011	Failed to open the file handle
QCSCLI_ERR_CANNOT_SET_IPADDR	1012	Failed to set IP address
QCSCLI_ERR_CANNOT_GET_NIC_PCI_INFO	1013	Failure retrieving NIC information
QCSCLI_ERR_RETRIEVE_IP_ADDR	1014	Error retrieving IP address information
QCSCLI_FAILED_GET_INFO	1015	Failed to get info from the data container
QCSCLI_WRONG_OPTION_FLAG	1016	General team configuration file parsing error
QCSCLI_EXCEEDMAXVLAN	1017	A maximum of 64 VLANs are allowed
QCSCLI_CANNOT_CREATE_LIVE_LINK	1018	LiveLink™ support applied only to the SLB team
QCSCLI_EXCEED_MAX_TARGET_IP	1019	LiveLink support allows up to four links
QCSCLI_TOO_MANY_PHY_NIC	1020	A maximum of eight NICs are allowed in a team
QCSCLI_CANNOT_CREATE_FECGEC_8023AD	1021	Cannot create FEC-GEC or 802.3ad team with standby adapter
QCSCLI_LL_IP_TARGET_IP_TYPE_MISMATCH	1022	Invalid IPv6 address
QCSCLI_INVALID_RANGE	1023	Value is out of range
QCSCLI_INVALID_INTERVAL	1024	Invalid probe retry frequency
QCSCLI_DUPLICATE_OPTION	1025	The same option has been specified previously
QCSCLI_DUPLICATE_MAC_ADDRESS	1026	Duplicate adapter physical MAC address
QCSCLI_DUPLICATE_VLANID	1027	Duplicate VLAN name
QCSCLI_TEAM_ALREADY_EXISTS	1028	Team with the specified name already exists

Table D-1. QLogic Control Suite CLI Exit Codes (Continued)

Code Text	Code Number	Meaning
QCCLI_CANNOT_MATCH_MAC_ADDR	1030	Cannot find device using the specified MAC address
QCCLI_NO_TEAM_TO_CONFIG	1031	No team to configure
QCCLI_CANNOT_CREATE_TEAM	1032	Failure while creating team
QCCLI_NO_LINK_FOR_IP_CFG	1033	No link is present in team to set IP address
QCCLI_ONE_OR_MORE_CREATE_FAILED	1034	Creation of one or more teams failed
QCCLI_TEAM_NO_MEMBER	1035	A team is required to have at least one valid member
QCCLI_NO_BRCM_NIC_IN_TEAM	1036	Team requires at least one Marvell NIC
QCCLI_ONLY_QLOGIC_NIC_FOR_VLAN	1037	Only Marvell-certified adapters are supported in VLAN
QCCLI_CANNOT_SET_IPADDR	1038	Failed to assign IP address on the virtual adapter
QCCLI_INVALID_CFG	1039	Invalid configuration
QCCLI_SET_ADVANCE_PARAM_FAILED	1040	Failed to set advanced parameter
QCCLI_INTERNAL_ERROR_INVALID_DATA	1041	Invalid or NULL data found
QCCLI_INVALID_PARAMETER	1042	Invalid parameter; parameter is too few
QCCLI_NIC_NOT_SUPPORTED	1043	The current NIC is not supported for this operation
QCCLI_SET_ASF_FAILED	1044	Failed to set the ASF table
QCCLI_SET_POWER_MGMT_FAILED	1045	Failed to set power management configuration
QCCLI_INVALID_TARGET_CMD	1046	Command or target identifier is invalid
QCCLI_NO_ISCSI_SESSIONS	1047	No iSCSI sessions exists on the system

Table D-1. QLogic Control Suite CLI Exit Codes (Continued)

Code Text	Code Number	Meaning
QCCLI_SET_MGMT_OBJ_FAILED	1048	Failed to set management object
QCCLI_SHOW_USAGE	1049	Displayed usage help message
QCCLI_CANNOT_CONNECT_WM	1050	Cannot connect WMI
QCCLI_EXCEEDMAXTAGGEDVLAN	1051	Only a maximum of 63 tagged VLANs are allowed
QCCLI_NO_TEAM_AVAILABLE	1052	There is no team to save
QCCLI_NOT_CONFIGURABLE	1053	Not configurable
QCCLI_CANNOT_SUSPEND_RESUME_DRIVER	1054	Failed to suspend or resume the driver
QCCLI_CANNOT_RESTART_DRIVER	1055	Failed to restart the driver
QCCLI_GET_CONFIG_FAILED	1056	Failed to get configuration
QCCLI_SET_CONFIG_FAILED	1057	Failed to save configuration
QCCLI_ACTION_FAILED	1058	Current action failed
QCCLI_FILE_TOO_BIG	1059	File is too big
QCCLI_FILE_READ_ERR	1060	Error read file
QCCLI_PORT_DIAG_FAILED	1061	Any port diagnostic test failed
QCCLI_PORT_DIAG_UNSUPPORTED_TEST	1062	Unsupported port diagnostic test
QCCLI_PORT_DIAG_BREAK	1063	User break port diagnostic test
QCCLI_PORT_DIAG_TIMEOUT	1064	Port diagnostic test timeout
QCCLI_NPAR_OUT_OF_SYNC	1065	NPAR is out of sync; need to reboot

Table D-1. QLogic Control Suite CLI Exit Codes (Continued)

Code Text	Code Number	Meaning
QCSCLI_ISCSI_CFG_NEED_SESSION_LOGOUT	1066	Cannot change iSCSI configuration with session
QCSCLI_ISCSI_CFG_NEED_RELOGIN_SESSION	1067	iSCSI configuration needs session to re-login to take effect
QCSCLI_CLI_MODE_EXIT	1068	CLI mode exit
QCSCLI_DATA_NOT_SUPPORTED	1069	Feature is not supported
QCSCLI_DATA_INVALID	1070	Data is invalid
QCSCLI_ACTION_FAILED_BECAUSE_ACCESS_DENIED	1071	Action failed because of access denied
QCSCLI_CLI_MODE_HOST_SPECIFIED	1074	Host is not specified to connect

E Revision History

Document Revision History	
Revision A, March 27, 2015	
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Changes	Sections Affected
Add support for the following OSs: Window Server 2022 VMware ESXi 7.0 U2 RHEL 8.4 SLES 15 SP3 VMware EXSi 7.0 U2 Removed support for the following OSs: Windows Server 2012, 2016 RHEL 7.4 RHEL 8.2 SLES 15 SP1 VMware ESXi 7.0 U1	All
Removed QCC GUI text; this application is no longer supported.	All
Updated steps for setting up a Marvell technical support account.	“Technical Support” on page xiv
Updated steps for locating and downloading product firmware, software, and documentation from the Marvell Web site.	“Downloading Updates and Documentation” on page xiv

Removed Knowledgebase section.	“Preface” on page x
Updated Figure 1-1 .	“Windows System Architecture” on page 2
Updated Figure 1-2 .	“Linux Systems Architecture” on page 2
In the bullets for RHEL 5 and SLES 10, added Linux kernel information.	“Linux Feature Limitations” on page 8
Corrected Step 6 ; updated the table listing the installer files.	“QCS CLI and Agent Installation on Windows and Linux” on page 13
Corrected Step 2 .	“Agent Installation on Windows and Linux” on page 14
Removed references to netqlremote, qlremote, iqlremote, and SuperInstaller	
Removed reference to qlremote.	“Agent Installation on Ubuntu” on page 14
Added capability for changing the DCBX mode.	“Configure Advanced Port Settings” on page 45
Added a NOTE indicating that this command was deprecated by Microsoft in the OS.	“Configure System TOE” on page 77
In the first paragraph, removed references to the Windows OS version.	
In the definition of the value keyword: Changed the first sentence to “Enables, disables, or automates TOE ...” Removed the second sentence (now incorporated into first sentence.)	
Added a NOTE indicating that this command was deprecated by Microsoft in the OS.	“Configuring Systoe for the System” on page 154
Changed the first two bullets to “For ... CLI and agents:”	“Windows Firewall Configuration” on page 168
Removed subsection for “SLES 11.x and SLES 10.x”; these OSs are no longer supported.	“Linux Firewall Configuration” on page 170
Added a description of how to connect to an iSCSI target.	“Connecting an iSCSI Target” on page 163
Removed Index.	Index

Glossary

adapter

The board that interfaces between the host system and the target devices. Adapter is synonymous with *Host Bus Adapter*, *Host Channel Adapter*, *host adapter*, and *adapter board*.

adapter port

A port on the adapter board.

adapter port beacon

An [LED](#) on the adapter. Flashing it enables you to locate the adapter.

bandwidth

A measure of the volume of data that can be transmitted at a specific transmission rate. A 1Gbps or 2Gbps Fibre Channel port can transmit or receive at nominal rates of 1 or 2Gbps, depending on the device to which it is connected. This corresponds to actual bandwidth values of 106MB and 212MB, respectively.

challenge-handshake authentication protocol

See [CHAP](#).

CHAP

Challenge-handshake authentication protocol (CHAP) is used for remote logon, usually between a client and server or a Web browser and Web server. A challenge/response is a security mechanism for verifying the identity of a person or process without revealing a secret password that is shared by the two entities. Also referred to as a *three-way handshake*.

CLI

Command line interface. A program interface driven by entering commands and parameters.

command line interface

See [CLI](#).

data center bridging

See [DCB](#).

data center bridging exchange

See [DCBX](#).

DCB

Data center bridging. Provides enhancements to existing 802.1 bridge specifications to satisfy the requirements of protocols and applications in the data center. Because existing high-performance data centers typically comprise multiple application-specific networks that run on different link layer technologies (Fibre Channel for storage and Ethernet for network management and LAN connectivity), DCB enables 802.1 bridges to be used for the deployment of a converged network where all applications can be run over a single physical infrastructure.

DCBX

Data center bridging exchange. A protocol used by [DCB](#) devices to exchange configuration information with directly connected peers. The protocol may also be used for misconfiguration detection and for configuration of the peer.

device

A computer subsystem, such as an adapter card, that mediates data in a computer network. The term device is used interchangeably with *target* and *target device*.

DHCP

Dynamic host configuration protocol. Enables computers on an IP network to extract their configuration from servers that have information about the computer only after it is requested.

driver

The software that interfaces between the file system and a physical data storage device or network media.

dynamic host configuration protocol

See [DHCP](#).

enhanced transmission selection

See [ETS](#).

Ethernet

The most widely used LAN technology that transmits information between computer, typically at speeds of 10 and 100 million bits per second (Mbps).

ETS

Enhanced transmission selection. A standard that specifies the enhancement of transmission selection to support the allocation of bandwidth among traffic classes. When the offered load in a traffic class does not use its allocated bandwidth, enhanced transmission selection allows other traffic classes to use the available bandwidth. The bandwidth-allocation priorities coexist with strict priorities. ETS includes managed objects to support bandwidth allocation.

FCoE

Fibre Channel over Ethernet. A new technology defined by the T11 standards body that allows traditional Fibre Channel storage networking traffic to travel over an Ethernet link by encapsulating Fibre Channel frames inside [Layer 2](#) Ethernet frames. For more information, visit www.fcoe.com.

Fibre Channel over Ethernet

See [FCoE](#).

firmware

Low-level software typically loaded into read-only memory and used to boot and operate an intelligent device.

graphical user interface

See [GUI](#).

GUI

A user interface that is based upon icons and visual relationships rather than text.

host

One or more adapters governed by a single memory or CPU complex.

Host Bus Adapter

An adapter that connects a host system (the computer) to other network and storage devices.

initiator

System component, such as a network interface card, that originates an I/O operation.

Internet Protocol

See [IP](#).

Internet simple name service

See [iSNS](#).

Internet small computer system interface

See [iSCSI](#).

Internet wide area RDMA protocol

See [iWARP](#).

IP

Internet protocol. A method by which data is sent from one computer to another over the Internet. IP specifies the format of packets, also called *datagrams*, and the addressing scheme.

IPv4

Internet protocol version 4. A data-oriented protocol used on a packet switched inter-network (Ethernet, for example). It is a best-effort delivery protocol: it does not guarantee delivery, ensure proper sequencing, or avoid duplicate delivery. These aspects are addressed by an upper layer protocol (TCP, and partly by UDP). IPv4 does, however, provide data integrity protection through the use of packet checksums.

IPv6

Internet protocol version 6. Next-generation version of IP that, among other things, lengthens the IP address from 32 bits to 128 bits.

iSNS

Internet simple name service is used for discovery and management of IP-based SANs.

iWARP

Internet wide area RDMA protocol. A networking protocol that implements RDMA for efficient data transfer over IP networks. iWARP is designed for multiple environments, including LANs, storage networks, data center networks, and WANs.

iQN

iSCSI qualified name. iSCSI node name based on the initiator manufacturer and a unique device name section.

iSCSI

Internet small computer system interface. Protocol that encapsulates data into IP packets to send over Ethernet connections.

Layer 2

Refers to the data link layer of the multilayered communication model, Open Systems Interconnection (OSI). The function of the data link layer is to move data across the physical links in a network, where a switch redirects data messages at the layer 2 level using the destination MAC address to determine the message destination.

LED

Light-emitting diode. Status indicator on a switch, router, adapter, or other device.

light-emitting diode

See [LED](#).

logical unit number

See [LUN](#).

LUN

Logical unit number, a subdivision of a SCSI target. It is the small integer handle that differentiates an individual disk drive or partition (volume) within a common SCSI target device such as a disk array.

Technically, a LUN can be a single physical disk drive, multiple physical disk drives, or a portion (volume) of a single physical disk drive. However, LUNs are typically not entire disk drives but rather virtual partitions (volumes) of a RAID set.

Using LUNs, the Fibre Channel host can address multiple peripheral devices that may share a common controller.

message signaled interrupts

See [MSI](#), [MSI-X](#).

MSI, MSI-X

Message signaled interrupts. One of two PCI-defined extensions to support message signaled interrupts (MSIs), in PCI 2.2 and later and PCI Express. MSIs are an alternative way of generating an interrupt through special messages that allow emulation of a pin assertion or deassertion.

MSI-X (defined in PCI 3.0) allows a device to allocate any number of interrupts between 1 and 2,048 and gives each interrupt separate data and address registers. Optional features in MSI (64-bit addressing and interrupt masking) are mandatory with MSI-X.

N_Port

Node port. A port that connects by a point-to-point link to either a single N_Port or a single F_Port. N_Ports handle creation, detection, and flow of message units to and from the connected systems. N_Ports are end ports in virtual point-to-point links through a fabric, for example, N_Port to F_Port to F_Port to N_Port using a single Fibre Channel fabric switch.

N_Port ID virtualization

See [NPIV](#).

network interface card

See [NIC](#).

NIC

Computer card installed to enable a dedicated network connection.

NIC partitioning

See [NPAR](#).

node port

See [N_Port](#).

NPAR

NIC partitioning. The division of a single NIC port into multiple physical functions or partitions, each with a user-configurable bandwidth and personality (interface type). Personalities include **NIC**, **FCoE**, and **iSCSI**.

NPIV

N_Port ID virtualization. The ability for a single physical Fibre Channel end point (**N_Port**) to support multiple, uniquely addressable, logical end points. With NPIV, a host Fibre Channel Adapter is shared in such a way that each virtual adapter is assigned to a virtual server and is separately identifiable within the fabric. Connectivity and access privileges within the fabric are controlled by identification of each virtual adapter and, hence, the virtual server using each virtual adapter.

operating system

See **OS**.

OS

Operating system. After being loaded by the boot program, the program that manages all other programs on a computer.

path

A path to a device is a combination of a adapter **port instance** and a target port as distinct from internal paths in the fabric network. A fabric network appears to the operating system as an opaque network between the adapter (initiator) and the target.

Because a path is a combination of an adapter and a target port, it is distinct from another path if it is accessed through a different adapter and/or it is accessing a different target port. Consequently, when switching from one path to another, the driver might be selecting a different adapter (initiator), a different target port, or both.

This is important to the driver when selecting the proper method of failover notification. It can make a difference to the target device, which might have to take different actions when receiving retries of the request from another initiator or on a different port.

PCI Express (PCIe)

A third-generation I/O standard that allows enhanced Ethernet network performance beyond that of the older peripheral component interconnect (PCI) and PCI extended (PCI-x) desktop and server slots.

ping

A computer network administration utility used to test whether a specified host is reachable across an IP network, and to measure the round-trip time for packets sent from the local host to a destination computer.

port

Access points in a device where a link attaches. The four types of ports are:

- **N_Port**—a Fibre Channel port that supports point-to-point topology.
- **NL_Port**—a Fibre Channel port that supports loop topology.
- **FL_Port**—a port in a fabric where an **N_Port** can attach.
- **FL_Port**—a port in a fabric where an **NL_Port** can attach.

port instance

The number of the port in the system. Each adapter may have one or multiple ports, identified with regard to the adapter as port 0, port 1, and so forth. To avoid confusion when dealing with a system containing numerous ports, each port is assigned a port instance number when the system boots up. So port 0 on an adapter might have a port instance number of 8, for example, if it is the eighth port discovered by the system.

RDMA

Remote direct memory access. The ability for one node to write directly to the memory of another (with address and size semantics) over a network. This capability is an important feature of VI networks.

RDMA over Converged Ethernet

See [RoCE](#), [RoCEv2](#).

remote direct memory access

See [RDMA](#).

RoCE, RoCEv2

RDMA over Converged Ethernet. A network protocol that allows remote direct memory access (RDMA) over a converged or a non-converged Ethernet network. RoCE is a link layer protocol that allows communication between any two hosts in the same Ethernet broadcast domain.

secure socket layer

See [SSL](#).

simple network management protocol

See [SNMP](#).

single root input/output virtualization

See [SR-IOV](#).

SNMP

Simple network management protocol. SNMP is a networking protocol that enables you to monitor the router using third-party applications that use SNMP.

SSL

Secure socket layer. A protocol that secures connections to the switch for Enterprise Fabric Suite, QuickTools, the API, and SMI-S.

SR-IOV

Single root input/output virtualization. A specification by the PCI SIG that enables a single PCIe device to appear as multiple, separate physical PCIe devices. SR-IOV permits isolation of PCIe resources for performance, interoperability, and manageability.

target

The storage-device endpoint of a SCSI session. Initiators request data from targets. Targets are typically disk-drives, tape-drives, or other media devices. Typically a SCSI peripheral device is the target but an adapter may, in some cases, be a target. A target can contain many LUNs.

A target is a device that responds to a requested by an initiator (the host system). Peripherals are targets, but for some commands (for example, a SCSI COPY command), the peripheral may act as an initiator.

TCP

Transmission control protocol. A set of rules to send data in packets over the Internet protocol.

TCP/IP offload engine

See [TOE](#).

TOE

TCP/IP offload engine. Technology used in iSCSI to optimize throughput by accelerating the performance of TCP/IP.

transmission control protocol

See [TCP](#).

vLAN

Virtual logical area network (LAN). A group of hosts with a common set of requirements that communicate as if they were attached to the same wire, regardless of their physical location. Although a VLAN has the same attributes as a physical LAN, it allows for end stations to be grouped together even if they are not located on the same LAN segment. VLANs enable network reconfiguration through software, instead of physically relocating devices.



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